Gated Action Dissipation for Bayesian Inverse Problems

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Abstract

In a nonlinear Bayesian inverse problem, the observations can constrain some directions in parameter space much more strongly than others, and these directions can change from one region of the posterior to another. We propose Gated Action Dissipation (GAD) to sample such posteriors. GAD estimates the score of the noise-perturbed target using samples from a current law and target derivatives evaluated at those samples. It adds a correction to the score of the noise-perturbed current law and weights this correction with a matrix formed from target curvature. Simulating reverse diffusion with this score estimate produces new unweighted samples. The next score estimate uses these samples, and the process repeats. GAD recovers a Gaussian target's exact noisy score in one round from any nonempty set of reference samples, provided the target derivatives are exact. We bound the forward Kullback–Leibler divergence after each round with exact conditional expectations for strongly log-concave perturbations of Gaussian targets. The contribution from the initial error decays geometrically when the perturbation derivatives vary sufficiently little. Synthetic tests show how the choice of gate affects sampling accuracy and how separation slows the correction of mixture weights. The inverse-problem tests use GAD with Gauss–Newton curvature and a gradually introduced likelihood for acoustic trajectory inversion and joint earthquake location. GAD follows the curved structure in the benchmark posterior projections and has the lowest mean error among the tested samplers on each problem's primary sliced Wasserstein metric at the final request allowance.

Keywords. Bayesian inverse problems, diffusion sampling, score estimation, matrix gates, Gauss–Newton approximation.

1 Introduction

An observation can tell us a great deal about one combination of parameters and very little about another. The posterior can then concentrate near a lower-dimensional set, with substantial uncertainty along the set. For a nonlinear forward model, the combinations that the data constrain most strongly can change along the set. To sample the posterior, we need to resolve both the narrow directions and how probability is distributed along the set [4, 36].

Suppose we draw samples from a broad distribution and weight them by a sharply concentrated likelihood. A few samples can account for most of the weight [1]. A sharply concentrated posterior can also make the sampling dynamics stiff because the target score $s_0 = \nabla \log p_0$, where p_0 is the posterior density, can change rapidly across the narrow directions. We identify the locally

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constrained directions using the negative derivative of the target score, the observed information $H_0 = -\nabla^2 \log p_0$, which measures local target curvature. For a Bayesian inverse problem, H_0 can be approximated by a Gauss–Newton (GN) matrix formed from derivatives of the forward model [4, 27]. Computing s_0 and H_0 requires only the unnormalized target density.

Gated Action Dissipation (GAD) begins with samples from a current law q. We evaluate the target derivatives at the samples from q, which we call the *reference samples* for the round. GAD estimates the noisy target score field from the reference samples and cached target derivatives, then simulates reverse diffusion to transport fresh draws from a Gaussian terminal law. The unweighted endpoints serve as the reference samples for the next round. Each round can start from samples of q even when its density is unavailable. The method accesses the target through derivatives of its unnormalized density.

Reverse diffusion requires $s_t = \nabla \log p_t$, where p_t is the target density after adding noise for time t [3, 35]. Consider a noisy state of the Ornstein–Uhlenbeck (OU) process and the clean states that could have produced it. Tweedie's identity gives s_t from the conditional mean of the clean state under the target posterior, given the noisy state [15]. The target-score identity (TSI) gives the same score from the conditional mean of the clean score under the same target posterior [10]. In our earlier work, we combined the Tweedie and target-score estimates with a scalar weight that depends on the noisy state and time [13]. The Laplace–Fisher Gate Identity (LFGI) gives a matrix weight determined by the conditional mean of H_0 under the target law. Its name reflects the relation between the curvature used in a Laplace approximation and Fisher information under the target OU posterior [14].

GAD averages the Tweedie and target-score signals over clean states from the current law q, conditioned on the noisy state. The q-LFGI matrix is formed from the mean of the observed information H_0 under this conditional law. Multiplying the difference between the target-score and Tweedie means by this matrix gives the correction to the noisy score of q. In the name GAD, "gated action" denotes this matrix-weighted score correction, and "dissipation" refers to the decrease in Kullback–Leibler divergence along the forward diffusion used in our analysis. For a Gaussian target, the matrix gives less weight to the correction in directions of high curvature. Each reference sample yields the same corrected score when the Gaussian target derivatives are exact. Averaging over any nonempty reference set recovers the noisy target score (Section 6). For a target whose curvature varies with position, the estimated noisy score can depend on where the reference samples lie.

We ask how many target requests are needed to reach a given sampling accuracy. A target request evaluates the energy $U = -\log \widetilde{p}_0$ (with $\widetilde{p}_0$ the unnormalized target density), its gradient, or a curvature matrix at a point. GAD reuses the same cached derivatives to evaluate the score field throughout reverse transport in each round.

1. We derive the q-LFGI correction from a conditional Poisson equation for the difference between the target and current scores. The corrected-score error depends on the conditional covariance between the target curvature and the gradient of the Poisson solution (Sections 4 to 6).
2. We use the correction to construct the iterative sampler in Algorithm 1. With exact conditional expectations, we bound the forward Kullback–Leibler (KL) divergence of the iteration for strongly log-concave perturbations of Gaussian targets whose perturbation derivatives vary sufficiently little (Section 8). The contribution from the initial error decays geometrically for any initial density with finite forward KL.
3. We test Gaussian score recovery, the correction of mode weights, and the effect of the matrix gate on synthetic targets. On two-dimensional targets, we compare GAD with the Metropolis-adjusted Langevin algorithm (MALA) and Hamiltonian Monte Carlo (HMC).

For two nonlinear inverse problems, we compare GAD with Markov chain Monte Carlo (MCMC) methods at common target-request allowances (Sections 11 to 13). Annealed Gauss–Newton GAD (GN-GAD) has the lowest mean error on each inverse problem's primary sliced Wasserstein metric at the final allowance. GAD corrects the component weights slowly in the most separated mixtures tested.

2 Related work

Tweedie's identity and the target-score identity both express the diffused target score $s_t = \nabla \log p_t$ as a conditional expectation. In both identities, the averaging law is the target posterior over clean states given the noisy state. Tweedie's identity averages a signal formed from the clean state [15], as in denoising score matching [38]. The target-score identity instead averages the scaled clean target score [10]. Our earlier work combines sample estimates of the Tweedie and target-score identities with a scalar weight that varies with state and time [13]. A coefficient depending only on time is derived in control-variate score matching [23]. LFGI chooses a matrix weight to minimize conditional mean squared score error under the target posterior [14]. The Tweedie and target-score signals have equal means under that posterior, so the LFGI gate changes the variance of their blend. GAD instead averages the signals and target curvature under the current law conditioned on the noisy state. The Tweedie and target-score means can then differ. The q-LFGI matrix, computed from the conditional mean of target curvature, weights the difference between the target-score and Tweedie means to correct the noisy current score (Sections 4 and 5).

Conditional expectations can also be estimated during sampling. For a noisy state, Reverse Diffusion Monte Carlo estimates the score by sampling a conditional distribution [22], while Stochastic Localization via Iterative Posterior Sampling estimates a conditional denoising mean by MCMC [18]. Alternating Gaussian noising with Metropolis-within-Gibbs denoising gives the mode-crossing updates in Diffusive Gibbs Sampling [7]. In Conditional Diffusion Sampling, parallel tempering provides the initial law, after which samples are transported with a closed-form drift [5]. Intermediate score estimation can also be coupled to slow and fast diffusions, as in Sampling by Averaging [8]. Within a GAD round, every score-field evaluation point reuses the same reference samples and the target derivatives already evaluated there to estimate the Tweedie and target-score conditional means.

Sampling and score training need not be separate stages. Iterated Denoising Energy Matching alternates draws from a current model with further score training [2]. Other learned approaches parameterize controls or reverse dynamics, including Path Integral Samplers, controlled diffusion methods, and Adjoint Sampling [20, 32, 37, 41]. A learned score guides the sequential Monte Carlo (SMC) sampler in the Particle Denoising Diffusion Sampler [31]. In Diffusion Path Samplers via Sequential Monte Carlo (DPSMC), by contrast, SMC evolves auxiliary variables for score estimation itself. The DPSMC construction also yields control-variate schedules [40]. For GAD, the score field is built directly from the current samples, and the equally weighted transport endpoints supply the next round. The gate experiment compares coefficients adapted from earlier work within the GAD sampling iteration (Section 11.3).

Curvature information is also used to design MCMC proposals for Bayesian inverse problems. Local geometry determines the proposals in Riemann-manifold Langevin and Hamiltonian methods [4, 17]. In Stochastic Newton MCMC, gradients and Hessians specify local Gaussian proposals [27], with a variant reusing a low-rank Hessian approximation at the maximum a posteriori (MAP) estimate [30]. Adaptation can also use a subspace identified by Hessian information, as in dimension-independent likelihood-informed MCMC [9]. In GAD, the conditional average of target

curvature under the current law determines the matrix weighting the diffusion-score correction. Our inverse-problem comparisons include MCMC methods with a fixed MAP Hessian or local Gauss–Newton matrices.

For particle transport, a kernel-based functional gradient of reverse KL gives the update in Stein variational gradient descent [26]. Taking the second variation of the same objective instead gives the Newton update in Stein variational Newton (SVN), which also describes curvature-informed kernels [12]. With added noise and a drift correction, stochastic SVN has the product target law as the invariant law of its continuous-time system [25]. Ensemble information enters differently in affine-invariant interacting Langevin dynamics, where the ensemble covariance preconditions the particle updates [16]. Rather than updating the reference samples by variational or interacting-Langevin dynamics, a GAD round uses the reference samples to estimate a diffusion score and transports fresh draws with that field. The GAD forward-KL contraction bound concerns the law of the transported draws at the end of a round.

Importance weights can concentrate when the proposal and target differ. The concentration of importance weights depends on the intrinsic dimension [1]. GAD gives equal weight to the endpoints passed between rounds. The OU kernel can assign most of the weight in a conditional average to a few reference samples during a round. For a Gaussian target, the corrected noisy score remains exact even when the OU conditional weights concentrate (Section 6). For separated mixtures, the clean score can change little when the component weights change [39]. We vary component separation and measure the clean-score discrepancy and the reduction in forward KL after one round (Section 11.2).

Girsanov comparisons relate reverse-diffusion error to integrated squared score error and the mismatch at the terminal law [6, 24]. We bound the integrated error of the corrected score by the KL dissipated along the forward OU diffusion for the perturbed Gaussian targets covered by our theorem. Combining the integrated score-error bound with the Girsanov comparison gives the forward-KL bound after one GAD round with exact conditional expectations (Section 8).

3 Relative scores and reverse diffusion

The OU process smooths the target and current laws with the same Gaussian kernel. The difference $d_t = s_t - s_t^q$ between the noisy target and current scores controls how quickly the KL divergence decreases along the forward diffusion. Adding the difference between the noisy target and current scores to the noisy current score recovers the noisy target score. Table 1 collects the notation for the OU laws and score correction.

3.1 Target and current OU paths

Let $\widetilde{p}_0$ be a positive unnormalized target density, and let $p_0 = \widetilde{p}_0/Z_{\mathrm{tar}}$ where $Z_{\mathrm{tar}} = \int_{\mathbb{R}^d} \widetilde{p}_0(x)\,\mathrm{d}x$. The target derivatives are

$$s_0(x) = \nabla_x \log \widetilde{p}_0(x), \qquad H_0(x) = -\nabla_x^2 \log \widetilde{p}_0(x).$$

Here s_0 is the clean score and H_0 is the observed information. Both are independent of Z_{tar}. The convergence theorem uses H_0, and the inverse-problem implementation uses the Gauss–Newton matrix in (9.1).

Let X_0 have law q on $\mathbb{R}^d$. The OU state at $t > 0$ is given by the following representation, where $\xi \sim \mathcal{N}(0, I_d)$ is independent of X_0.

$$Y_t = \alpha_t X_0 + \sqrt{\gamma_t}\,\xi, \qquad \alpha_t = e^{-t}, \qquad \gamma_t = 1 - e^{-2t}.$$

Table 1: **Notation.** Target OU path, current OU path, and GAD iteration. Scores use s, observed information matrices use H, and matrix gates use G. Hats indicate quantities computed from reference samples.

Notation	Meaning
d, I_d	Parameter dimension and $d \times d$ identity matrix.
X_0, Y_t, x, y	Clean random state, noisy random state, and their realizations.
α_t, γ_t	OU mean factor e^{-t} and noise variance $1 - e^{-2t}$.
$K_t^{\mathrm{OU}}(y \mid x)$	OU transition density.
$\widetilde{p}_0, p_0, q$	Unnormalized target density, normalized target density, and current law.
p_t, q_t	Target and current OU marginal densities.
$\rho_{y,t}, \rho_{y,t}^q$	Laws of $X_0 \mid Y_t = y$ when X_0 has law p_0 or q.
s_0, s_t, s_t^q	Clean target score, noisy target score, and noisy current score.
H_0, H^q	Target observed information and its mean under $\rho_{y,t}^q$.
$H_{0,\beta}^{\mathrm{GN}}$	Gauss–Newton approximation to the observed information at likelihood power β.
R_t, d_t	Density ratio p_t/q_t and score difference $\nabla \log R_t$.
$m_t^q, G_{\mathrm{LFGI}}^q$	Conditional mean of the target-score signal minus the Tweedie signal under $\rho_{y,t}^q$, and the matrix gate.
a_t^q, e_t^q	Correction $G_{\mathrm{LFGI}}^q m_t^q$ and score error $d_t - a_t^q$.
$s_t^{a,q}, \widehat{s}_t^{a,q}$	Corrected score with exact expectations and its sample-based version.
$D(q), E(q)$	Path norms of d_t and e_t^q weighted by p_t on $[0, T]$.
ϕ_d, Φ_T, Φ	Standard Gaussian density, exact-expectation GAD map at horizon T, and its infinite-horizon limit.
N, K	Number of reference samples per round and number of rounds.
α, α_t	Inverse-problem parameter vector and scalar OU attenuation, respectively; α_i is a parameter component.
$T, t_{\min}$	Terminal OU time and numerical endpoint time; the analysis uses endpoint zero.
U_{like}, β	Data misfit and likelihood power in Section 9.

Write $K_t^{\mathrm{OU}}(y \mid x)$ for the $\mathcal{N}(\alpha_t x, \gamma_t I_d)$ transition density evaluated at y. The OU marginal density q_t is the integral of this kernel against q. The law of X_0 given $Y_t = y$ is

$$\rho_{y,t}^q(\mathrm{d}x) := \frac{K_t^{\mathrm{OU}}(y \mid x) q(\mathrm{d}x)}{q_t(y)}.$$

When q has a density we also write it as $q(x)$. We use p_0 for the target density and, when used as a law, the measure $p_0(x)\,\mathrm{d}x$. Replacing q by p_0 gives the target OU marginal p_t and the target OU posterior

$$\rho_{y,t}(\mathrm{d}x) = \frac{K_t^{\mathrm{OU}}(y \mid x) p_0(x) \mathrm{d}x}{p_t(y)}.$$

The target and current scores are $s_t(y) = \nabla_y \log p_t(y)$ and $s_t^q(y) = \nabla_y \log q_t(y)$, respectively. Given a specified field $s(y, t)$, the reverse OU stochastic differential equation (SDE) is [3, 35]

$$\mathrm{d}Z_t = \{-Z_t - 2s(Z_t, t)\}\mathrm{d}t + \sqrt{2}\,\mathrm{d}\overline{B}_t, \qquad t : T \downarrow t_{\min}. \tag{3.1}$$

Here $\overline{B}_t$ is standard Brownian motion in the reverse time direction. The initial condition for (3.1) is specified at T. GAD uses ϕ_d, the $\mathcal{N}(0, I_d)$ density, as the terminal density. In the analysis, the interval is $[0, T]$, and the difference between ϕ_d and the exact terminal density p_T contributes the additive error $\mathrm{KL}(p_T \| \phi_d)$.

3.2 Relative density, relative score, and KL dissipation

For a fixed current law q, we suppress its superscript in the density ratio and relative score, writing

$$R_t(y) = \frac{p_t(y)}{q_t(y)}, \qquad d_t(y) = \nabla \log R_t(y) = s_t(y) - s_t^q(y).$$

The target and current laws evolve under the same OU diffusion, so the relative de Bruijn identity gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(p_t\|q_t) = -\mathbb{E}_{p_t}[\|d_t\|^2].$$

The expectation $\mathbb{E}_{p_t}$ is over $Y_t \sim p_t$, and the norm inside it is Euclidean. For a vector field $u_t(y)$ on a horizon $T > 0$, write

$$\|u\|_p^2 := \int_0^T \mathbb{E}_{p_t}[\|u_t\|^2]\mathrm{d}t.$$

Write $D(q) := \|d\|_p$, with the horizon T implicit in $D(q)$ and in the error norm $E(q)$ below. Integrating the derivative gives

$$\boxed{D(q)^2 = \mathrm{KL}(p_0\|q) - \mathrm{KL}(p_T\|q_T).} \tag{3.2}$$

The Girsanov bound (8.7) in Section 8 uses the same p_t weighting. Under Assumption 8.1, Appendix B.3 justifies (3.2) for every current density with $\mathrm{KL}(p_0\|q) < \infty$.

3.3 Corrected score and its error

Reverse diffusion requires the noisy target score s_t, which we must estimate using reference samples and clean target derivatives. The noisy current score s_t^q can be estimated from the reference samples by Tweedie's identity in (4.1). Since $s_t = s_t^q + d_t$, we seek an approximation a_t^q to the score difference d_t. Adding a_t^q to s_t^q gives the corrected score, an estimate of s_t with error $d_t - a_t^q$:

$$s_t^{a,q} := s_t^q + a_t^q, \qquad e_t^q := d_t - a_t^q, \qquad s_t^{a,q} = s_t - e_t^q. \tag{3.3}$$

Let $E(q) := \|e^q\|_p$. The two reverse SDEs driven by $s^{a,q}$ and s_t have the same noise coefficient and differ in drift by $2e_t^q$. The Girsanov bound (8.7) bounds the forward KL from the target to the reverse-SDE endpoint law by $E(q)^2$ plus the terminal-law mismatch. The forward KL after one round is at most θ^2 times its starting value plus the terminal error when $E(q)^2 \le \theta^2 D(q)^2$ with $0 \le \theta < 1$, by (3.2). We therefore seek a correction built from reference samples and target derivatives for which $E(q)$ is small relative to $D(q)$.

4 Conditional score identities

To construct the correction a_t^q in (3.3), we compare the target-score and Tweedie conditional means under the current OU posterior, the law of the clean state given the noisy state. Both means can be estimated from reference samples and the clean target score s_0. A conditional Poisson equation relates their difference to the noisy score difference $d_t = s_t - s_t^q$.

4.1 Tweedie and target-score signals

At a noisy state $y \in \mathbb{R}^d$ and time $t > 0$, we write the Tweedie and target-score signals as

$$b(x; y, t) = \frac{\alpha_t x - y}{\gamma_t}, \qquad c(x; t) = \frac{s_0(x)}{\alpha_t}, \qquad \delta(x; y, t) = c(x; t) - b(x; y, t).$$

Tweedie's identity gives $s_t = \mathbb{E}_{\rho_{y,t}}[b]$, and the target-score identity gives $s_t = \mathbb{E}_{\rho_{y,t}}[c]$ [13, 15]. These expectations are over X_0 conditional on $Y_t = y$, with X_0 drawn initially from the target. Thus the disagreement signal δ has zero mean under $\rho_{y,t}$. Under the current OU posterior $\rho^q_{y,t}$, the Tweedie mean is s^q_t. We denote the mean target-score-minus-Tweedie signal by m^q_t, so

$$\mathbb{E}_{\rho^q_{y,t}}[b] = s^q_t(y), \qquad m^q_t(y) := \mathbb{E}_{\rho^q_{y,t}}[\delta]. \tag{4.1}$$

When q has a differentiable positive density and integration by parts is valid, the conditional mean m^q_t in (4.1) satisfies

$$m^q_t(y) = \frac{1}{\alpha_t}\mathbb{E}_{\rho^q_{y,t}}[\nabla_x \log(p_0/q)(X_0)]\,.$$

Appendix A.2 derives the density-ratio formula.

4.2 Conditional Poisson equation and its moments

We relate the conditional mean m^q_t to the unknown score difference d_t by expressing both through moments of the gradient of a conditional Poisson solution. The assumptions below concern this representation and exclude empirical current laws, for which (4.1) still defines m^q_t.

For fixed y and $t > 0$, we assume $\rho_{y,t} \ll \rho^q_{y,t}$. The centered density ratio

$$\zeta = \frac{\mathrm{d}\rho_{y,t}}{\mathrm{d}\rho^q_{y,t}} - 1, \qquad \mathbb{E}_{\rho^q_{y,t}}[\zeta] = 0,$$

represents the difference between the target OU posterior $\rho_{y,t}$ and current OU posterior $\rho^q_{y,t}$. Let $u = u_{y,t}$ solve the conditional Stein–Poisson equation in weak form

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla u \cdot \nabla \varphi] = \mathbb{E}_{\rho^q_{y,t}}[\zeta\varphi] \quad \text{for all centered } \varphi \in H^1(\rho^q_{y,t}). \tag{4.2}$$

We assume a centered weak solution exists. The components of x and $\nabla_x \log \rho_{y,t}(x)$ must also belong to $H^1(\rho^q_{y,t})$ after centering. The centered coordinate and target-posterior-score components are the test functions used for (4.3). A finite Poincaré constant for $\rho^q_{y,t}$ and $\zeta \in L^2(\rho^q_{y,t})$ suffice for solvability (Appendix A.1). The convergence proof uses the residual identity (8.2), which holds independently of Poisson solvability.

The right-hand side of (4.2) is the difference of conditional means

$$\mathbb{E}_{\rho^q_{y,t}}[\zeta\varphi] = \mathbb{E}_{\rho_{y,t}}[\varphi] - \mathbb{E}_{\rho^q_{y,t}}[\varphi].$$

The weak equation (4.2) expresses this difference as an average involving ∇u, the Poisson flux. The shifted observed-information matrix is $A_0(x;t) = \alpha_t^2 I_d + \gamma_t H_0(x)$. Testing (4.2) with the components of x and $\nabla_x \log \rho_{y,t}(x)$ gives

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla u] = \frac{\gamma_t}{\alpha_t} d_t, \qquad \mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = \alpha_t \gamma_t\, m^q_t. \tag{4.3}$$

The mean of ∇u determines d_t, and the mean of $A_0 \nabla u$ is proportional to m^q_t. Appendix A.3 derives both relations.

5 Reconstructing the relative score

The moments in (4.3) give an equation for d_t in terms of m^q_t once the mean of $A_0\nabla u$ is separated into a product of conditional means and a covariance remainder. The conditional averages of the

observed information H_0 and shifted matrix A_0 under the current OU posterior are

$$H^q(y,t) := \mathbb{E}_{\rho^q_{y,t}}[H_0(X_0)], \qquad A^q(y,t) := \alpha_t^2 I_d + \gamma_t H^q(y,t) = \mathbb{E}_{\rho^q_{y,t}}[A_0(X_0;t)].$$

We assume $A^q(y,t)$ is positive definite wherever it is inverted. If $H_0(x)$ is positive semidefinite for all x, then $A^q \succeq \alpha_t^2 I_d$.

Subtracting the current-posterior mean from the Poisson flux gives the centered vector $v_\perp = \nabla u - \mathbb{E}_{\rho^q_{y,t}}[\nabla u]$. The vector $r_H(y,t) = \mathbb{E}_{\rho^q_{y,t}}[(H_0 - H^q)v_\perp] \in \mathbb{R}^d$ is the mean product of the centered curvature matrix and the centered flux evaluated at the same state. Since $A_0 - A^q = \gamma_t(H_0 - H^q)$, expanding the product yields

$$\mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = A^q \, \mathbb{E}_{\rho^q_{y,t}}[\nabla u] + \gamma_t r_H. \tag{5.1}$$

Inserting the two means from (4.3) into (5.1) gives

$$\alpha_t^2 m_t^q = A^q d_t + \alpha_t r_H. \tag{5.2}$$

We can estimate m_t^q and A^q from the reference samples using target derivatives. Computing r_H would require the conditional Poisson solution. Omitting the Poisson covariance term r_H from (5.2) gives the correction a_t^q. Its error e_t^q follows from (3.3):

$$a_t^q = \alpha_t^2 (A^q)^{-1} m_t^q, \qquad e_t^q = -\alpha_t (A^q)^{-1} r_H. \tag{5.3}$$

If H_0 is constant, r_H vanishes and $a_t^q = d_t$. The matrix multiplying m_t^q in (5.3) is the q-LFGI gate.

6 The q-LFGI gate

For a symmetric $d \times d$ matrix P such that $\alpha_t^2 I_d + \gamma_t P$ is positive definite, define

$$\Psi_t(P) = \alpha_t^2 (\alpha_t^2 I_d + \gamma_t P)^{-1}.$$

The q-LFGI gate is obtained by evaluating the LFGI matrix function Ψ_t from [14] at H^q:

$$G^q_{\mathrm{LFGI}}(y,t) = \Psi_t(H^q(y,t)) = \alpha_t^2 (A^q(y,t))^{-1}. \tag{6.1}$$

The gated signal is $z_G(x;y,t) = b(x;y,t) + G(y,t)\delta(x;y,t)$, where the matrix $G(y,t)$ is fixed at each y,t. By (4.1) its mean under $\rho^q_{y,t}$ is $s_t^q + G m_t^q$. Using (6.1) in (5.3) gives

$$a_t^q = G^q_{\mathrm{LFGI}} m_t^q, \qquad s_t^{a,q} = \mathbb{E}_{\rho^q_{y,t}}[z_{G^q_{\mathrm{LFGI}}}]. \tag{6.2}$$

Equation (6.2) defines the corrected score without a Poisson solve.

A Gaussian target has constant curvature, as in the Gaussian case of LFGI [14]. If $p_0 = \mathcal{N}(\mu_G, P^{-1})$ with $P \succ 0$, then $H_0 = P$, and averaging H_0 under any current OU posterior still gives P. The gate coefficient in an eigendirection of P with eigenvalue λ_j is

$$G_j(t) = \frac{\alpha_t^2}{\alpha_t^2 + \gamma_t \lambda_j}.$$

At a given $t > 0$, $G_j(t)$ decreases with λ_j. The gate makes the derivative of z_G with respect to x vanish in this Gaussian case, by (A.3). Thus z_G is constant in x and equals its target OU posterior mean s_t. Averaging it over any nonempty set of reference samples gives s_t, provided the Tweedie and target-score signals use the same normalized weights. GAD samples the Gaussian

target in one round when reverse diffusion starts from the exact terminal law p_T and is integrated exactly.

When $q = p_0$, the gated signal z_G has mean s_t under the target OU posterior for every fixed matrix G. The LFGI risk in (6.3) compares the conditional mean squared fluctuations of z_G around s_t. The conditional curvature is $H^{p_0}(y,t) = \mathbb{E}_{\rho_{y,t}}[H_0(X_0)]$. The gate $G^q_{\mathrm{LFGI}} = \Psi_t(H^{p_0})$ then coincides with $G_\star$, the LFGI minimizer of conditional mean squared score error over $G \in \mathbb{R}^{d\times d}$ at fixed y, t [14]. The minimizer is given by the following expression under the finite moment and positive-definite conditional covariance assumptions in Appendix A.2:

$$G_\star(y,t) = \arg\min_{G\in\mathbb{R}^{d\times d}} \mathcal{R}(G;y,t), \qquad \mathcal{R}(G;y,t) := \mathbb{E}_{\rho_{y,t}}[\|z_G(X_0;y,t) - s_t(y)\|^2]. \tag{6.3}$$

For $q \neq p_0$ we use the reconstruction (6.2). Its error is given by (5.3) under the solvability assumptions stated after (4.2). For the convergence analysis, Lemma 8.2 gives a separate error identity that uses only conditional expectations.

7 Gated Action Dissipation

With exact conditional expectations, a GAD round fixes the score $s_t^{a,q}$ from (6.2) and integrates (3.1) from $Z_T \sim \phi_d$ at $t = T$ to $t = 0$. Write $\Phi_T(q)$ for the law of Z_0. The next round starts from this law according to

$$q_{k+1} = \Phi_T(q_k). \tag{7.1}$$

The correction $a_t^q = G^q_{\mathrm{LFGI}} m_t^q$ vanishes at $q = p_0$ because the Tweedie and target-score signals have equal means under the target OU posterior.

Algorithm 1 approximates each conditional expectation by an average over the reference samples with the normalized OU kernel weights in (I.1). During each round, the reference samples and their target derivatives stay fixed. At each score-field evaluation point (y,t), we recompute the weights and conditional means. The synthetic experiments in Sections 11 and 12 use the reverse SDE. The inverse problems use the probability-flow ordinary differential equation (ODE) in (10.1).

The conditional weights in Algorithm 1 are used only to evaluate the score field. All endpoints have the same weight when they become the next round's reference samples. For the inverse problems, (I.2) gives the GN score field. Appendix I.3 gives the likelihood schedules and request accounting.

1. **Input.** Provide N unweighted reference samples $X_{1:N}^{(0)} = (X_1^{(0)}, \dots, X_N^{(0)})$, a grid $0 < t_{\min} = t_0 < \cdots < t_{L_t} = T$, a terminal distribution, a reverse SDE integrator (3.1) or probability-flow ODE integrator (10.1), a clean target score evaluator, a curvature evaluator, and a round count K.
2. **For $k = 0, \dots, K-1$, perform the following steps.**
   (a) Cache $s_0(X_i^{(k)})$ and the observed information $H_i = H_0(X_i^{(k)})$, or its Gauss–Newton approximation (9.1), at each reference state. Hold the cache fixed for the round. For annealed GN-GAD, use the derivatives at power β_{k+1} in (9.4).
   (b) At each score-field evaluation point (y, t), compute the normalized weights $w_i(y, t)$ from (I.1). For the current empirical law q_k, average the Tweedie signal b, target-score signal c, and cached curvature. The score signals are defined in Section 4.1.

$$\widehat{b}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) b(X_i^{(k)}; y, t),$$
$$\widehat{c}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) c(X_i^{(k)}; t),$$
$$\widehat{H}^{q_k}(y,t) = \sum_{i=1}^{N} w_i(y,t) H_i.$$

   Use the score field

$$\widehat{m}_t^{q_k}(y) = \widehat{c}^{q_k} - \widehat{b}^{q_k}, \qquad \widehat{s}_t^{a,q_k}(y) = \widehat{b}^{q_k} + \Psi_t(\widehat{H}^{q_k})\widehat{m}_t^{q_k}(y).$$

   Apply Ψ_t as in (6.1) when the empirical shifted matrix is positive definite. For the toy targets with indefinite observed information, use the eigenvalue safeguards in Appendix F.3.
   (c) Draw N fresh states at T from the terminal distribution. Integrate with $\widehat{s}_t^{a,q_k}$ down to $t_{\min}$ and set $X_{1:N}^{(k+1)}$ to the unweighted endpoints.
3. **Output.** Return the unweighted samples $X_{1:N}^{(K)}$.

Algorithm 1: **Gated Action Dissipation with the q-LFGI gate.**

7.1 A one-dimensional illustration

For a one-dimensional illustration, we use a target with three Gaussian components. Each round starts from a fitted Gaussian mixture, uses quadrature for the conditional means, and fits another mixture to the transported endpoints. The sampler in Algorithm 1 instead uses the endpoints directly. The illustration starts from a broad, nearly Gaussian mixture and uses exact target derivatives with a gate safeguard for negative curvature.

In Figure 1, the first round moves most of the missing mass to the separated left mode, and later rounds make smaller corrections. Estimated forward KL falls from 0.572 initially to 0.076 after one round, and later rounds bring it down close to the error from the finite diffusion horizon. The separated mixture serves as a numerical illustration beyond the strongly log-concave target class of Assumption 8.1. Appendix G.1 gives the target, initial law, mixture fit, gate safeguard, and construction of the displayed densities.

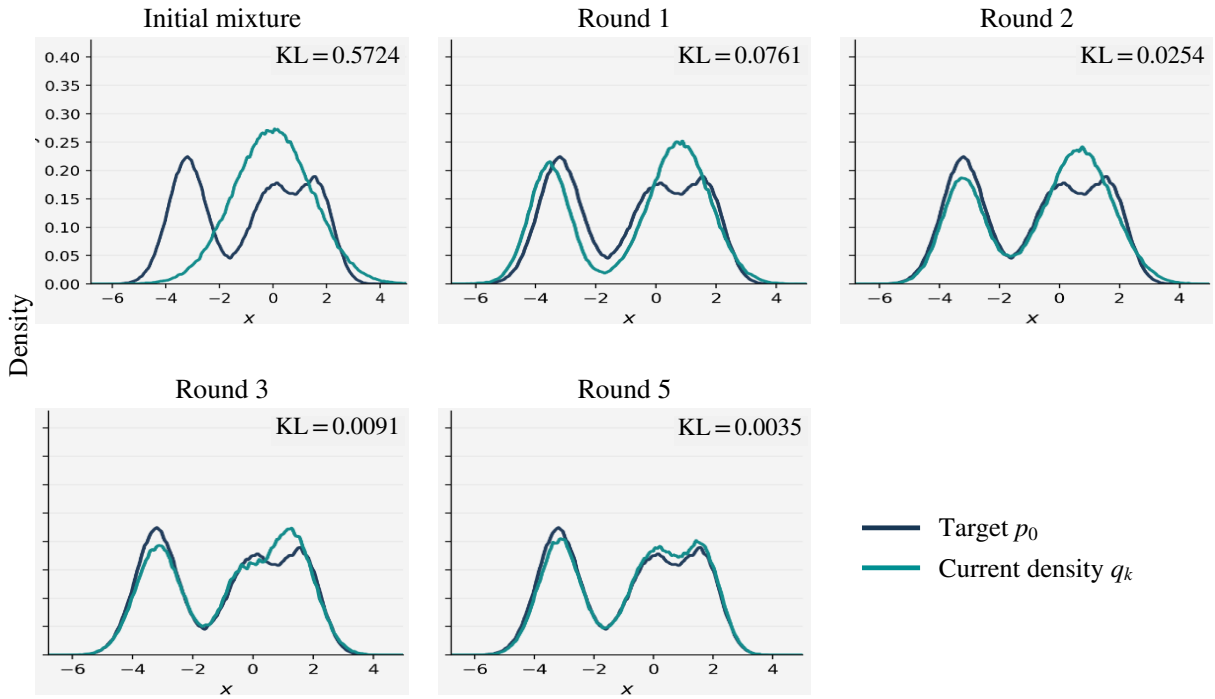


Figure 1: **GAD rounds on a one-dimensional Gaussian mixture.** The target p_0 (dark blue) and the fitted current density q_k (teal) are shown initially and after rounds 1, 2, 3, and 5. The horizontal axis is the state x, and the vertical axis is density. The curves are smoothed histograms of independent draws from the target and fitted mixtures. Each panel reports an estimate of $\mathrm{KL}(p_0\|q_k)$, and lower values indicate closer agreement. Appendix G.1 gives the display construction.

8 Convergence with exact expectations for perturbed Gaussian targets

We bound the integrated squared score error $E(q)^2$ for strongly log-concave perturbations of Gaussian targets using bounds on the perturbation's curvature and gradient. This gives a bound on forward KL after one round of (7.1). Appendix B gives the estimates used in the proof.

8.1 Target class and score error

The difference between the target observed-information matrix and its constant quadratic part determines one term of the score error. The following assumptions bound that difference and the variation of the perturbation gradient. The notation $\succeq$ denotes the Loewner order on symmetric matrices, while $\|\cdot\|_{\mathrm{op}}$ is the Euclidean operator norm.

Assumption 8.1 (Strongly log-concave perturbations of Gaussian targets)**.** Up to an additive constant,

$$-\log \widetilde{p}_0(x) = \tfrac{1}{2}x^\top \Lambda x + V(x), \qquad \Lambda = \Lambda^\top \succ 0, \qquad V \in C^2(\mathbb{R}^d),$$

and there are constants $\kappa > 0$, $L_V \geq 0$, and $\Delta_V < \infty$ such that, for all $x, x' \in \mathbb{R}^d$,

$$H_0(x) \succeq \kappa I_d, \qquad \|\nabla^2 V(x)\|_{\mathrm{op}} \leq L_V, \qquad \|\nabla V(x) - \nabla V(x')\| \leq \Delta_V. \tag{8.1}$$

Under (8.1), $\kappa I_d \preceq H_0 \preceq (\|\Lambda\|_{\mathrm{op}} + L_V) I_d$. Hence p_0 is normalizable with Gaussian tails and has moments of all orders. The gradient ∇V is bounded because its variation is bounded, and a

linear term in $-\log \widetilde{p}_0$ can be absorbed into V without changing L_V or Δ_V. For every current law, $H^q = \mathbb{E}_{\rho^q_{y,t}}[H_0] \succeq \kappa I_d$, so $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d$ is invertible and the q-LFGI gate is defined. The admissible current densities form the domain

$$\mathcal{Q} := \left\{ q : \ q \text{ is a probability density on } \mathbb{R}^d \text{ and } \mathrm{KL}(p_0 \| q) < \infty \right\}.$$

A density in $\mathcal{Q}$ is positive almost everywhere, but it need not be smooth, have moments, or have a bounded ratio to p_0.

Equation (5.3) expresses the score error through the covariance of curvature and the Poisson flux. For the target class in Assumption 8.1, Lemma 8.2 expresses the same error through $H^q - \Lambda$ and the difference between the means of ∇V under the target and current OU posteriors $\rho_{y,t}$ and $\rho^q_{y,t}$. The derivative bounds in (8.1) control these terms without requiring a Poisson solution.

Lemma 8.2 (q-LFGI residual identity)**.** *Under Assumption 8.1, let q be a probability law, $t > 0$, and $y \in \mathbb{R}^d$. Then*

$$e_t^q(y) = \gamma_t (A^q)^{-1} (H^q - \Lambda)\, d_t(y) + \alpha_t (A^q)^{-1} \big(\mathbb{E}_{\rho^q_{y,t}}[\nabla V] - \mathbb{E}_{\rho_{y,t}}[\nabla V] \big). \tag{8.2}$$

Consequently,

$$\| e_t^q(y) \| \le \frac{\gamma_t}{\alpha_t^2 + \kappa\gamma_t} \left\{ L_V \| d_t(y) \| + \frac{\alpha_t \Delta_V}{\gamma_t} \sqrt{\tfrac{1}{2} \mathrm{KL}(\rho_{y,t} \| \rho^q_{y,t})} \right\}. \tag{8.3}$$

Proof. All averages are finite because $\rho^q_{y,t}$ has moments of all orders for $t > 0$ (Appendix B.1). The Gaussian tails of the strongly log-concave target OU posterior justify taking the mean of its score in (A.1) to obtain $\mathbb{E}_{\rho_{y,t}}[\delta] = 0$. Tweedie's identity gives $d_t = (\alpha_t/\gamma_t)(\mathbb{E}_{\rho_{y,t}}[X_0] - \mathbb{E}_{\rho^q_{y,t}}[X_0])$. Since $-s_0(x) = \Lambda x + \nabla V(x)$, the signal $-\alpha_t \delta(x) = (\Lambda + \alpha_t^2 \gamma_t^{-1} I_d) x + \nabla V(x) - \alpha_t \gamma_t^{-1} y$ is affine in x apart from ∇V. Therefore

$$\alpha_t m_t^q = \alpha_t (\mathbb{E}_{\rho^q_{y,t}}[\delta] - \mathbb{E}_{\rho_{y,t}}[\delta]) = \frac{1}{\alpha_t} (\alpha_t^2 I_d + \gamma_t \Lambda) d_t - (\mathbb{E}_{\rho^q_{y,t}}[\nabla V] - \mathbb{E}_{\rho_{y,t}}[\nabla V]).$$

Multiplying by $\alpha_t (A^q)^{-1}$ gives a_t^q. Subtracting a_t^q from d_t and using $A^q - \alpha_t^2 I_d - \gamma_t \Lambda = \gamma_t (H^q - \Lambda)$ proves (8.2). For (8.3), use $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d$ and $\| H^q - \Lambda \|_{\mathrm{op}} = \| \mathbb{E}_{\rho^q_{y,t}}[\nabla^2 V] \|_{\mathrm{op}} \le L_V$. For a unit vector v, the function $v \cdot \nabla V$ has oscillation at most Δ_V. The means of $v \cdot \nabla V$ under the target and current OU posteriors therefore differ by at most Δ_V times the total-variation distance between the posteriors, which Pinsker's inequality bounds by $\{\frac{1}{2} \mathrm{KL}(\rho_{y,t} \| \rho^q_{y,t})\}^{1/2}$. □

For a Gaussian target, V is affine, both terms in (8.2) vanish, and the corrected score equals the target score for every current law. In general, the posterior KL in (8.3) is controlled by the forward KL already dissipated along the OU path. For $q \in \mathcal{Q}$ and $t > 0$,

$$\mathbb{E}_{p_t} \Big[\mathrm{KL}(\rho_{Y_t,t} \| \rho^q_{Y_t,t}) \Big] = \mathrm{KL}(p_0 \| q) - \mathrm{KL}(p_t \| q_t) = \int_0^t \mathbb{E}_{p_u}[\| d_u \|^2] \mathrm{d}u. \tag{8.4}$$

The first equality is the chain rule for relative entropy applied to the joint laws of (X_0, Y_t), which share the OU kernel. The second is (3.2) on $[0, t]$ (Appendix B.3).

8.2 Uniform score-error bound and contraction

To bound KL after each round, we seek $E(q)^2 \le \theta^2 D(q)^2$ uniformly over $q \in \mathcal{Q}$ and $T > 0$. Contraction requires $\theta^2 < 1$. We obtain a closed-form choice for θ^2 from the curvature and

gradient variation bounds in Assumption 8.1:

$$\eta_H := \frac{L_V^2}{\kappa^2}, \qquad \eta_V := \frac{\Delta_V^2}{4\kappa}, \qquad C_{\mathrm{closed}} := \frac{\eta_H + \eta_V + \sqrt{\eta_H^2 + \eta_V^2}}{2}. \tag{8.5}$$

Since $\eta_H, \eta_V \geq 0$, we have $\sqrt{\eta_H^2 + \eta_V^2} \leq \eta_H + \eta_V$, so $C_{\mathrm{closed}} \leq \eta_H + \eta_V$.

Proposition 8.3 (Uniform bound for the integrated score error)**.** *Under Assumption 8.1, let C_{closed} be defined by (8.5). Every $q \in \mathcal{Q}$ and every horizon $T > 0$ satisfy*

$$E(q)^2 \leq C_{\mathrm{closed}} D(q)^2 = C_{\mathrm{closed}} \{\mathrm{KL}(p_0 \| q) - \mathrm{KL}(p_T \| q_T)\}. \tag{8.6}$$

Proof. By (8.4), the mean conditional KL at time t is the integral of $\mathbb{E}_{p_u} \|d_u\|^2$ from 0 to t. The proof in Appendix B.4 multiplies the mean conditional KL by a function of time chosen from the target bounds (8.1). In the transformed time, the derivative of that product bounds the integrand of $E(q)^2$. Integrating gives (8.6). □

Remark (Sharper bounds). The appendix gives alternative coefficients in $E(q)^2 \leq \theta^2 D(q)^2$ for the same score correction. An admissible nondecreasing function $S : [0,1] \to [0,\infty)$ satisfying the scalar differential inequality in Lemma B.5 gives $\theta^2 = S(1)$ by retaining the conditional curvature matrix inside the inverse. When $\Lambda = \lambda I_d$, let M be an upper bound on the eigenvalues of $H_0(x)$ for all x, so that $\kappa I_d \preceq H_0(x) \preceq M I_d$. Then (B.8) reduces the scalar criterion to the curvature endpoints κ and M, and (B.9) gives another closed-form coefficient. These target-dependent alternatives can give $\theta^2 < 1$ even when $C_{\mathrm{closed}} \geq 1$, as in Appendix B.5.

Under Assumption 8.1, the apparent singular terms in the corrected score cancel in (B.2). Lemma B.2 bounds the score by $C_q(1 + \|y\|)$ uniformly for $t > 0$ for each current probability law q. Lemma B.3 then gives a unique nonexplosive strong solution of (3.1) for any terminal law, up to and including $t = 0$. If the terminal law has a density ν_T with $\mathrm{KL}(p_T \| \nu_T) < \infty$, applying the localized Girsanov comparison of Lemma B.4 on $[0, T]$ gives an endpoint density q_{out} satisfying

$$\mathrm{KL}(p_0 \| q_{\mathrm{out}}) \leq \mathrm{KL}(p_T \| \nu_T) + E(q)^2. \tag{8.7}$$

The proof uses a stopped change of drift in place of a Novikov condition.

Theorem 8.4 (Geometric forward-KL contraction)**.** *Assume Assumption 8.1, and let $0 \leq \theta^2 < 1$ satisfy $E(q)^2 \leq \theta^2 D(q)^2$ for every $q \in \mathcal{Q}$ and $T > 0$. Such a coefficient is supplied by Proposition 8.3 or the sharper bounds in the preceding remark. Set $\varepsilon_T := \mathrm{KL}(p_T \| \phi_d)$.*

(i) *For every $q \in \mathcal{Q}$ and $T > 0$,*

$$\mathrm{KL}(p_0 \| \Phi_T(q)) \leq \varepsilon_T + \theta^2 \mathrm{KL}(p_0 \| q). \tag{8.8}$$

Hence $\Phi_T(\mathcal{Q}) \subseteq \mathcal{Q}$, and the iteration (7.1) started from $q_0 \in \mathcal{Q}$ satisfies

$$\mathrm{KL}(p_0 \| q_k) \leq \theta^{2k} \mathrm{KL}(p_0 \| q_0) + \frac{1 - \theta^{2k}}{1 - \theta^2} \varepsilon_T. \tag{8.9}$$

(ii) *For every $q \in \mathcal{Q}$, the limit $\Phi(q) := \lim_{T \to \infty} \Phi_T(q)$ exists in total variation and has a density. Moreover,*

$$\Phi(\mathcal{Q}) \subseteq \mathcal{Q}, \qquad \mathrm{KL}(p_0 \| \Phi(q)) \leq \theta^2 \mathrm{KL}(p_0 \| q). \tag{8.10}$$

Consequently, $q_{k+1} = \Phi(q_k)$ satisfies $\mathrm{KL}(p_0 \| q_k) \leq \theta^{2k} \mathrm{KL}(p_0 \| q_0)$ for every $q_0 \in \mathcal{Q}$, and p_0 is the unique fixed point of Φ in $\mathcal{Q}$.

Proof. For part (i), apply (8.7) with $\nu_T = \phi_d$, then the assumed score-error bound and $D(q)^2 \le \mathrm{KL}(p_0\|q)$. This gives $\mathrm{KL}(p_0\|\Phi_T(q)) \le \varepsilon_T + \theta^2 D(q)^2$, hence (8.8). The right side is finite and $\Phi_T(q)$ has a density, so $\Phi_T(q) \in \mathcal{Q}$. Induction gives (8.9). Part (ii) is proved in Appendix B.6. □

The condition $\theta < 1$ in Theorem 8.4 involves only the target derivatives. Any initial density with $\mathrm{KL}(p_0\|q_0) < \infty$ is admissible, including a broad Gaussian whose density ratio to p_0 is unbounded. The terminal error vanishes as the horizon increases, by (B.10). The infinite-horizon map reaches a Gaussian target in one round. For a Gaussian target, only the terminal-law mismatch remains in the finite-horizon bound of Theorem 8.4.

For the non-Gaussian target with $-\log \widetilde{p}_0(x) = \frac{1}{2}\|x\|^2 + \epsilon \cos x_1$ and $0 \le \epsilon < 1$, we have $\Lambda = I_d$ and $V(x) = \epsilon \cos x_1$. In Assumption 8.1, one may take $\kappa = 1 - \epsilon$, $L_V = \epsilon$, and $\Delta_V = 2\epsilon$. Then the closed-form choice (8.5) gives a bound below one for $\epsilon < 0.4639$, and $\theta^2 < 0.2683$ at $\epsilon = 0.3$. Appendix B.5 gives examples with coordinate coupling and with a sharper bound. Theorem 8.4 assumes a strongly log-concave target with bounded perturbation gradient. The iteration in the theorem uses exact observed information and conditional expectations in continuous-time reverse diffusion.

9 Computational extensions for inverse problems

In the acoustic and earthquake experiments, we form the gate from a Gauss–Newton approximation to the observed information. We compare direct GN-GAD, which uses the full likelihood from the start, with annealed GN-GAD, which multiplies the log likelihood by a power that increases to one. Theorem 8.4 applies to exact observed information and a fixed target.

9.1 Gauss–Newton information

Let $\alpha \in \mathbb{R}^d$ have standard Gaussian prior, and let $F\colon \mathbb{R}^d \to \mathbb{R}^{n_{\mathrm{obs}}}$ be the forward map to the observations. The data model is $y_{\mathrm{obs}} = F(\alpha_\star) + \eta$, where $\alpha_\star$ is the data-generating parameter and $\eta \sim \mathcal{N}(0, \Gamma)$ is independent Gaussian noise with $\Gamma \succ 0$. Write $r(\alpha) = \Gamma^{-1/2}(F(\alpha) - y_{\mathrm{obs}})$, $U_{\mathrm{like}}(\alpha) = \|r(\alpha)\|^2/2$, and let $J(\alpha) = \nabla r(\alpha) \in \mathbb{R}^{n_{\mathrm{obs}} \times d}$ be the Jacobian of the whitened residual. The posterior has unnormalized density $\widetilde{p}_0(\alpha) = \exp\{-\|\alpha\|^2/2 - U_{\mathrm{like}}(\alpha)\}$. We retain its exact clean score and use the standard Gauss–Newton approximation to the observed information [4, 27]

$$H_0^{\mathrm{GN}}(\alpha) = I_d + J(\alpha)^\top J(\alpha). \tag{9.1}$$

The matrix in (9.1) is positive definite, while the exact observed information is

$$H_0(\alpha) = H_0^{\mathrm{GN}}(\alpha) + R_{\mathrm{nl}}(\alpha), \qquad R_{\mathrm{nl}}(\alpha) = \sum_j r_j(\alpha) \nabla^2 r_j(\alpha). \tag{9.2}$$

The residual R_{nl} vanishes for an affine forward map. The Gauss–Newton approximation can still differ from the exact curvature for a nonlinear map, even when the clean scores are accurate and the approximation is positive definite.

For the gate in (6.1), we replace the conditional mean of H_0 by that of H_0^{GN}. The shifted Gauss–Newton matrix $\alpha_t^2 I_d + \gamma_t \mathbb{E}_{\rho^q_{y,t}}[H_0^{\mathrm{GN}}]$ stays positive definite. Equation (I.3) separates the effect of substituting Gauss–Newton curvature from the Poisson covariance remainder. The experiments evaluate the resulting sampler's combined error, including conditional estimation, numerical transport, and any likelihood annealing.

9.2 Likelihood annealing

A concentrated likelihood can leave few reference samples in regions of high posterior probability, affecting the conditional score estimates. Likelihood annealing lets the reference law adapt over successive rounds as the likelihood is introduced.

Intermediate target distributions are a standard device in annealed importance sampling and sequential Monte Carlo [11, 28]. For likelihood annealing, we use the family

$$p_{0,\beta}(\alpha) = Z_\beta^{-1} \exp\{-\tfrac{1}{2}\|\alpha\|^2 - \beta U_{\mathrm{like}}(\alpha)\}, \qquad 0 \le \beta \le 1, \tag{9.3}$$

where Z_β normalizes the density. Thus $p_{0,0}$ is the prior and $p_{0,1} = p_0$. The likelihood power β in (9.3) is distinct from OU time t. At each fixed power the clean score and GN observed-information approximation are

$$s_{0,\beta}(\alpha) = -\alpha - \beta J(\alpha)^\top r(\alpha), \qquad H^{\mathrm{GN}}_{0,\beta}(\alpha) = I_d + \beta J(\alpha)^\top J(\alpha). \tag{9.4}$$

At full power, $H^{\mathrm{GN}}_{0,1} = H^{\mathrm{GN}}_0$. The likelihood power multiplies the likelihood terms and leaves the prior terms fixed.

Choose a nondecreasing sequence $\beta_1, \dots, \beta_K = 1$. Round k in Algorithm 1 evaluates (9.4) at power β_{k+1}, holds the reference samples and their derivatives fixed during transport, and passes the unweighted endpoints to the next round. This is the annealed GN-GAD implementation. Direct GN-GAD uses $\beta = 1$ throughout. Once $\beta = 1$, any additional rounds keep the posterior as their target.

In both inverse problems, we compare the specified likelihood schedules with direct $\beta_k = 1$ rounds at the same allowances for target requests. The annealed MCMC methods use the same tempered targets (9.3).

10 Samplers and comparison settings

Each GAD round computes the score field $\widehat{s}_t^{a,q}$ of Algorithm 1 from the current N unweighted reference samples. The field stays fixed while fresh Gaussian terminal samples are transported. Their endpoints become the next reference samples. No density fit or acceptance step is applied to these endpoints. The mixture-fitted illustration in Section 7.1 is a separate diagnostic. The synthetic experiments in Sections 11 and 12 use the reverse OU SDE (3.1), exact target scores, and exact observed information. Both inverse problems in Section 13 use the Gauss–Newton score estimate (I.2) with the score and curvature at the likelihood power in (9.4). The probability-flow ODE in decreasing OU time is

$$\frac{\mathrm{d}Z_t}{\mathrm{d}t} = -Z_t - \widehat{s}_t^{a,q}(Z_t), \qquad t : T \downarrow t_{\min}, \qquad Z_T \sim \phi_d. \tag{10.1}$$

The reverse SDE and probability flow have the same marginal laws when both use the exact score and the same exact terminal law [35]. Theorem 8.4 analyzes the reverse SDE (3.1). Transport grids, initial reference samples, and safeguards are given in Appendices C to G.

MALA applies the Hastings correction to a Langevin proposal using the unnormalized target and its clean score [33, 34]. We write $U = -\log \widetilde{p}_0$ for the inverse-problem potential, α_{MAP} for its minimizer, and $C_{\mathrm{MAP}} = [\nabla^2 U(\alpha_{\mathrm{MAP}})]^{-1}$ for the inverse Hessian when it is positive definite. MAP–Laplace denotes the Gaussian $\mathcal{N}(\alpha_{\mathrm{MAP}}, C_{\mathrm{MAP}})$. The MAP-preconditioned Metropolis-adjusted Langevin algorithm (MAP-pMALA) uses the proposal

$$\alpha' = \alpha + hC_{\mathrm{MAP}} s_0(\alpha) + \sqrt{2h}\, C_{\mathrm{MAP}}^{1/2} \xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

with step size $h > 0$ and fixed covariance preconditioning [30]. The MAP Hessians in the two main applications are positive definite. Each MAP-pMALA chain starts from MAP–Laplace, adapts h during warmup, and fixes it for production. Additional inverse-problem methods use fixed-metric MALA, the Gauss–Newton simplified manifold Metropolis-adjusted Langevin algorithm (GN-smMALA) with a local Gauss–Newton matrix [17], and fixed-mass HMC. Appendix I.4 gives their likelihood powers, initialization, configuration selection, and the chain states retained for evaluation.

The two-dimensional comparison uses isotropic MALA and unit-mass HMC, each with four chains started from the same broad initial law as GAD (Section 12). HMC uses Gaussian momentum and leapfrog trajectories with a Metropolis correction [29]. Appendix F.4 gives the step-size adaptation and the other settings.

All methods within a study access the same unnormalized target. GAD also uses exact observed information or the stated Gauss–Newton approximation. The two-dimensional and inverse-problem comparisons use matched allowances for target requests, as described in Appendix F.4 and Section 13.1. For evaluation, we use independent target samples when available. The inverse problems use samples from the No-U-Turn Sampler (NUTS) [21]. In the inverse problems, the NUTS samples also define the metric used to select configurations on the calibration seed, a separate sampler seed used to choose settings before the evaluation runs. No sampler uses NUTS draws for its initialization, preconditioning, or adaptation.

11 Numerical tests of the score estimator and gate

The Gaussian score-recovery test uses a Gaussian target and changes the law of the reference samples to check the cancellation in (A.3). Next, we increase the separation of two mixture components while holding the initial weight mismatch fixed. The mixture-weight test measures the discrepancy between the clean target and current scores and the forward-KL reduction after one GAD round. Finally, we vary the gate on an eight-dimensional target while holding the sample count and number of rounds fixed.

The Gaussian score-recovery experiment, gate comparison, and two-dimensional sampler benchmarks compare equal-size sample sets by the empirical sliced Wasserstein-2 distance (SW_2). Using one-dimensional projections, we compute the empirical distance (E.1) and report it as a percentage of the target's root mean squared coordinate spread. Lower values indicate closer agreement. Distances between two independent target samples give a scale for comparison. Here sample standard deviation (SD) is computed for the metric over the indicated runs. The mixture study uses forward KL because the evolving densities are available.

11.1 Gaussian score recovery with finitely many reference samples

The score-recovery experiment uses an anisotropic Gaussian target with two reference laws, a broad Gaussian and a bimodal mixture. We compare each estimated score with the exact noisy score and compare the endpoints after one round. The two reference laws yield the exact score and the same paired endpoints to numerical precision. The endpoint distance to independent target draws is comparable to the distance between two independent target samples (Table 2). The endpoint-to-target distance includes numerical integration and finite-sample error, even when the score estimate is exact. Appendix E.2 specifies the target, reference laws, and numerical settings.

Table 2: **Gaussian cancellation and one-round sample agreement.** The broad Gaussian and bimodal-mixture reference laws each use 4,096 reference samples and 4,096 transported endpoints. Score residuals are maximum absolute errors before field interpolation. Normalized SW_2 is in percent, mean $\pm$ sample SD over three seeds, against an independent 4,096-point target sample. Lower values are better. The final column compares two independent target samples. Appendix E.2 gives the paired-run design.

Reference law	Max score residual	One-round SW_2 (%)	Target/target (%)
Gaussian	2.80×10^{-11}	3.69 ± 0.49	4.05 ± 0.63
Bimodal	1.27×10^{-10}	3.69 ± 0.49	4.05 ± 0.63

11.2 Sensitivity of the clean score to mixture weights

Suppose the current law has the right mixture components but gives them the wrong weights. Where one component dominates, changing its weight has little effect on the clean score. As the components separate, the clean scores differ appreciably mainly in the low-probability region where the components overlap [39]. For a smooth current density q, define

$$K_{\mathrm{mix}}(q) := \mathrm{KL}(p_0 \| q), \qquad F_{\mathrm{mix}}(q) := \mathbb{E}_{p_0} \| s_0 - \nabla \log q \|^2, \qquad \chi(q) := \frac{\sigma^2 F_{\mathrm{mix}}(q)}{K_{\mathrm{mix}}(q)}. \tag{11.1}$$

The target and initial mixtures share two Gaussian components with the same standard deviation σ. In this test, we vary their separation in units of σ while holding the initial weight mismatch fixed. The target weights are equal and the initial law assigns weights $0.7, 0.3$ to the same components (Appendix E.3). For each fixed target, we measure fractional KL progress by

$$g_k := 1 - \frac{K_{\mathrm{mix}}(q_{k+1})}{K_{\mathrm{mix}}(q_k)}, \qquad q_{k+1} = \Phi_T(q_k). \tag{11.2}$$

Larger g_k in (11.2) means greater progress. The clean-score discrepancy $F_{\mathrm{mix}}(q)$ in (11.1) compares scores at time zero. The integral $D(q)^2$ in (3.2) measures the squared difference between the noisy target and current scores over the OU path.

The density calculation uses exact clean target derivatives and conditional expectations computed by quadrature. At each round, we obtain the next density by solving the Fokker–Planck equation numerically. The calculation uses neither reference samples nor a Gaussian mixture fit. The initial clean-score sensitivity χ and first-round fractional forward-KL reduction each fall by about three orders of magnitude as separation increases from 5 to 9. After 30 rounds, GAD has almost recovered the weights of the less separated mixture. Forward KL remains appreciable for the more separated mixture. The iteration corrects the weight imbalance more slowly for mixtures whose clean score changes less with the component weights (Figure 2). Appendix E.3 gives the values at each separation and the numerical checks.

11.3 Effect of the gate over several rounds

We vary the gate in Algorithm 1 on a smooth eight-dimensional target with positive, position-dependent curvature. Appendix E.4 defines the density and the coordinates used in the figure. The gates share the same initial reference samples, sample counts, and reverse-SDE random draws within each seed. The constant choices 0 and I_d recover Tweedie and TSI. Beyond these constant choices, the spatially uniform scalar and matrix gates vary with t, while the local scalar and local matrix blend gates also depend on y through conditional moments under q.

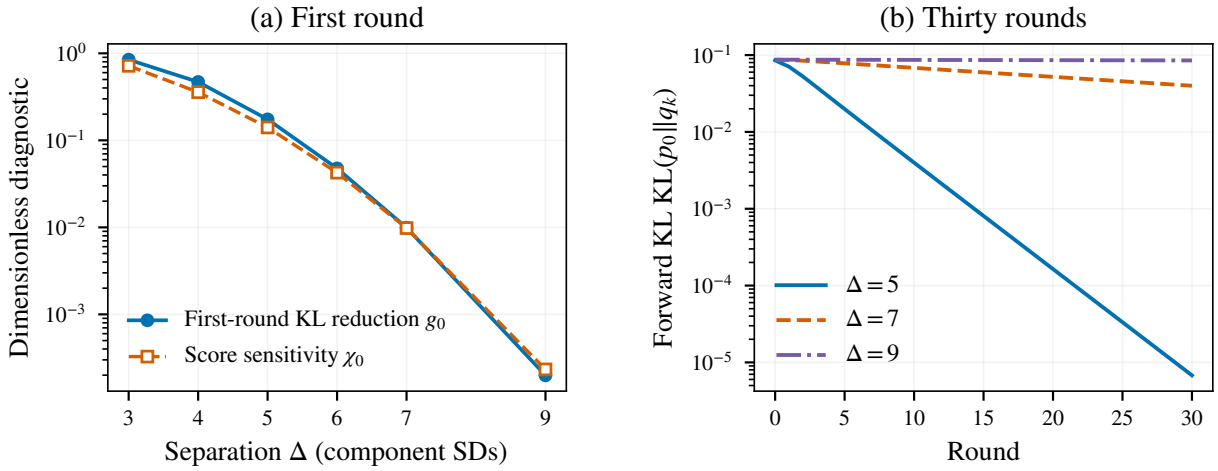


Figure 2: **Clean-score sensitivity and KL progress for separated mixtures.** The target has equal component weights and the initial law has weights $0.7, 0.3$ on the same components. (a) First-round fractional forward-KL reduction g_0 and clean-score sensitivity χ_0 versus separation in component standard deviations. Larger g_0 means a greater KL reduction. (b) Forward KL over 30 exact-expectation rounds, with no mixture fit, for separations 5, 7, and 9. Lower KL indicates closer agreement. These are deterministic density calculations, so there are no sampling bands. Appendix E.3 gives the grid resolution checks.

Adapting the scalar control-variate (CV) coefficient to the current law gives the uniform gate labeled q-CV. Its local counterpart adapts the control-variate score identity (CVSI) and is labeled q-CVSI [23]. For a uniform matrix coefficient, we adapt the DPSMC schedule and label the gate q-DPSMC [40]. The local matrix blend instead uses a conditional covariance quotient [14]. Each coefficient is evaluated within Algorithm 1, with formulas and numerical safeguards given in Appendix E.5.

q-LFGI has the lowest mean SW_2 at both rounds three and four when sample and round counts are held fixed (Tables 3 and 16). Its round-three error is 8.56 ± 0.40, near the target/target scale 7.87 ± 0.29. The local scalar gate q-CVSI has the next lowest round-three SW_2. None of the TSI runs completes three rounds under the stopping rule in Appendix E.5. Figure 3 shows the first two transformed coordinates, and the metric uses all eight physical coordinates.

Table 3: **Gate comparison at equal sample and round counts.** Normalized full eight-dimensional SW_2 in percent, mean ± sample SD over five seeds, using 1,024 GAD endpoints and 1,024 independent target samples per run. Lower is better. Every gate uses 1,024 reference samples per round. The target/target comparison scale is 7.87 ± 0.29. "Runs" counts completed round-three runs. No TSI run completes round three under the rule in Appendix E.5.

Gate	Round 1	Round 2	Round 3	Runs
Tweedie	201.81 ± 2.56	200.61 ± 2.65	199.41 ± 2.18	5/5
TSI	31.22 ± 2.21	$(1.2 \pm 1.1) \times 10^6$	failed	0/5
Uniform scalar (q-CV)	160.54 ± 2.17	116.82 ± 2.05	71.06 ± 2.13	5/5
Uniform matrix (q-DPSMC)	144.32 ± 2.75	99.54 ± 1.36	63.46 ± 2.53	5/5
Local scalar (q-CVSI)	74.39 ± 2.13	30.98 ± 1.85	12.12 ± 0.75	5/5
Local matrix blend	57.13 ± 3.67	40.50 ± 2.94	30.05 ± 3.47	5/5
q-LFGI	22.57 ± 1.50	9.45 ± 1.60	8.56 ± 0.40	5/5

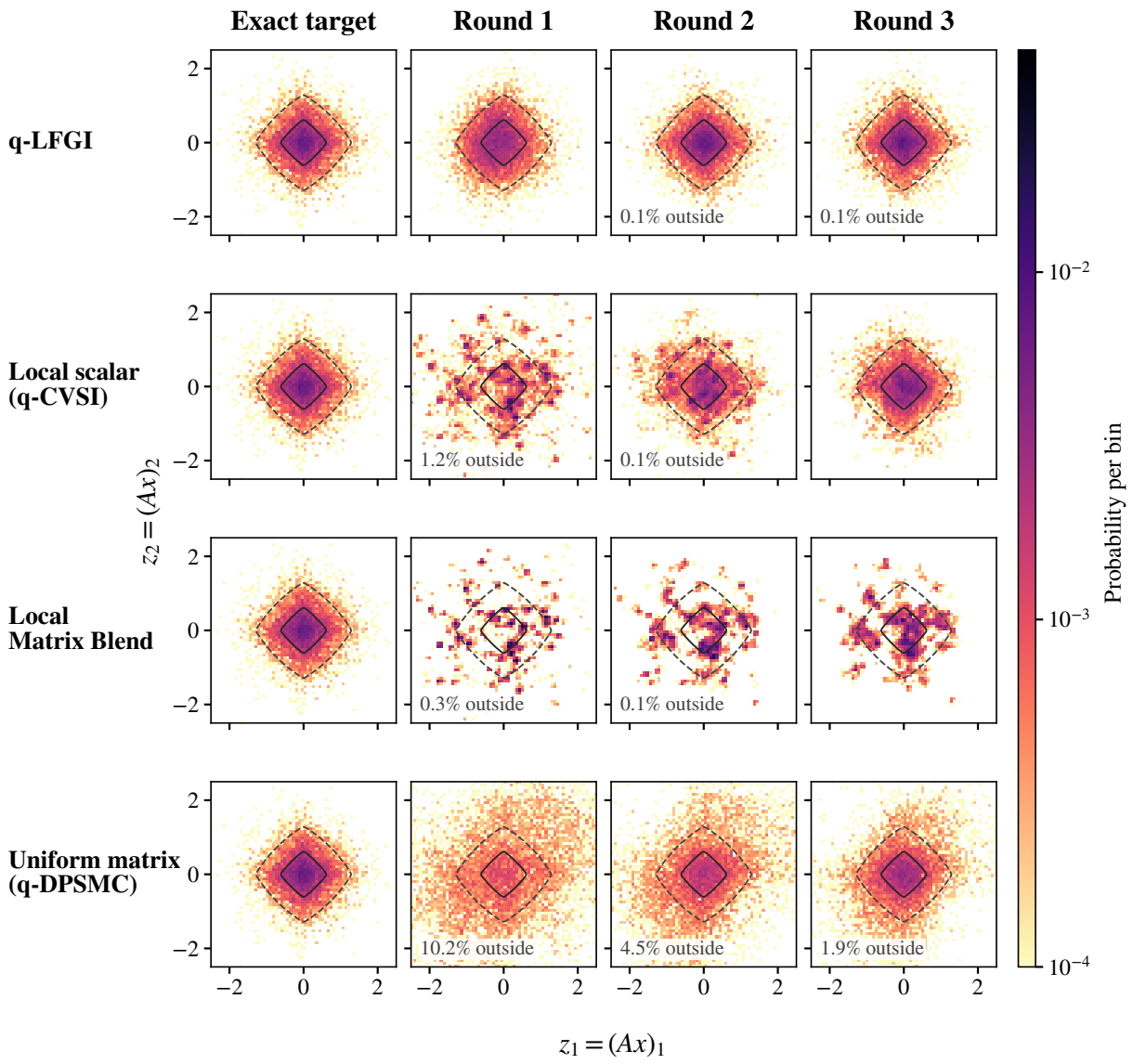


Figure 3: **Gate comparison over three rounds on an eight-dimensional target.** Rows show q-LFGI, local scalar (q-CVSI), local matrix blend, and uniform matrix (q-DPSMC) gates. Columns show the exact target and GAD rounds 1–3. Axes are the first two transformed coordinates $z = Ax$ defined in Appendix E.4. Panels share axes and a logarithmic probability-per-bin scale normalized by the full sample count. Contours enclose 50% and 90% of the exact projected target. The histograms use samples generated only for plotting, which we call display draws, from a single run. The errors in Table 3 use all eight physical coordinates and five seeds. Appendix E.5 gives the display construction.

12 Low-dimensional sampler benchmarks

We compare GAD with isotropic MALA and unit-mass HMC on an anisotropic Gaussian, unequal overlapping lobes, a ring, and an S-shaped chain. The densities and parameters are given in Appendix F.1 in the order shown in Figure 4. All methods start from the same broad Gaussian law. The comparison uses a common allowance for energy, gradient, and curvature-matrix requests, including MCMC initialization and warmup. The request counts and settings are given

in Appendices F.3 to F.5.

Table 4 compares 1,536 samples from each method with the same 1,536 independent target samples within each run. The MCMC samples retain chain dependence and states repeated after rejections. GAD has the lowest mean error on the Gaussian, lobes, and S-chain. HMC has the lowest mean error on the ring. The between-seed SDs are comparable to or larger than many of the differences in mean error.

Table 4: **Two-dimensional comparison with MALA and HMC.** Normalized empirical SW_2 in percent, mean $\pm$ sample SD over three seeds. Lower values are better. Each method contributes 1,536 samples, compared with 1,536 independent target samples shared by all methods. GAD uses 4,096 reference samples per round and three rounds. Every method receives 24,576 energy, gradient, and curvature-matrix requests in total, including MCMC initialization and warmup. Neither MCMC method uses preconditioning. Target/target compares two independent target samples of the same size. The normalization is given in (E.1).

Target	GAD	MALA	HMC	Target/target
Gaussian	6.09 ± 1.17	6.93 ± 2.79	7.29 ± 2.79	6.12 ± 0.13
Unequal lobes	5.41 ± 0.29	6.82 ± 1.38	5.60 ± 1.98	4.55 ± 0.26
Ring	5.42 ± 1.02	6.27 ± 1.54	4.94 ± 1.70	4.23 ± 0.83
S-shaped chain	5.29 ± 2.45	7.65 ± 3.27	5.40 ± 3.22	5.29 ± 1.82

Once the GAD score field is computed, we can transport fresh draws without further target requests. These additional display draws do not enter the accuracy metrics or the next GAD round. The GAD display in Figure 4 uses these fresh draws, while the MCMC display resamples the request-limited chains. The number of independent draws affects the smoothness of the histograms. Figures 12 and 13 repeat the display with equal base samples, meaning the same number of samples before resampling and jitter for every method, and show the samples used in Table 4.

13 Posterior sampling in nonlinear inverse problems

We compare direct and annealed GN-GAD at common allowances for target requests on two nonlinear inverse problems. The acoustic example infers a trajectory from 16 coefficients. The earthquake example infers locations, origin times, and three shared velocity parameters for five events, giving 23 unknowns in total. Appendix H gives two initial-field inversions whose reported posterior summaries are well described by MAP–Laplace.

13.1 Experimental setup and evaluation

The acoustic and earthquake problems use prior-whitened coefficients $\alpha \sim \mathcal{N}(0, I_d)$ and observations $y_{\mathrm{obs}} = F(\alpha) + \eta$, with $\eta \sim \mathcal{N}(0, \Gamma)$. The posterior target is

$$\widetilde{p}_0(\alpha) = \exp\left\{-\frac{1}{2}\|\alpha\|^2 - \frac{1}{2}\|\Gamma^{-1/2}(F(\alpha) - y_{\mathrm{obs}})\|^2\right\}, \qquad p_0(\alpha) = \frac{\widetilde{p}_0(\alpha)}{Z_{\mathrm{tar}}}. \tag{13.1}$$

Appendices C.1 and D.1 specify the forward map F and noise covariance Γ. The clean score comes from differentiating the log posterior in (13.1). The gate uses the Gauss–Newton approximation (9.1) to the observed information. Under annealing, we multiply the likelihood terms in the score and observed-information approximation by the likelihood power as in (9.4).

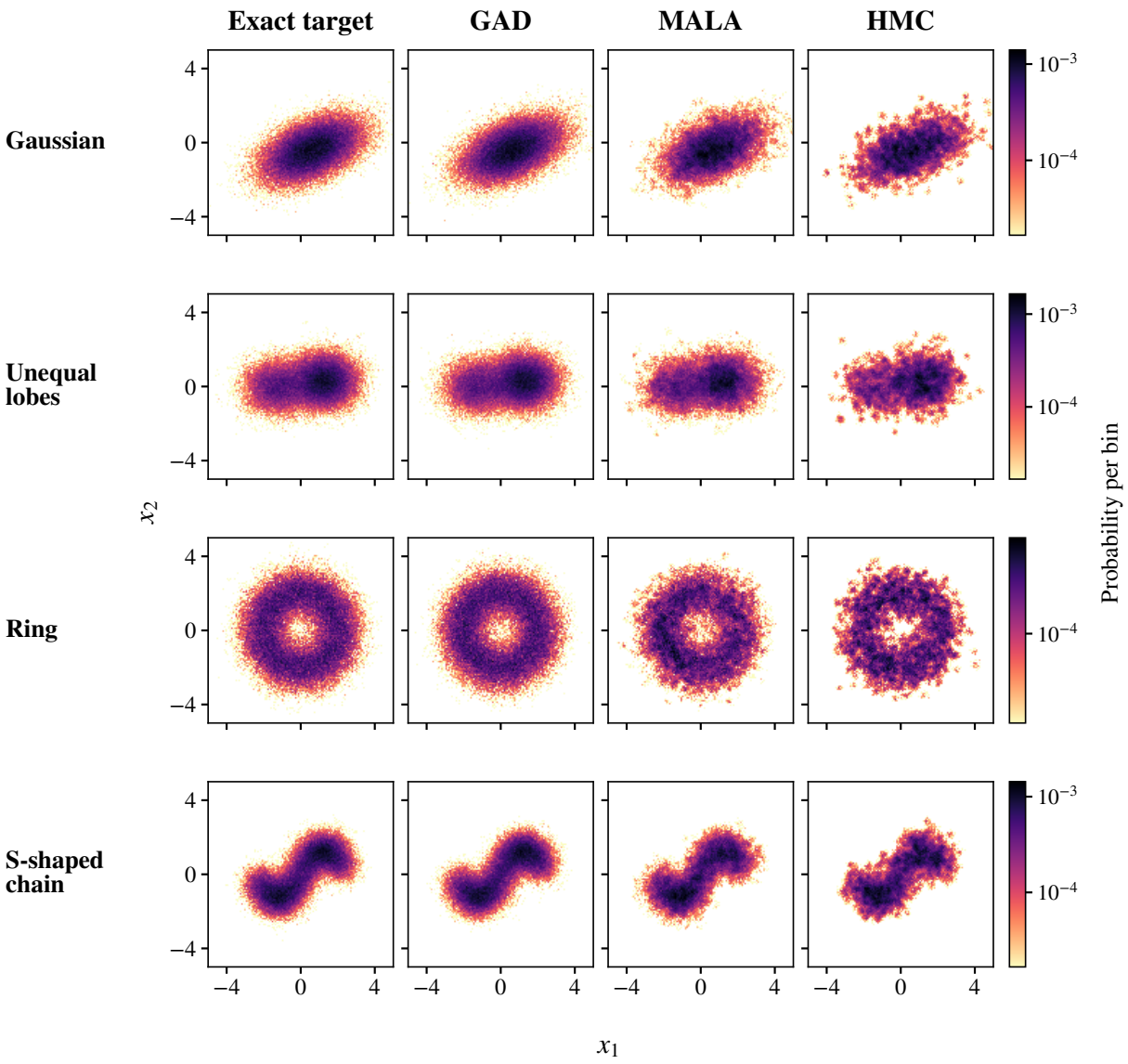


Figure 4: **Two-dimensional distributions at the selected request allowance.** Rows show the Gaussian, unequal-lobe, ring, and S-shaped targets. Columns show samples from the target, three-round GAD, MALA, and HMC in the original coordinates for one seed. The target and GAD panels start from independent draws, with the GAD draws transported by the third-round score field. The MCMC panels start from the request-limited production states. All panels use the same resampling and Gaussian jitter for display. Colors show probability per bin on a shared log scale within each target. Appendix F.2 gives the display construction and identifies the samples used for the metrics.

Each problem uses one fixed synthetic dataset. For each problem, we select the MCMC settings by the final primary error on a calibration seed, then hold the settings fixed for five evaluation seeds (Appendix I.4). Within each problem the seeds share the dataset and the NUTS benchmark samples.

Direct and annealed GN-GAD share their initial samples and terminal Gaussian draws within each seed. Both use the same sample count per round, round count, and request allowance. The comparison varies the likelihood schedule. The MCMC methods in the main tables start from

the same initial law as GN-GAD, except for MAP-pMALA, which starts from MAP–Laplace and uses the inverse exact posterior Hessian at the MAP. The MAP-pMALA comparison includes the effect of starting from MAP–Laplace. Appendix I.3 gives the starting laws, schedules, and allowances.

We measure how sampling accuracy changes with cumulative access to the target. The request count is

$$W = N_E + N_G + N_H, \tag{13.2}$$

where the three terms count pointwise energy, gradient, and curvature-matrix requests. We use E+G+H for this sum. A curvature-matrix request returns either the exact observed-information matrix or its Gauss–Newton approximation. Each curvature-matrix request counts once in either case. The count includes sampler setup and warmup. Each GAD round reuses the derivatives cached at its start to evaluate the score field and integrate the reverse dynamics. Appendix I.3 gives the accounting for each method.

The primary acoustic metric is joint sliced Wasserstein distance in prior-whitened coefficients, normalized by the root mean squared coefficient spread of the NUTS samples. The primary earthquake metric averages the normalized sliced Wasserstein distances for the location and origin-time blocks of four sparsely observed earthquakes, labeled E2–E5. Event E1 is the densely observed anchor. Maximum event error is the largest absolute difference from the NUTS event probabilities, in percentage points. The data-fit ratio compares the median noise-normalized squared residual with the NUTS median, so the target is one. Appendices C and D give the exact metric definitions. For acoustic inversion, the events are midpoint height below zero, entry into a specified disk, and crossing the horizontal axis during the observed time interval. For earthquake location, the events are depth below 5 km or above 10 km for each earthquake. Tables and uncertainty bands report the mean and one sample SD over the evaluation seeds. The physical histograms illustrate a single seed.

13.2 Acoustic trajectory inversion

In the acoustic problem, we infer the planar path of a moving source using time-difference-of-arrival (TDOA) observations from two fixed receivers. The path is represented by eight modes of a zero-mean Gaussian process in each coordinate. At each known pulse time, the forward map takes the difference in range to the receivers and divides by the sound speed. The likelihood uses Gaussian timing noise. Appendix C.1 specifies the path basis, forward map, and observation covariance.

Each pulse constrains the difference in range to the two receivers. Many source positions can give the same range difference. The temporal prior links these positions into paths, whose uncertainty can follow a curve that a local Gaussian may describe poorly. Figure 5 shows the generating path, receivers, and arrival differences. The data are generated from a path with a nonzero mean and a larger basis than the common inference prior (Appendix C.1).

Both GAD variants follow the broad curve in the NUTS midpoint samples at the final request allowance. In the run shown in Figure 6, the annealed HMC samples are less evenly spread along the NUTS midpoint curve. Appendix C gives the errors by coefficient block and summaries of the paths.

Annealed GN-GAD has a joint SW_2 of 0.134 ± 0.012 at the final allowance. The mean joint SW_2 is 37.1% below that of annealed HMC (0.213 ± 0.004), which has the lowest error among the MCMC methods on this measure (Table 5). Annealing reduces GAD's joint error in every paired seed. Annealed GN-smMALA has the lowest mean maximum event error, and MAP-pMALA comes closest to NUTS on median data misfit.

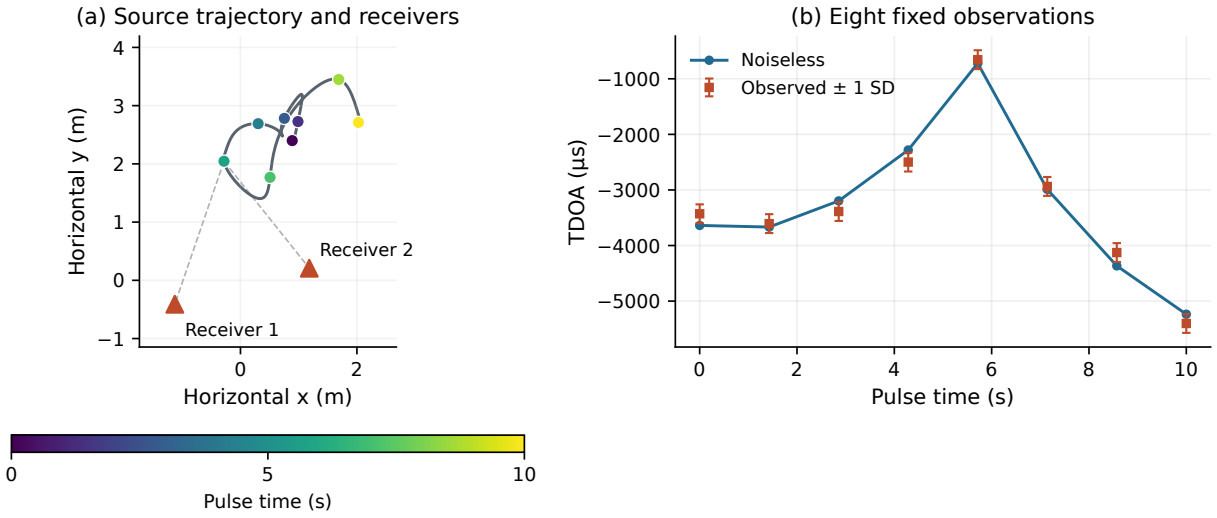


Figure 5: **Acoustic trajectory acquisition and fixed observations.** (a) Generating path in meters, with eight pulses (circles) and two receivers (triangles). Pulse color gives emission time. Dashed segments show the two ranges for one pulse. (b) Noiseless and observed time differences of arrival in microseconds at the same pulse times. Error bars show one noise SD. Appendix C.1 gives the forward map and the fixed observations.

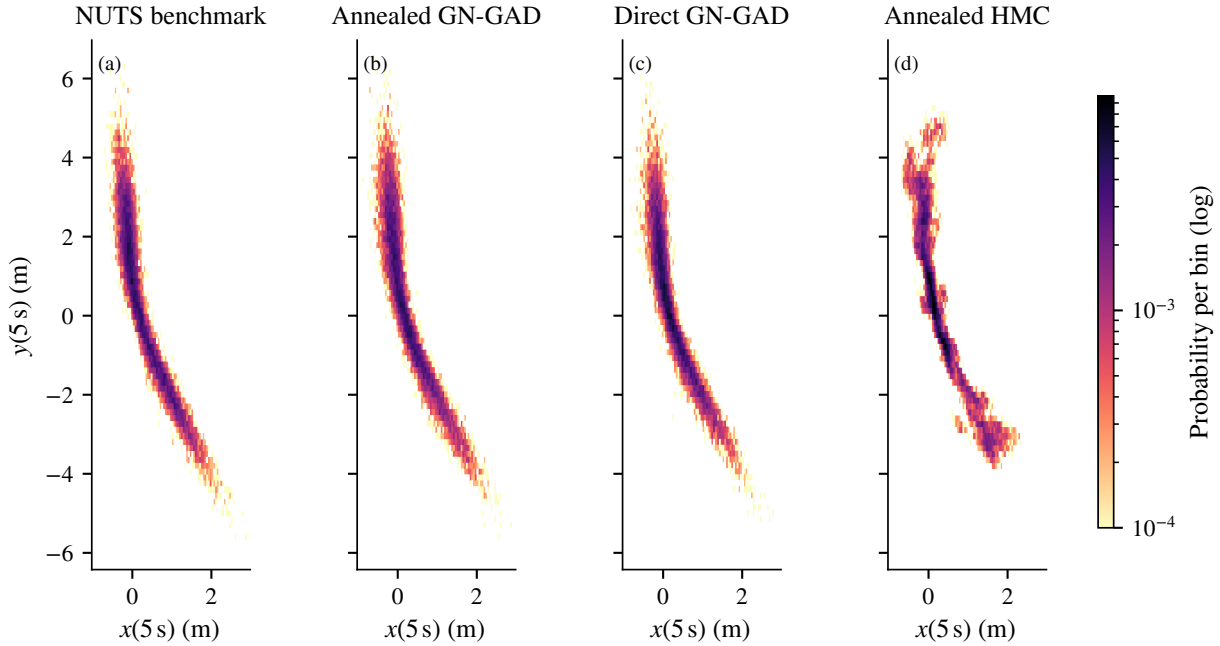


Figure 6: **Acoustic midpoint geometry at the final allowance.** The axes give the two coordinates of the path at its temporal midpoint, 5 s, in meters. Panels show samples from NUTS, annealed GN-GAD, direct GN-GAD, and annealed HMC at the final request allowance. The three sampler panels use one fixed seed. Panels use common bins and a shared logarithmic probability-per-bin scale, with counts divided by the full sample size. Appendix C.4 gives the display construction.

Table 5: **Acoustic endpoint comparison.** Five-seed means ± sample SD under an allowance of 27,517 E+G+H requests. Event error is the largest absolute probability error in percentage points (pp). Lower SW_2 and event errors are better, and a fit ratio of one matches NUTS. Appendix C gives the metric definitions, actual request counts, and full comparison.

Method	Joint $SW_2 \downarrow$	Event error (pp) $\downarrow$	Fit ratio $\to 1$
Annealed GN-GAD	0.134 ± 0.012	15.2 ± 2.3	2.03 ± 0.12
Direct GN-GAD	0.154 ± 0.009	19.3 ± 2.3	2.15 ± 0.07
MAP-pMALA	0.630 ± 0.020	47.3 ± 7.7	0.98 ± 0.07
Annealed MALA	0.218 ± 0.001	16.7 ± 3.5	2.33 ± 0.25
GN-smMALA	0.312 ± 0.113	16.4 ± 4.5	7.36 ± 1.61
Annealed GN-smMALA	0.239 ± 0.026	13.2 ± 3.6	6.71 ± 1.27
Annealed HMC	0.213 ± 0.004	14.6 ± 4.3	2.16 ± 0.12

13.3 Joint earthquake location and velocity inference

For the earthquake problem, we jointly infer locations, origin times, and three shared velocity parameters for five earthquakes using sparse arrival-time observations. Each event contributes east, north, depth, and origin-time coordinates. The velocity model describes how the primary (P) and secondary (S) wave speeds change with depth. The unknowns are the primary wave speed at the surface, its vertical gradient, and the ratio of P- to S-wave speeds. A pick is modeled as the event origin time plus the travel time to the station, with phase-dependent Gaussian noise. Appendix D.1 gives the priors, the analytic travel-time formula, and its domain.

E1 serves as an anchor because we observe both P and S arrivals at every station. For each of E2–E5 we have just two P arrivals and one S arrival, for a total of 24 observations with fixed station associations (Figure 7). S–P arrival-time differences and differences between pairs of P arrivals constrain distances to a pair of stations. Many locations along a curved set can explain those differences. The likelihood depends on each event's origin time and on velocities shared across events. The set of compatible event locations remains curved after the affine change from event to whitened coordinates. Appendix D.1 lists the generating parameters and picks.

Figure 8 compares the location–depth tradeoffs for events E2 and E4 at the final allowance.

Annealed GN-GAD has an event-block SW_2 of 0.154 ± 0.019 at the final allowance. The mean event-block SW_2 is 42.5% below that of direct GN-smMALA, which has the lowest block error among the MCMC methods (Table 6). Annealing roughly halves GAD's mean block error and reduces it in all five paired seeds. Annealed GN-GAD also has the lowest mean event-probability error, and MAP-pMALA comes closest to the NUTS median data misfit. Applying the annealed GN-GAD likelihood schedule to GN-smMALA increases its event-block error at the final request allowance.

Figure 9 plots primary error against the common request allowance at each checkpoint, a saved sampler output at a specified cumulative request allowance. Full-power samples are retained at likelihood power one, where the target is the final posterior. Both annealed GAD runs reach full likelihood power for their final two rounds and continue to improve at full power. Intermediate-power points show distance to the final posterior.

14 Discussion

The correction derived in Section 5 uses target derivatives averaged under the current OU posterior. The error of the corrected score depends on how target curvature varies across the clean states that could have produced the noisy state (Sections 5 and 6). The contraction argument bounds

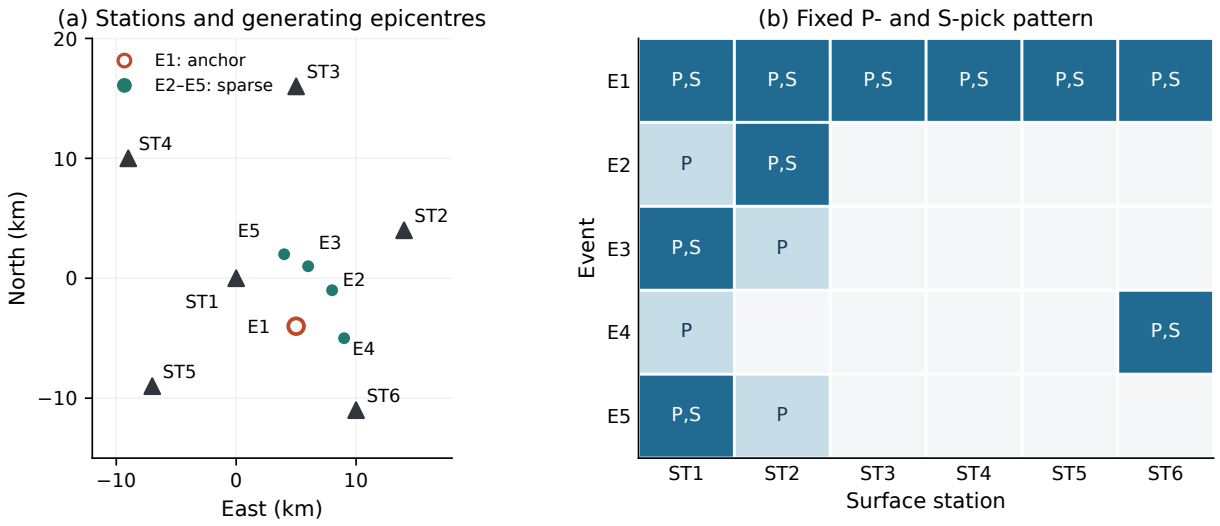


Figure 7: **Earthquake acquisition geometry and observed phases.** (a) Surface stations (triangles) and generating epicenters (circles) in east and north coordinates in km. (b) P- and S-wave arrivals observed for each event and station. The anchor event E1 has both phases at all six stations. A blank cell indicates no pick. Appendix D.1 gives the generating parameters and fixed arrival times.

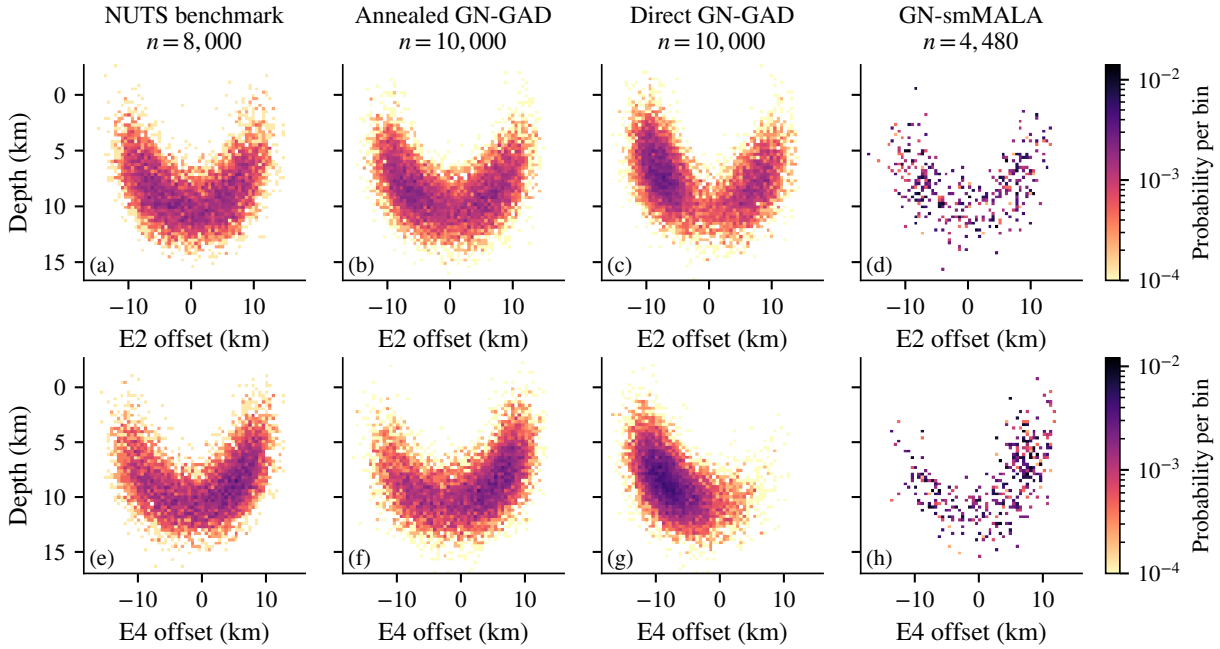


Figure 8: **Earthquake location and depth at the final allowance.** For events E2 and E4, horizontal offset is measured along the leading horizontal principal axis of the NUTS samples and centered at their mean. Offset and depth are in km, with depth positive downward. Columns show NUTS, annealed GN-GAD, direct GN-GAD, and GN-smMALA. Sampler panels use one seed. Each row shares bins and a logarithmic probability-per-bin scale normalized by the full sample count. Appendix D.4 gives the display construction.

the integrated corrected-score error for perturbed Gaussian targets using conditional averages.

Table 6: **Earthquake-location endpoint comparison.** Five-seed means ± sample SD under an allowance of 27,000 E+G+H requests. The block metric averages the normalized SW_2 distances for E2–E5, and event error is the largest absolute probability error in percentage points (pp). Lower errors are better, and a fit ratio of one matches NUTS. Appendix D gives the metric definitions, actual request counts, and full comparison.

Method	Event-block $SW_2 \downarrow$	Event error (pp) $\downarrow$	Fit ratio $\to 1$
Annealed GN-GAD	0.154 ± 0.019	7.7 ± 1.2	1.45 ± 0.05
Direct GN-GAD	0.311 ± 0.104	8.9 ± 3.3	1.36 ± 0.04
MAP-pMALA	0.491 ± 0.025	15.3 ± 4.3	1.00 ± 0.02
Annealed MALA	0.653 ± 0.020	41.7 ± 4.5	2.73 ± 0.07
GN-smMALA	0.267 ± 0.028	16.5 ± 2.0	3.76 ± 0.12
Annealed GN-smMALA	0.374 ± 0.021	29.0 ± 2.7	4.92 ± 0.25
Annealed HMC	0.359 ± 0.028	28.6 ± 5.3	1.28 ± 0.06

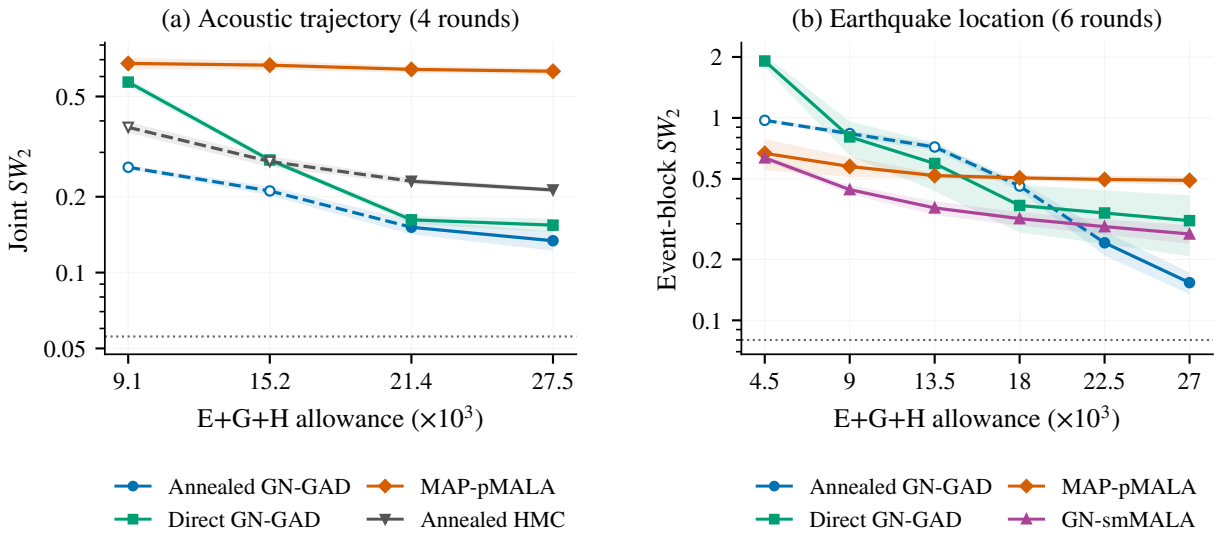


Figure 9: **Geometric accuracy under common request allowances.** The left panel shows joint normalized SW_2 for acoustic inversion, and the right panel shows the mean normalized SW_2 over event blocks E2–E5. Lower values indicate closer agreement with NUTS. Markers give means over five sampler seeds and shading gives one sample SD. The horizontal axis gives each checkpoint's common allowance in thousands of energy, gradient, and curvature-matrix requests (Appendix I.3). Hollow markers denote checkpoints without full-power samples, and dashed segments touch those checkpoints. Dotted lines show discrepancies between disjoint groups of NUTS chains. The vertical axes are logarithmic.

The contraction analysis requires neither a Poisson solve nor a smooth initial density (Section 8).

Constant curvature makes the Gaussian score estimate exact regardless of where the reference samples lie, provided the target derivatives are exact. Changing the gate on the eight-dimensional target with varying curvature changes the sample error after the same number of rounds (Table 3). The clean score changes less with the component weights as the mixture components separate, and forward KL improves more slowly (Figure 2). GAD has the lowest mean error on three of the two-dimensional benchmarks, and HMC has the lowest mean error on the ring (Table 4).

Annealed GN-GAD comes closest to NUTS on the primary sliced Wasserstein measure in both inverse problems, while MAP-pMALA comes closest on median data misfit (Tables 5 and 6). Different trajectories or event locations can predict similar arrival times. The median residual measures agreement with the observations. The distributional distances and physical projections assess where the samplers place probability among the trajectories or locations that can produce those observations.

In the Allen–Cahn and Navier–Stokes initial-field inversions, MAP–Laplace agrees closely with the NUTS benchmark on the displayed projections and gives the smallest reported field-moment errors. GN-GAD improves on MAP-pMALA in sliced Wasserstein distance, with a smaller gain over MAP–Laplace (Appendix H). In the acoustic and earthquake examples, the NUTS projections instead show curved range-difference and location–depth tradeoffs. The orientation of uncertainty changes along those projected sets (Figures 6 and 8).

15 Limitations and future work

Theorem 8.4 concerns exact conditional expectations and exact curvature in reverse diffusion for a fixed, strongly log-concave target with sufficiently small variation of the perturbation derivatives. The separated Gaussian-mixture targets lie outside the strongly log-concave class covered by Theorem 8.4. In the inverse problems, we estimate the conditional means from samples, approximate the curvature by Gauss–Newton, and integrate a deterministic flow numerically. We choose a likelihood schedule for each problem and change the target between rounds as the likelihood power increases. The earthquake schedule is also tested with GN-smMALA (Table 6 and Appendix I.2).

Storing dense curvature matrices and repeatedly evaluating the score field add to the cost of transport (Appendix I.5). Estimating densities along the probability flow may also allow importance weighting to correct errors in component probabilities. The density calculations in [14] use posterior pilot samples. We would need to evaluate the density accurately using the samples generated by GAD itself.

16 Conclusion

GAD estimates a diffusion score using the current samples and derivatives of the unnormalized target density. Each round constructs a corrected noisy score from those samples, then uses reverse diffusion to generate the next unweighted sample set. The corrected score equals a Gaussian target's noisy score for any nonempty set of reference samples when target derivatives are exact. The contribution of the initial error to the forward KL bound decays geometrically across rounds with exact conditional expectations for strongly log-concave perturbations of Gaussian targets whose perturbation derivatives vary sufficiently little.

Annealed GN-GAD follows the curved posterior structure seen in the acoustic and earthquake NUTS benchmarks. It gives the lowest mean primary sliced Wasserstein error among the methods

compared on both problems at the final request allowance. Each round reuses its target derivatives throughout transport. Structured curvature and faster reverse integration could reduce the cost of evaluating the score field along those trajectories.

A Poisson equation and q-LFGI identities

A.1 Weak Poisson solvability

For the weak problem (4.2), assume $\zeta \in L^2(\rho^q_{y,t})$ and a finite current-posterior Poincaré constant $C_{\mathrm{P}}(\rho^q_{y,t})$. For centered $\varphi \in H^1(\rho^q_{y,t})$,

$$|\mathbb{E}_{\rho^q_{y,t}}[\zeta\varphi]| \le \|\zeta\|_{L^2(\rho^q_{y,t})}\|\varphi\|_{L^2(\rho^q_{y,t})} \le \sqrt{C_{\mathrm{P}}(\rho^q_{y,t})}\|\zeta\|_{L^2(\rho^q_{y,t})}\|\nabla\varphi\|_{L^2(\rho^q_{y,t})}.$$

The bilinear form $\mathbb{E}_{\rho^q_{y,t}}[\nabla u \cdot \nabla\varphi]$ is coercive on mean-zero $H^1(\rho^q_{y,t})$ modulo constants. Lax–Milgram gives a unique mean-zero weak solution under these conditions, which justify the Poisson representation used in Sections 4 and 5. The convergence proof uses the score-error identity (8.2).

A.2 Paired identities and calibrated q-LFGI risk

Because

$$\nabla_x K_t^{\mathrm{OU}}(y \mid x) = -\alpha_t b(x; y, t) K_t^{\mathrm{OU}}(y \mid x),$$

differentiation of the OU marginals gives $\mathbb{E}_{\rho^q_{y,t}}[b] = s^q_t$ and $\mathbb{E}_{\rho_{y,t}}[b] = s_t$. If q has a positive differentiable density and the boundary term vanishes, integration by parts in x gives $\mathbb{E}_{\rho^q_{y,t}}[\nabla \log q(X_0)] = \alpha_t \mathbb{E}_{\rho^q_{y,t}}[b]$. Subtracting the mean of b from the mean of $c = s_0/\alpha_t$ gives the expression for m^q_t in terms of the density ratio following (4.1). The target OU posterior score is

$$\nabla_x \log \rho_{y,t}(x) = s_0(x) - \frac{\alpha_t}{\gamma_t}(\alpha_t x - y) = \alpha_t\{c(x;t) - b(x;y,t)\}. \tag{A.1}$$

Its mean under $\rho_{y,t}$ vanishes under the boundary condition for the target, giving $\mathbb{E}_{\rho_{y,t}}[c] = s_t$. For a matrix gate $G(y,t)$, set $a^G_t = Gm^q_t$ and $e^G_t = d_t - a^G_t$. The Tweedie and target-score signals both have mean $s_t(y)$ under the target OU posterior, so

$$e^G_t(y) = \mathbb{E}_{\rho_{y,t}}[z_G] - \mathbb{E}_{\rho^q_{y,t}}[z_G] = \mathbb{E}_{\rho^q_{y,t}}\left[g\{z_G - \mathbb{E}_{\rho^q_{y,t}}[z_G]\}\right]. \tag{A.2}$$

Direct differentiation gives

$$J_G(x;y,t) := \nabla_x z_G(x;y,t) = \frac{1}{\alpha_t\gamma_t}\{\alpha_t^2 I_d - G(y,t)A_0(x;t)\}. \tag{A.3}$$

Averaging (A.3) under the current OU posterior gives

$$\mathbb{E}_{\rho^q_{y,t}}[J_G] = \frac{1}{\alpha_t\gamma_t}\{\alpha_t^2 I_d - GA^q(y,t)\}.$$

Hence

$$\mathbb{E}_{\rho^q_{y,t}}[J_G] = 0 \quad\Longleftrightarrow\quad G = \alpha_t^2 (A^q)^{-1} = \Psi_t(H^q) = G^q_{\mathrm{LFGI}}.$$

At $q = p_0$, assume the second moments and integration by parts used in the following identities are valid. Integrating the derivative of the target OU posterior score (A.1) under $\rho_{y,t}$ gives the covariance of the zero-mean disagreement signal

$$\mathbb{E}_{\rho_{y,t}}[(c-b)(c-b)^\top] = \frac{1}{\gamma_t} I_d + \frac{1}{\alpha_t^2} H^{p_0}(y,t),$$

and integration by parts with $b - s_t$ gives

$$\mathbb{E}_{\rho_{y,t}}[(b - s_t)(c - b)^\top] = -\frac{1}{\gamma_t} I_d.$$

Assume the disagreement covariance is positive definite, as holds when $\alpha_t^2 I_d + \gamma_t H^{p_0}$ is positive definite. The conditional squared-error risk in (6.3) has normal equation

$$G\left(\frac{1}{\gamma_t} I_d + \frac{1}{\alpha_t^2} H^{p_0}(y,t)\right) = \frac{1}{\gamma_t} I_d.$$

Its unique solution is

$$G = \Psi_t(H^{p_0}) = G_\star,$$

the LFGI population-optimal gate [14]. This proves (6.3).

A.3 Conditional moments of the Poisson flux

The weak Poisson equation (4.2) implies, componentwise for every sufficiently regular vector-valued f,

$$\mathbb{E}_{\rho_{y,t}}[f] - \mathbb{E}_{\rho^q_{y,t}}[f] = \mathbb{E}_{\rho^q_{y,t}}[(\nabla f)\nabla u].$$

Taking $f = z_G$ and using (A.2) gives

$$e_t^G = \mathbb{E}_{\rho^q_{y,t}}[J_G \nabla u].$$

Taking $f = \nabla\varphi$ yields the differentiated Stein identity

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla^2\varphi\, \nabla u] = \mathbb{E}_{\rho_{y,t}}[\nabla\varphi] - \mathbb{E}_{\rho^q_{y,t}}[\nabla\varphi].$$

For $\varphi(x) = \|x\|^2/2$,

$$\mathbb{E}_{\rho^q_{y,t}}[\nabla u] = \mathbb{E}_{\rho_{y,t}}[X_0] - \mathbb{E}_{\rho^q_{y,t}}[X_0] = \frac{\gamma_t}{\alpha_t} d_t.$$

For $\varphi = \log \rho_{y,t}$, the target OU posterior satisfies

$$\nabla^2 \log \rho_{y,t} = -\frac{1}{\gamma_t} A_0, \qquad \mathbb{E}_{\rho_{y,t}}[\nabla \log \rho_{y,t}] = 0, \qquad \mathbb{E}_{\rho^q_{y,t}}[\nabla \log \rho_{y,t}] = \alpha_t m_t^q.$$

Substitution gives

$$\mathbb{E}_{\rho^q_{y,t}}[A_0 \nabla u] = \alpha_t \gamma_t\, m_t^q,$$

proving (4.3).

B Proofs for convergence with exact expectations

Throughout, Assumption 8.1 holds and q is a probability law on $\mathbb{R}^d$. A density and finite $\mathrm{KL}(p_0\|q)$ are assumed only where stated. At a score-evaluation point (y,t) with $t > 0$, write $\mu^q := \mathbb{E}_{\rho^q_{y,t}}[X_0]$. Statements about (3.1) on a window $[t_1, t_2]$ use the reverse clock $u = t_2 - t$, in which the equation driven by a score s reads

$$\mathrm{d}Z_u = \{Z_u + 2s(Z_u, t_2 - u)\}\mathrm{d}u + \sqrt{2}\,\mathrm{d}B_u, \qquad 0 \le u \le t_2 - t_1. \tag{B.1}$$

B.1 Posterior moments and growth of the corrected score

For $t > 0$, the kernel $x \mapsto K_t^{\mathrm{OU}}(y \mid x)$ is positive, and every polynomial in x times it is bounded. Hence $\rho_{y,t}^q$ has moments of all orders even when q has none. Since H_0 is bounded and s_0 has linear growth, all posterior averages in Sections 4 to 6 are finite. Because $\nabla_y \log \rho_{y,t}^q(x) = b(x; y, t) - s_t^q(y)$, differentiation under the integral gives, componentwise, $\nabla_y \mathbb{E}_{\rho_{y,t}^q}[F] = \mathrm{Cov}_{\rho_{y,t}^q}(F, b)$ for every $F(x)$ of polynomial growth. On each compact set $[\varepsilon, T] \times \{\|y\| \le n\}$ with $0 < \varepsilon < T$, the Gaussian kernel bounds the polynomial-weighted numerators uniformly, and the positive continuous density $q_t(y)$ has a positive minimum. Together, the positive minimum of the density and the uniform bounds on the numerators bound the derivatives on each compact set. In particular, $s^{a,q}$ is continuous on $(0, \infty) \times \mathbb{R}^d$ and locally Lipschitz in y, uniformly for t in compact subsets of $(0, \infty)$.

Lemma B.1 (Posterior moment bound)**.** *Choose $R < \infty$ with $\beta_R := q(\{\|x\| \le R\}) > 0$. For all $t > 0$ and $y \in \mathbb{R}^d$,*

$$\mathbb{E}_{\rho_{y,t}^q} \|\alpha_t X_0 - y\|^2 \le (\|y\| + R)^2 + 2 \log(1/\beta_R), \qquad \|\alpha_t \mu^q\| \le 2\|y\| + R + \sqrt{2 \log(1/\beta_R)}.$$

Proof. Let $\ell := \int \exp\{-\|\alpha_t x - y\|^2/(2\gamma_t)\}\, q(\mathrm{d}x)$, so that $\mathrm{d}\rho_{y,t}^q/\mathrm{d}q = \exp\{-\|\alpha_t x - y\|^2/(2\gamma_t)\}/\ell$. Nonnegativity of $\mathrm{KL}(\rho_{y,t}^q \| q)$ gives $\mathbb{E}_{\rho_{y,t}^q} \|\alpha_t X_0 - y\|^2 \le -2\gamma_t \log \ell$. Restricting the integral to $\{\|x\| \le R\}$ gives $\ell \ge \beta_R \exp\{-(\|y\| + \alpha_t R)^2/(2\gamma_t)\}$. The squared-displacement bound in Lemma B.1 follows from $\alpha_t, \gamma_t \le 1$. The conditional-mean bound follows from Jensen's inequality and the triangle inequality. □

Lemma B.2 (Score cancellation and linear growth)**.** *For $t > 0$ and $y \in \mathbb{R}^d$,*

$$s_t^{a,q}(y) = -(A^q)^{-1}\{H^q y + \alpha_t(\mathbb{E}_{\rho_{y,t}^q}[\nabla V] - (H^q - \Lambda)\mu^q)\}. \tag{B.2}$$

With R and β_R as in Lemma B.1, $\|s_t^{a,q}(y)\| \le C_q(1 + \|y\|)$ for all $t > 0$ and y, where

$$C_q := \frac{\|\Lambda\|_{\mathrm{op}} + 3L_V + \sup_x \|\nabla V(x)\| + L_V\{R + \sqrt{2\log(1/\beta_R)}\}}{\min\{1, \kappa\}}.$$

The linear growth bound also holds for the target score s_t, with constant C_{p_0}.

Proof. Tweedie's identity gives $s_t^q = (\alpha_t \mu^q - y)/\gamma_t$. Since $-s_0(x) = \Lambda x + \nabla V(x)$, $m_t^q = -\alpha_t^{-1}(\Lambda \mu^q + \mathbb{E}_{\rho_{y,t}^q}[\nabla V]) - (\alpha_t \mu^q - y)/\gamma_t$. Insert $a_t^q = \alpha_t^2 (A^q)^{-1} m_t^q$ into $s_t^{a,q} = s_t^q + a_t^q$ and use $I_d - \alpha_t^2 (A^q)^{-1} = \gamma_t (A^q)^{-1} H^q$. The factors $1/\gamma_t$ cancel, and (B.2) follows. The growth bound uses $A^q \succeq (\alpha_t^2 + \kappa\gamma_t) I_d \succeq \min\{1, \kappa\} I_d$, $\|H^q\|_{\mathrm{op}} \le \|\Lambda\|_{\mathrm{op}} + L_V$, $\|H^q - \Lambda\|_{\mathrm{op}} \le L_V$, and Lemma B.1. At $q = p_0$, $m_t^q = 0$ by (4.1), so $s_t^{a,p_0} = s_t$. □

B.2 Well-posedness and path-entropy comparison

Lemma B.3 (Well-posedness)**.** *Let $0 \le t_1 < t_2$, and let s be $s^{a,q}$ or the target score. For every law of Z_0, (B.1) has a unique strong solution on $[0, t_2 - t_1]$, including the endpoint, and the solution does not explode.*

Proof. For $0 < \varepsilon < t_2$, the drift is continuous and locally Lipschitz on $[0, t_2 - \max\{t_1, \varepsilon\}]$ (Appendix B.1), so a pathwise unique strong solution exists there up to explosion. For $s = s^{a,q}$, Lemma B.2 and Gronwall's inequality give, pathwise,

$$\sup_u \|Z_u\| \le (\|Z_0\| + 2C_q t_2 + \sqrt{2} \sup_{u \le t_2} \|B_u\|) e^{(1+2C_q)t_2},$$

and the target score satisfies Lemma B.2 with C_{p_0}. Hence the solution does not explode, with no moment condition on Z_0. When $t_1 = 0$, the pathwise bound on $\sup_u \|Z_u\|$ makes the drift integrable along each path up to $u = t_2$. Thus Z_u converges as $u \uparrow t_2$, and uniqueness on every $[0, t_2 - \varepsilon]$ gives uniqueness through the endpoint. □

For the exact target score, apply the diffusion time-reversal theorem [19, Theorem 2.1] on the part of the forward interval with $t \geq \varepsilon > 0$. The OU coefficients are Lipschitz and the marginals there are smooth and bounded, with locally bounded scores. The reversed OU process solves (B.1) weakly [see also 3]. Pathwise uniqueness (Lemma B.3) and the Yamada–Watanabe theorem identify its law with that of the solution started from p_{t_2}. Path continuity extends the identification to $t_1 = 0$. In particular,

$$\begin{aligned} Z_u &\sim p_{t_2-u}, \qquad 0 \leq u \leq t_2 - t_1, \\ \mathrm{Law}(Z_{t_2-t_1} \mid Z_0 = z) &= \mathrm{Law}(Y_{t_1} \mid Y_{t_2} = z) \qquad \text{for } p_{t_2}\text{-almost every } z. \end{aligned} \tag{B.3}$$

The second relation follows by disintegrating the common law with respect to its initial point in reverse time. The forward-OU conditional law in (B.3) has a density by Bayes' rule with the OU transition density.

Lemma B.4 (Interval path-entropy comparison)**.** *Let $0 \leq t_1 < t_2$, let ν be a probability density with $\mathrm{KL}(p_{t_2}\|\nu) < \infty$, and let q_{out,t_1} be the endpoint law of (B.1) with $s = s^{a,q}$ and $Z_0 \sim \nu$. Then q_{out,t_1} has a density and*

$$\mathrm{KL}(p_{t_1}\|q_{\mathrm{out},t_1}) \leq \mathrm{KL}(p_{t_2}\|\nu) + \int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t. \tag{B.4}$$

Proof. Write $\tau := t_2 - t_1$, and let $\mathbf{P}^z$ and $\mathbf{R}^z$ be the path laws on $C([0,\tau];\mathbb{R}^d)$ of the target and corrected solutions started at z. By (3.3), their drifts differ by $2e^q_{t_2-u}$. Let σ_n be the exit time from the open ball of radius n, and let $A_n := \{\sigma_n > \tau\}$. By Lemma B.2, the drift difference is bounded before $\sigma_n \wedge \tau$. Girsanov's theorem with this stopped bounded drift change gives a law $\mathbf{R}^{z,n}$ equivalent to $\mathbf{P}^z$ under which the coordinate process solves the corrected equation up to σ_n. Pathwise uniqueness up to σ_n gives $\mathbf{R}^{z,n} = \mathbf{R}^z$ on measurable subsets of A_n. Thus $\mathbf{P}^z$ and $\mathbf{R}^z$ are equivalent on A_n, with the stopped Girsanov density. Paths are continuous and neither process explodes, so $\mathbf{P}^z(A_n) \to 1$ and $\mathbf{R}^z(A_n) \to 1$. Hence $\mathbf{P}^z$ and $\mathbf{R}^z$ are equivalent and, $\mathbf{P}^z$-almost surely,

$$\log \frac{\mathrm{d}\mathbf{P}^z}{\mathrm{d}\mathbf{R}^z} = \sqrt{2}\int_0^\tau e^q_{t_2-u}(Z_u) \cdot \mathrm{d}B_u + \int_0^\tau \|e^q_{t_2-u}(Z_u)\|^2 \mathrm{d}u,$$

where B is the Brownian motion of (B.1) under $\mathbf{P}^z$. This localization replaces a Novikov condition [6, Section 5.2].

Let $\mathbf{P}$ and $\mathbf{R}$ mix the initial points over p_{t_2} and ν, respectively. By (B.3), $Z_u \sim p_{t_2-u}$ under $\mathbf{P}$. The expected quadratic variation of the stochastic integral is $2\int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t$, which is finite by Lemma B.2 and the finite second moments of p_t. The stochastic integral thus has mean zero under $\mathbf{P}^z$ for p_{t_2}-almost every z. The chain rule for relative entropy [24] gives

$$\mathrm{KL}(\mathbf{P}\|\mathbf{R}) = \mathrm{KL}(p_{t_2}\|\nu) + \int_{t_1}^{t_2} \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t.$$

The data-processing inequality for the endpoint map proves (B.4). Finally, $\nu \ll p_{t_2}$ because $p_{t_2} > 0$. For p_{t_2}-almost every z, the endpoint law of $\mathbf{P}^z$ has a density by (B.3), and the endpoint law of $\mathbf{R}^z$ is equivalent to it. Mixing over ν shows that q_{out,t_1} has a density. □

B.3 The relative de Bruijn identity

To justify the relative de Bruijn identity and its integral (3.2) for finite-forward-KL current densities, fix $q \in \mathcal{Q}$, $T > 0$, and $0 < \varepsilon < T$. For $t \in [\varepsilon, T]$, the marginals p_t and q_t are smooth and positive. By Tweedie's identity, $\nabla^2 \log q_t = (\alpha_t^2/\gamma_t^2)\,\mathrm{Cov}_{\rho^q_{y,t}}(X_0) - \gamma_t^{-1} I_d$. With β_R as in Lemma B.1,

$$\beta_R (2\pi\gamma_t)^{-d/2} \exp\{-(\|y\| + R)^2/(2\gamma_t)\} \le q_t(y) \le (2\pi\gamma_t)^{-d/2}.$$

The bounds on q_t and the posterior moment estimate of Lemma B.1 show that $|\log q_t|$, $\|\nabla \log q_t\|$, and $\|\nabla^2 \log q_t\|$ grow at most quadratically in y, uniformly for $t \in [\varepsilon, T]$. The target log density $\log p_t$ and its first two derivatives satisfy the same quadratic-growth bounds, and p_t also has Gaussian tails. These bounds justify differentiation under the integral and integration by parts without boundary terms in the common Fokker–Planck equation $\partial_t p = \nabla \cdot (yp) + \Delta p$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(p_t \| q_t) = -\mathbb{E}_{p_t}[\|d_t\|^2]$ on $[\varepsilon, T]$, and

$$\mathrm{KL}(p_\varepsilon \| q_\varepsilon) - \mathrm{KL}(p_T \| q_T) = \int_\varepsilon^T \mathbb{E}_{p_t}[\|d_t\|^2] \mathrm{d}t.$$

Data processing gives $\mathrm{KL}(p_\varepsilon \| q_\varepsilon) \le \mathrm{KL}(p_0 \| q)$. Joint lower semicontinuity of relative entropy under the weak convergence $(p_\varepsilon, q_\varepsilon) \to (p_0, q)$ gives the reverse inequality in the limit. Monotone convergence as $\varepsilon \downarrow 0$ proves (3.2). For (8.4), the joint laws of (X_0, Y_t) under p_0 and q share the OU kernel. Disintegrating their relative entropy over X_0 gives $\mathrm{KL}(p_0 \| q)$, and disintegrating over Y_t gives $\mathrm{KL}(p_t \| q_t) + \mathbb{E}_{p_t}[\mathrm{KL}(\rho_{Y_t,t} \| \rho^q_{Y_t,t})]$. Combining with (3.2) at horizon t gives (8.4) for $t > 0$. At $t = 0$ the conditional laws given the clean state coincide, so both sides are zero.

B.4 Proof of the uniform score-error bound

After a change of time, we choose a scalar function whose product with the mean conditional KL in (8.4) has a derivative that bounds the score-error integrand from above. Integrating then gives a multiple of $D(q)^2$ as a bound for $E(q)^2$.

Set $\omega_t := \alpha_t^2/\gamma_t$ and $u := \kappa/(\omega_t + \kappa)$, so u increases from zero to one and $\mathrm{d}\omega_t/\mathrm{d}t = -2(\alpha_t/\gamma_t)^2$. For fixed $q \in \mathcal{Q}$, write

$$D_{\mathrm{cond}}(u) := \mathbb{E}_{p_{t(u)}}\left[\mathrm{KL}(\rho_{Y_{t(u)},t(u)} \| \rho^q_{Y_{t(u)},t(u)})\right].$$

By (8.4), $D_{\mathrm{cond}}(0) = 0$, $D'_{\mathrm{cond}} \ge 0$ almost everywhere, and $D_{\mathrm{cond}}(u_T) = D(q)^2$, where $u_T = \kappa/(\omega_T + \kappa)$. The functions S in Lemma B.5 and (B.8) are chosen from the target before q is specified.

Lemma B.5 (Matrix bound depending only on the target)**.** *Under Assumption 8.1, define the bounded target sets*

$$\mathcal{H} := \overline{\mathrm{conv}}\{H_0(x) : x \in \mathbb{R}^d\}, \qquad \mathcal{F} := \{\nabla V(x) : x \in \mathbb{R}^d\}.$$

At $\omega = \kappa(1-u)/u$, let

$$R_H := (\omega I_d + H)^{-1}, \qquad C_H := R_H(H - \Lambda).$$

For $w > \sup_{H \in \mathcal{H}} \|C_H\|_{\mathrm{op}}^2$, define $B_H(u,w) := (I_d - C_H C_H^\top/w)^{-1/2} R_H$ and set

$$J(u,w) := \frac{\kappa}{4u^2} \sup_{H \in \mathcal{H}} \mathrm{diam}\{B_H(u,w)\mathcal{F}\}^2. \tag{B.5}$$

Suppose $S : [0,1] \to [0,\infty)$ is nondecreasing, $S(0) = 0$, continuous at zero, and piecewise C^1 with finitely many nonnegative jumps. If, for $0 < u \le 1$,

$$S(u) > \sup_{H \in \mathcal{H}} \|C_H\|_{\mathrm{op}}^2, \qquad S'(u) \ge J(u, S(u)) \tag{B.6}$$

away from the jumps, then every $q \in \mathcal{Q}$ and every $T > 0$ satisfy $E(q)^2 \le S(u_T) D(q)^2 \le S(1) D(q)^2$.

Proof. At each score-evaluation point (y,t), let $P = \rho_{y,t}$, $Q = \rho^q_{y,t}$, $H = H^q = \mathbb{E}_Q H_0 \in \mathcal{H}$, and $\delta f = \mathbb{E}_Q \nabla V - \mathbb{E}_P \nabla V$. The residual identity (8.2) becomes

$$e = C_H d + (\alpha_t/\gamma_t) R_H \delta f.$$

For $w > \|C_H\|_{\mathrm{op}}^2$, completion of the square gives

$$\|C_H d + z\|^2 \le w\|d\|^2 + z^\top (I_d - C_H C_H^\top / w)^{-1} z.$$

Taking $z = (\alpha_t/\gamma_t) R_H \delta f$ in this inequality and applying Pinsker's inequality to the bounded function $B_H \nabla V$ gives

$$\|e\|^2 \le w\|d\|^2 + \tfrac{1}{2}(\alpha_t/\gamma_t)^2 \operatorname{diam}(B_H \mathcal{F})^2 \mathrm{KL}(P\|Q).$$

The inequality holds for every $H \in \mathcal{H}$, including $H^q(y,t)$ for the current law q. We use the envelope in (B.5), which depends only on the target, and average over p_t. With $\mathrm{d}u/\mathrm{d}t = 2u^2(\alpha_t/\gamma_t)^2/\kappa$, this gives

$$E(q)^2 \le \int_0^{u_T} \{S D'_{\mathrm{cond}} + J(u,S) D_{\mathrm{cond}}\} \mathrm{d}u \le S(u_T) D_{\mathrm{cond}}(u_T).$$

The last inequality follows from (B.6). Each jump of $S D_{\mathrm{cond}}$ is nonnegative. The initial product vanishes since $D_{\mathrm{cond}}(0) = S(0) = 0$. The bound requires neither optimization over current laws nor regularity of their density ratios. □

Proof of Proposition 8.3. For η_H, η_V from (8.5), the pointwise bound (8.3), Minkowski's inequality in $L^2(p_t)$, and the change of time $u = \kappa/(\omega_t + \kappa)$ give

$$E(q)^2 \le \int_0^{u_T} \left(\sqrt{\eta_H}\, u \sqrt{D'_{\mathrm{cond}}(u)} + \sqrt{\eta_V} \sqrt{D_{\mathrm{cond}}(u)} \right)^2 \mathrm{d}u. \tag{B.7}$$

For $\eta_H, \eta_V > 0$, set

$$c := \frac{\eta_H - \eta_V + \sqrt{\eta_H^2 + \eta_V^2}}{2}, \qquad S(u) := \eta_V u + c u^2.$$

Then $0 < c < \eta_H$, $S > \eta_H u^2$, $S' \ge \eta_V$, and

$$(S - \eta_H u^2)(S' - \eta_V) - \eta_H \eta_V u^2 = 2c(\eta_H - c) u^2 (1 - u) \ge 0.$$

These are the positivity conditions for the quadratic form showing that the integrand in (B.7) is at most $(S D_{\mathrm{cond}})'$. Thus $E(q)^2 \le S(u_T) D(q)^2 \le S(1) D(q)^2 = C_{\mathrm{closed}} D(q)^2$. The cases $\eta_H = 0$ or $\eta_V = 0$ follow by continuity. This proves Proposition 8.3. □

When the quadratic part is isotropic, the matrix condition (B.6) simplifies. If $\Lambda = \lambda I_d$ and $\kappa I_d \preceq H_0 \preceq M I_d$, then H commutes with $H - \lambda I_d$. Consequently the squared singular values of B_H are

$$\frac{1}{(\omega + \ell)^2 - (\ell - \lambda)^2 / S}, \qquad \ell \in [\kappa, M].$$

For $S < 1$, the denominator is concave in ℓ, so its minimum is at an endpoint. For $S \geq 1$, it is increasing in $\ell > 0$, with the same conclusion. Since $\operatorname{diam} \mathcal{F} \leq \Delta_V$, it suffices that S have the regularity in Lemma B.5 and satisfy

$$S'(u) \geq \frac{\Delta_V^2}{4\kappa} \max_{\ell \in \{\kappa, M\}} \frac{1}{[1 + (\ell/\kappa - 1)u]^2 - [(\ell - \lambda)/\kappa]^2 u^2 / S(u)}, \tag{B.8}$$

with both denominators positive. The scalar criterion applies in any dimension, including to nonseparable targets. The maximum in (B.8) evaluates the entire denominator separately at $\ell = \kappa$ and $\ell = M$. For general anisotropic Λ, use (B.6) or the closed-form bound.

For a computed piecewise affine S, the criterion (B.8) can be certified exactly. On each interval with slope $v > 0$, put $\eta_V = \Delta_V^2/(4\kappa)$ and check the two cubic polynomial inequalities

$$v\left\{[1 + (\ell/\kappa - 1)u]^2 S(u) - [(\ell - \lambda)/\kappa]^2 u^2\right\} - \eta_V S(u) \geq 0, \qquad \ell \in \{\kappa, M\},$$

together with positivity of the denominators. Nonnegative Bernstein coefficients give a bound valid on the interval. Rational coefficients and upward rounding allow exact verification. An initial segment reaching $u = 0$ must also be verified. Once the inequalities have been verified on every interval, $S(1)$ gives a uniform bound.

A second closed-form bound uses the coefficients

$$\eta_{H,\ell} := \frac{(\ell - \lambda)^2}{\ell^2}, \qquad \eta_{V,\ell} := \frac{\Delta_V^2}{4\ell}, \qquad C_{\text{joint}} := \sum_{\ell \in \{\kappa, M\}} \frac{\eta_{H,\ell} + \eta_{V,\ell} + \sqrt{\eta_{H,\ell}^2 + \eta_{V,\ell}^2}}{2}. \tag{B.9}$$

Then $E(q)^2 \leq C_{\text{joint}} D(q)^2$. To see this, put $z = (\alpha_t/\gamma_t)\delta f - (\omega + \lambda)d$. In this notation the residual identity (8.2) reads $e = d + (\omega I_d + H)^{-1} z$. In an eigenbasis of H, each squared component is convex as a function of $(\omega + \ell)^{-1}$. Bounding by the two endpoint values and summing components gives

$$\|e\|^2 \leq \sum_{\ell \in \{\kappa, M\}} \left\| \frac{\ell - \lambda}{\omega + \ell} d + \frac{\alpha_t/\gamma_t}{\omega + \ell} \delta f \right\|^2.$$

Apply the scalar bound (B.7) to each endpoint field with time scale ℓ and coefficients $\eta_{H,\ell}, \eta_{V,\ell}$, and add the bounds. Every term uses the conditional-KL identity (8.4). Whenever their respective hypotheses hold, C_{closed}, $S(1)$, and C_{joint} are valid for every $q \in \mathcal{Q}$. Their minimum is therefore an admissible choice of θ^2.

B.5 Further examples of the score-error bound

The potential $\frac{1}{2}\|x\|^2 + 0.34 \cos x_1 \cos x_2$ couples two coordinates. Its Hessian perturbation has eigenvalues $-0.34 \cos(x_1 \pm x_2)$, so we obtain $\kappa = 0.66$, $L_V = 0.34$, $\Delta_V = 0.68$, and the closed-form bound $\theta^2 < 0.380$. For the one-dimensional potential $U(x) = x^2/2 + 0.12 \log \cosh(5x)$, we may take $\kappa = \lambda = 1$, $M = 4$, and $\Delta_V = 1.2$. The bound in (B.9) gives $\theta^2 < 0.972$, whereas $\eta_H + \eta_V = 9.36$.

B.6 Terminal error and the infinite-horizon map

The terminal error $\varepsilon_T = \mathrm{KL}(p_T \| \phi_d)$ is nonincreasing in T by data processing, since the OU semigroup leaves ϕ_d invariant. To bound it, write p_T as the mixture of $\mathcal{N}(\alpha_T x, \gamma_T I_d)$ for $x \sim p_0$. Joint convexity of relative entropy bounds ε_T by the mean KL divergence from these Gaussians to ϕ_d. The Gaussian KL formula gives

$$\varepsilon_T \leq \tfrac{1}{2}\left\{\alpha_T^2 \mathbb{E}_{p_0}\|X_0\|^2 + d(\gamma_T - 1 - \log \gamma_T)\right\} = O(e^{-2T}). \tag{B.10}$$

Here $\alpha_T^2 = e^{-2T}$ and $\gamma_T - 1 - \log \gamma_T = O(e^{-4T})$.

Proof of Theorem 8.4(ii). Fix $q \in \mathcal{Q}$. The corrected field does not depend on the horizon. For $t_1 < t_2$, let $\mathcal{U}^q_{t_2,t_1}$ be the Markov kernel of the corrected solution from physical time t_2 to t_1, so that $\Phi_T(q) = \phi_d \mathcal{U}^q_{T,0}$. By the score-error bound assumed in Theorem 8.4 and monotone convergence, $\int_0^\infty \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t \le \theta^2 \mathrm{KL}(p_0\|q) < \infty$. Hence $E^2_{>T} := \int_T^\infty \mathbb{E}_{p_t}[\|e_t^q\|^2]\mathrm{d}t \to 0$. For $S > T$, the Markov property gives $\Phi_S(q) = \nu_{S,T}\mathcal{U}^q_{T,0}$ with $\nu_{S,T} := \phi_d \mathcal{U}^q_{S,T}$, and (B.4) on $[T, S]$ gives $\mathrm{KL}(p_T\|\nu_{S,T}) \le \varepsilon_S + E^2_{>T}$. Contraction of total variation (TV) under $\mathcal{U}^q_{T,0}$, the triangle inequality, and Pinsker's inequality give

$$\mathrm{TV}(\Phi_S(q), \Phi_T(q)) \le \mathrm{TV}(\nu_{S,T}, p_T) + \mathrm{TV}(p_T, \phi_d) \le \sqrt{\tfrac{1}{2}(\varepsilon_S + E^2_{>T})} + \sqrt{\tfrac{1}{2}\varepsilon_T}.$$

By (B.10), the right side tends to zero uniformly in $S > T$ as $T \to \infty$. Probability measures are complete in total variation, so $\Phi(q)$ exists. Each $\Phi_T(q)$ has a density, and total-variation limits preserve absolute continuity, so $\Phi(q)$ has a density. Lower semicontinuity of $\mathrm{KL}(p_0\|\cdot)$ under total-variation convergence and (8.8) give

$$\mathrm{KL}(p_0\|\Phi(q)) \le \liminf_{T\to\infty}\{\varepsilon_T + \theta^2 \mathrm{KL}(p_0\|q)\} = \theta^2 \mathrm{KL}(p_0\|q).$$

This proves (8.10), and induction gives the geometric bound. Taking $q = p_0$ gives $\Phi(p_0) = p_0$. A fixed point $q \in \mathcal{Q}$ satisfies $\mathrm{KL}(p_0\|q) \le \theta^2 \mathrm{KL}(p_0\|q)$, hence $q = p_0$. □

C Acoustic trajectory experiment details

C.1 Prior discretization, generating trajectory, and fixed data

A moving acoustic source has position $r_\alpha(\tau) \in \mathbb{R}^2$ during the physical-time interval $0 \le \tau \le 10$ s. The unknown is its complete trajectory, represented by two independent zero-mean Matérn–5/2 Gaussian processes with marginal SD 1.5 m and correlation time 2 s. Let (λ_k, ϕ_k) be the temporal covariance eigenpairs and $b_k = \sqrt{\lambda_k}\phi_k$. The trajectory uses eight Karhunen–Loève modes in each coordinate.

$$r_\alpha(\tau) = \left(\sum_{k=1}^{8} b_k(\tau)\alpha_k, \sum_{k=1}^{8} b_k(\tau)\alpha_{8+k}\right), \qquad \alpha \sim \mathcal{N}(0, I_{16}). \tag{C.1}$$

The retained modes are not rescaled to restore the variance of the full process. Physical time τ is distinct from diffusion time t.

Two stationary receivers $a_1, a_2 \in \mathbb{R}^2$ record eight associated pulses at $\tau_i = 10(i-1)/7$ s. With known sound speed $c = 343$ m/s, the time difference of arrival (TDOA), in microseconds, is

$$F_i(\alpha) = \frac{10^6}{c}\left(\|r_\alpha(\tau_i) - a_2\| - \|r_\alpha(\tau_i) - a_1\|\right), \qquad y_{\mathrm{obs}} = F(\alpha) + \eta. \tag{C.2}$$

Independent receiver timestamp errors with SD 120 μs give $\eta \sim \mathcal{N}(0, \Gamma)$, where $\Gamma = 2(120\,\mu\mathrm{s})^2 I_8$. Pulse association, the emission schedule, receiver locations, and sound speed are fixed and known.

For each coordinate, the temporal covariance is

$$k(\tau, \tau') = 1.5^2\left(1 + \sqrt{5}\,u + \frac{5u^2}{3}\right)e^{-\sqrt{5}u}, \qquad u = |\tau - \tau'|/2.$$

The discretization uses 401 equally spaced nodes on $[0, 10]$ at times τ_j^{grid} with trapezoidal weights w_j. Interior weights are 0.025 s and endpoint weights are half as large. If v_k is a unit eigenvector

Table 7: **Fixed acoustic observations.** Physical pulse times in seconds and observed arrival differences in microseconds, rounded for display. Independent TDOA noise SD is $\sqrt{2} \times 120\,\mu$s. These are the data in (C.2).

Pulse	Time (s)	Observed TDOA (μs)
1	0.000000	-3,429.083531
2	1.428571	-3,605.751643
3	2.857143	-3,387.756589
4	4.285714	-2,498.507053
5	5.714286	-656.299655
6	7.142857	-2,936.812510
7	8.571429	-4,126.443627
8	10.000000	-5,402.098042

of the symmetric matrix with entries $\sqrt{w_i}k(\tau_i^{\mathrm{grid}}, \tau_j^{\mathrm{grid}})\sqrt{w_j}$, with eigenvalue λ_k, the scaled mode in (C.1) is evaluated by

$$b_k(\tau) = \frac{1}{\sqrt{\lambda_k}} \sum_j k(\tau, \tau_j^{\mathrm{grid}})\sqrt{w_j}(v_k)_j.$$

Eigenvalues are ordered from largest to smallest, and the largest-magnitude entry of each eigenvector is positive. The leading eight modes are used for inference without variance rescaling. Thus every coefficient affects the source path, and there are no unused Gaussian coordinates.

The exact receiver locations, in meters, are

$$a_1 = (-1.1276311449, -0.4104241720), \qquad a_2 = (1.1817693036, 0.2083778132).$$

The range calculation uses $\sqrt{\|v\|^2 + 10^{-18}}$ in place of $\|v\|$, with distances in meters. The generating path is

$$r_\star(\tau) = c_\star + (2\tau/10 - 1)d_\star + \sum_{k=3}^{16} b_k(\tau)(\xi_{1k}, \xi_{2k}),$$
$$c_\star = (0.65, 2.5), \qquad d_\star = (0.20, -0.125),$$

where ξ is a 2-by-16 array of standard normals drawn in row order by NumPy's default generator with seed 20261101. For each spatial coordinate, the coefficients of modes one and two are set to zero. The linear path is added separately and is not projected into the inference basis. Figure 5 uses this generating path.

Table 7 records the eight observed TDOAs used by every method. The noise realization is generated as standard normals with seed 20261102 on the sorted union of the eight pulse times and the 24 equally spaced points of $[0, 10]$, giving 30 distinct times. The eight pulse entries are multiplied by $\sqrt{2} \times 120\,\mu$s and added to the noiseless range differences. Generating noise directly as a difference of receiver timestamps gives the same Gaussian model. The recorded data use the scaled standard-normal TDOA errors from the pulse entries of the 30-time draw. The posterior uses $\Gamma = 28{,}800 I_8$ in squared microseconds, with fixed noise variance. We keep the mismatch between the sixteen-mode, nonzero-mean generating trajectory and the eight-mode zero-mean prior used for inference.

C.2 Numerical transport and initialization

Each round stores the exact clean scores and GN matrices at its likelihood power using (9.4). For transport, we integrate the deterministic probability flow with 600 steps of the classical fourth-

order Runge–Kutta method (RK4) on a logarithmic OU-time grid from 10 to $6.352721252 \times 10^{-7}$. The 2,048 reference samples and their derivatives stay fixed during integration.

For annealed transport, the Gaussian integration check uses the unweighted means $\bar{x}$, $\bar{s}$, and $\bar{H}$ of the current reference positions, clean scores, and GN matrices. The test target is $\mathcal{N}(\bar{x} + \bar{H}^{-1}\bar{s}, \bar{H}^{-1})$, so the averages of its score and curvature at the reference positions equal $\bar{s}$ and $\bar{H}$, respectively. Annealed GN-GAD passes these checks at the intermediate powers without refining the grid. Appendix I.5 describes the error criteria. Direct and annealed GN-GAD retain their transported endpoints with equal weights.

The initial law is an equal mixture of the prior and MAP–Laplace. The MAP estimate is computed from eight starts that do not use the data-generating parameter. Its cost, 2,941 requests, is counted for each method and seed that uses it. The exact MAP Hessian is positive definite without eigenvalue clipping. Each GAD round adds $3 \times 2{,}048 = 6{,}144$ requests, giving allowances $9{,}085, 15{,}229, 21{,}373, 27{,}517$. The other MCMC methods start from the same initial mixture. MAP-pMALA uses eight chains initialized from MAP–Laplace. Each chain adapts its step size for 256 transitions targeting acceptance 0.574, then holds it fixed. At the final allowance, each MAP-pMALA chain has 1,279 production states, including rejection repeats. Setup and warmup count toward the allowance. Additional kernels and calibration are specified in Appendix I.4.

C.3 Reference and metric definitions

The acoustic NUTS benchmark sample set consists of four chains, each using 1,500 warmup transitions and 10,000 retained states, target acceptance 0.995, and maximum depth 12. NUTS operates on the exact posterior after an affine change to MAP coordinates. Lazy coefficient-reflection proposals use a Metropolis–Hastings (MH) correction at full likelihood power. The reflection proposals are used only for the NUTS run. The last two NUTS chains define the benchmark sample set used for evaluation. The split-benchmark discrepancy compares those chains with the first two NUTS chains. The maximum rank $\widehat{R}$ over the four chains is 1.000253. Minimum bulk and tail effective sample sizes (ESS) are 15,518/11,895. There are zero divergences or depth saturations. The minimum chain energy Bayesian fraction of missing information is 0.800. The maximum event Monte Carlo standard error (MCSE) over all NUTS benchmark samples is 0.00501285, slightly above the precision target of 0.005 set for this run. The held-out evaluation group's crossing-probability MCSE is 0.00706. Diagnostics and split discrepancies do not certify global coverage.

For a coefficient block of dimension d_b, normalized sliced distance is

$$S_b = \left[\frac{d_b}{\operatorname{tr}(C_{\mathrm{ref},b})} \frac{1}{1{,}024} \sum_{\ell=1}^{1{,}024} W_2^2\Big((v_\ell^\top)_{\#}\widehat{q}_b, (v_\ell^\top)_{\#}\widehat{p}_{\mathrm{ref},b} \Big) \right]^{1/2}. \tag{C.3}$$

Directions are fixed random unit vectors, with equal-size sample sets randomly subsampled to at most 2,048 states. The direction seed is 23351. The subsampling seed is 9531. Equation (C.3) uses scalar trace normalization in prior coefficient coordinates, not covariance whitening. The split-benchmark discrepancy is 0.05576. The low coefficient block contains modes 1–4 per coordinate. The high coefficient block contains modes 5–8 per coordinate. Each block has dimension eight. Relative covariance error is $\|C_q - C_{\mathrm{ref}}\|_F / \|C_{\mathrm{ref}}\|_F$, using metric samples. Table 8 reports these complementary checks for every selected acoustic comparator. At the final allowance, the mean joint SW_2 of annealed GN-GAD is 37.1% lower than that of annealed HMC, the best MCMC method on this measure, and 13.3% lower than that of direct GN-GAD. Annealing reduces the joint, covariance, and both block errors of GN-GAD in every paired seed.

Table 8: **Acoustic geometric checks at the final allowance.** Means ± sample SD over five seeds. For every error, smaller values indicate closer agreement. All errors use the metric samples. The coefficient blocks and normalization are defined in Appendix C.3.

Method	Joint SW_2	Low-mode SW_2	High-mode SW_2	Covariance error
Annealed GN-GAD	0.134 ± 0.012	0.179 ± 0.017	0.078 ± 0.006	0.335 ± 0.045
Direct GN-GAD	0.154 ± 0.009	0.215 ± 0.017	0.102 ± 0.009	0.395 ± 0.043
MAP-pMALA	0.630 ± 0.020	0.804 ± 0.012	0.436 ± 0.039	0.885 ± 0.006
Annealed MALA	0.218 ± 0.001	0.245 ± 0.005	0.204 ± 0.019	0.416 ± 0.048
Annealed MAP-pMALA	0.784 ± 0.042	0.984 ± 0.043	0.500 ± 0.068	1.732 ± 0.114
GN-smMALA	0.312 ± 0.113	0.341 ± 0.082	0.297 ± 0.146	0.609 ± 0.310
Annealed GN-smMALA	0.239 ± 0.026	0.259 ± 0.044	0.234 ± 0.037	0.486 ± 0.095
Annealed HMC	0.213 ± 0.004	0.216 ± 0.013	0.227 ± 0.018	0.365 ± 0.062

Table 9: **Acoustic physical-event probabilities at the final allowance.** Sampler entries are five-seed means ± sample SD. Agreement is measured against the NUTS value in each column. The SD reflects variation across sampler seeds.

Method	Midpoint below zero	Enters disk	Crosses zero
NUTS	0.442	0.752	0.690
Annealed GN-GAD	0.450 ± 0.028	0.781 ± 0.019	0.842 ± 0.023
Direct GN-GAD	0.448 ± 0.019	0.781 ± 0.009	0.883 ± 0.023
MAP-pMALA	0.462 ± 0.116	0.279 ± 0.077	0.981 ± 0.007
Annealed MALA	0.479 ± 0.019	0.708 ± 0.034	0.857 ± 0.035
Annealed MAP-pMALA	0.470 ± 0.041	0.522 ± 0.046	0.878 ± 0.039
GN-smMALA	0.477 ± 0.078	0.697 ± 0.033	0.849 ± 0.048
Annealed GN-smMALA	0.435 ± 0.053	0.720 ± 0.028	0.821 ± 0.036
Annealed HMC	0.474 ± 0.040	0.742 ± 0.032	0.836 ± 0.043

For the physical-event checks, we measure the probabilities that a path's midpoint height is below zero, that the path enters the disk centered at $(0, 2)$ m with radius 1 m, and that it crosses $y = 0$ during $[0, 10]$ s. Disk entry and crossing use a 1,601-point trajectory grid. A path crosses if the below-zero residence time lies strictly between zero and ten seconds. Maximum event error is the largest absolute error over these three probabilities, multiplied by 100. The data-fit statistic is the median, over a fixed subsample of at most 512 metric states, of $\|r(\alpha)\|^2/8$, divided by the NUTS benchmark median 0.93964. A ratio of one matches the NUTS median.

Table 9 separates the three physical events. Annealed GN-GAD estimates the probability of midpoint height below zero as 0.450 (NUTS 0.442) and the disk-entry probability as 0.781 (NUTS 0.752). All reported samplers estimate a higher whole-path crossing probability than the NUTS value 0.690. For example, annealed GN-GAD, annealed HMC, and annealed MALA give 0.842, 0.836, and 0.857, respectively. A small transverse displacement of a path near the crossing boundary can change its whole-path indicator. Transverse spread in the samples could also increase the arrival residuals.

C.4 Output-window sensitivity and display construction

The MCMC methods other than MAP-pMALA retain the final half of their $\beta = 1$ production states. MAP-pMALA retains all of its production states. To check output-window sensitivity, we recompute the errors in Table 10 from all saved full-power production states and from their latter half. Annealed GAD has lower joint SW_2 at the final allowance under either window. Annealed

Table 10: **Acoustic MCMC output-window sensitivity.** Joint normalized SW_2 (lower is better), five-seed mean ± sample SD at each method's actual final expenditure. All-production and latter-half windows use the same cumulative run. Annealed GAD is 0.134 ± 0.012 at this allowance.

Method	All $\beta = 1$ production	Latter half
MAP-pMALA	0.630 ± 0.020	0.620 ± 0.026
Annealed MALA	0.223 ± 0.005	0.218 ± 0.001
Annealed MAP-pMALA	0.701 ± 0.044	0.784 ± 0.042
GN-smMALA	0.352 ± 0.122	0.312 ± 0.113
Annealed GN-smMALA	0.237 ± 0.023	0.239 ± 0.026
Annealed HMC	0.201 ± 0.007	0.213 ± 0.004

HMC spends 25,725 requests because one more trajectory for all chains would exceed 27,517. All other acoustic endpoint rows spend the full allowance.

The seed-43 metric samples also provide the coordinate uncertainty summaries in Figure 10. Pointwise posterior intervals describe physical spread, not uncertainty over sampler seeds. The acoustic midpoint histogram in Figure 6 instead uses 10,000 display draws per panel from that same fixed seed. GAD transports fresh Gaussian draws with the score field from the final reference samples. For the MCMC displays, we draw a parent uniformly from the saved states at the beginning of the retained window and draw a production age uniformly to create each branch. Each method's fixed production kernel then advances its branches. MAP-pMALA uses the saved post-warmup states and ages $1, \ldots, 1{,}279$. Branches do not receive longer trajectories or a new warmup. They illustrate the law within the retained window conditional on its initial states. They do not expand the coverage of those initial states. The NUTS histogram has 2,500 evenly selected draws from each of its four chains.

All histogram panels use 89 bins per axis, common NUTS-based physical limits, and bin count divided by the full sample size. Empty bins remain empty, off-panel mass remains in the denominator, and no smoothing or in-panel renormalization is applied. Different experiment seeds are never pooled. Display requests are separately recorded and excluded from metrics, coordinate bands, and the comparison allowance.

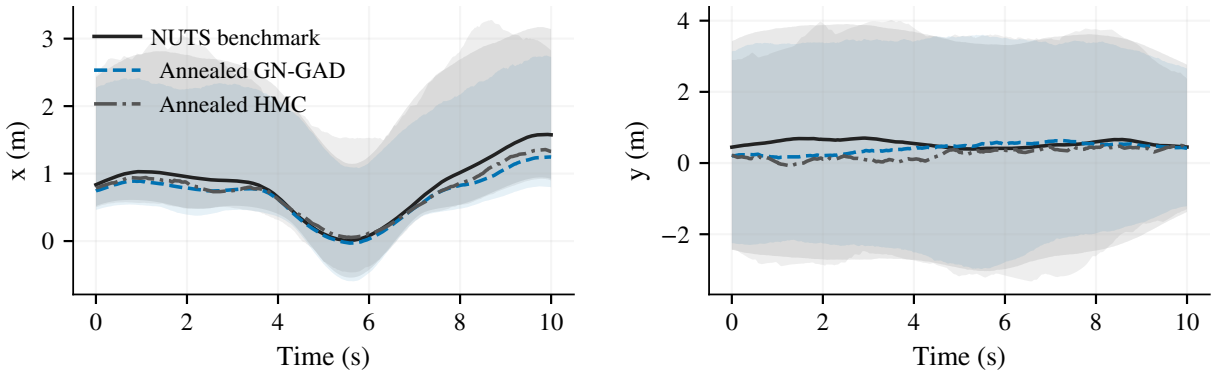


Figure 10: **Acoustic coordinate uncertainty for experiment seed 43.** Medians and pointwise 90% posterior intervals for the two spatial coordinates in meters over $[0, 10]$ s. The NUTS bands use the 20,000-state evaluation group. The final metric samples contain 2,048 states for annealed GN-GAD and 384 states for annealed HMC. No display-only draws or other experiment seeds enter the intervals. The intervals describe uncertainty at each time along the path. Whole-path event probabilities are reported in Table 9.

D Joint earthquake-location experiment details

D.1 Physical model, priors, and fixed observations

Units are km, seconds, and km/s. The velocity parameters are the surface P-wave speed v_0, the vertical gradient g, and the P- to S-wave speed ratio κ_{PS}. The event coordinates have independent Gaussian priors. The parameters v_0, g, and $\kappa_{\mathrm{PS}} - 1$ have independent lognormal priors. The six surface stations have horizontal coordinates

$$(0, 0),\ (14, 4),\ (5, 16),\ (-9, 10),\ (-7, -9),\ (10, -11),$$

in station-number order. For $\alpha \sim \mathcal{N}(0, I_{23})$, event $k = 1, \ldots, 5$ has

$$e_k = 3 + 9\alpha_{4k-3}, \quad n_k = 2 + 9\alpha_{4k-2},$$
$$z_k = 8 + 3\alpha_{4k-1}, \quad t_{0,k} = \alpha_{4k},$$
$$v_0 = 5.2e^{0.05\alpha_{21}},$$
$$g = 0.06e^{0.3\alpha_{22}},$$
$$\kappa_{\mathrm{PS}} = 1 + 0.73e^{0.1\alpha_{23}}.$$

Depth is positive downward. The Gaussian prior on depth extends above the surface.

The medium has depth-dependent speeds $v_P(z) = v_0 + gz$ and $v_S(z) = v_P(z)/\kappa_{\mathrm{PS}}$. For a source at depth z_s, receiver at depth $z_r = 0$, and Euclidean source–receiver separation R, the analytic travel times are

$$T_P = \frac{1}{g}\operatorname{arcosh}\left(1 + \frac{g^2R^2}{2(v_0 + gz_s)(v_0 + gz_r)}\right), \qquad T_S = \kappa_{\mathrm{PS}}T_P. \tag{D.1}$$

The formula in (D.1) accounts for rays curved by the velocity gradient. A pick from event k at station j has prediction $t_{0,k} + T_P$ or $t_{0,k} + T_S$. Independent Gaussian pick errors have SD 0.08 s for P and 0.15 s for S, defining the diagonal covariance in (13.1). Derivatives are obtained from this analytic forward map.

The likelihood is zero where a source has nonpositive wave speed when the likelihood power is positive. The travel-time calculation is nonfinite outside its domain, and the MCMC implementations reject nonfinite proposals. Travel times use (D.1), evaluated in its stable $\log(1+a+\sqrt{a(a+2)})$

Table 11: **Generating earthquake parameters.** Positions are in km, depth is positive downward, and origin times are in seconds. E1 is the anchor.

Event	East	North	Depth	Origin time
E1	5.0	-4.0	6.0	0.00
E2	8.0	-1.0	8.5	0.31
E3	6.0	1.0	7.0	-0.22
E4	9.0	-5.0	10.0	0.12
E5	4.0	2.0	5.5	-0.41

Table 12: **Fixed earthquake arrival picks.** Recorded arrival times in seconds, rounded for display. A dash denotes an unobserved phase. Independent noise SDs are 0.08 s (P) and 0.15 s (S).

Event	Station	P arrival (s)	S arrival (s)
E1	ST1	1.582452	2.855981
E1	ST2	2.458755	4.333243
E1	ST3	3.806598	6.602302
E1	ST4	3.667507	6.489049
E1	ST5	2.601111	4.316244
E1	ST6	1.877515	3.306574
E2	ST1	2.411565	–
E2	ST2	2.323569	3.965978
E3	ST1	1.346819	2.779552
E3	ST2	1.720495	–
E4	ST1	2.562834	–
E4	ST6	2.192577	3.807015
E5	ST1	0.965825	1.690106
E5	ST2	1.748654	–

form, where $a = g^2R^2/[2(v_0 + gz_s)(v_0 + gz_r)]$. Automatic differentiation of this analytic map supplies Jacobians, GN matrices, exact scores, and the MAP Hessian. No discretized ray solver or implicit root differentiation is used.

Table 11 lists the generating event parameters. The shared velocity parameters are $(v_0, g, \kappa_{\mathrm{PS}}) = (5.35, 0.07, 1.76)$, with g in s^{-1}. The station numbering is the same in Figure 7 and Table 12. Sparse-event picks were assigned to the nearest and second-nearest stations in horizontal distance at the generating locations, with S observed only at the nearest. The selected event–station–phase associations are held fixed. Missing phases and stations do not contribute detection or censoring terms to the likelihood.

Table 12 gives all 24 recorded arrival times. To reproduce the realization, independent standard normals from NumPy's default generator with data seed 71 are drawn first for E1, P then S at stations $1, 6, 2, 5, 4, 3$. For E2–E5, the order is P then S at the nearest station, followed by P at the second-nearest station. Each draw is multiplied by its phase SD, 0.08 s or 0.15 s, and added to the generating origin time plus the travel time in (D.1). The likelihood uses these recorded values and independent errors. The model omits station corrections, unknown phase associations, and additional pick uncertainty.

The acquisition setting was selected by screening NUTS benchmark samples for departure from MAP–Laplace and a moment-matched Gaussian. An exploratory sampler run then compared two likelihood schedules before the six-round schedule was fixed. With the dataset held fixed, we selected comparator settings on seed 42 and repeated the samplers on five further seeds. These

repeats measure variation across sampler runs on the selected problem, rather than performance across unseen datasets.

D.2 Transport controls and NUTS benchmark

Direct and annealed GN-GAD start from the prior and use 1,500 reference samples in each of six rounds. Each round costs 4,500 requests. Neither variant uses the MAP estimate, so no MAP cost is counted. Annealed powers are $(0.02, 0.07, 0.2, 0.45, 1, 1)$, versus six ones for direct GAD. Within each seed, the two variants share initial samples and terminal Gaussian draws. Round scores and GN matrices scale only the likelihood as in (9.4).

The integration grid starts with OU terminal time 5 and 600 RK4 steps on a logarithmic grid. Each round checks transport on $\mathcal{N}(\bar{x}, \bar{H}^{-1})$, where $\bar{x}$ is the mean of the reference positions and $\bar{H}$ is the mean GN matrix at the current likelihood power. The terminal time is extended and the grid is refined as required to meet the 1% error criteria in Appendix I.5. The final OU time is at most 10^{-5}, also constrained by a 1% variance-ratio criterion. Refinement permits up to 64 times the initial step count. The shifted resolvent eigenvalues are guarded below by $\max(10^{-3}, \alpha_t^2/2)$, bounding gate gain by two. The saved control settings specify the grid selection rule. Exact realized grids are available for only some of the earthquake runs.

MAP-pMALA uses four chains initialized from MAP–Laplace and preconditioned by the inverse exact MAP Hessian. Each chain has 200 warmup transitions targeting 0.574 acceptance. The MAP estimate uses four starts that do not use the data-generating parameter, and its cost is counted for each method that uses it. The MAP Hessian is used without clipping. The other MCMC methods start from the prior and run populations of chains. Their selected settings are given in Appendix I.4.

The NUTS run uses four chains with a dense mass matrix. Each chain has 1,500 warmup transitions, 2,000 retained states, target acceptance 0.95, and maximum depth ten. Starts are independently dispersed by half a MAP–Laplace SD. Maximum rank $\widehat{R}$ is 1.001316, minimum ESS 1,951, mean acceptance 0.9551, with zero divergences and depth saturations. Comparing the first two with the last two chains gives a discrepancy of 0.07997 for event-block SW_2 and 0.07873 for full-space SW_2. The primary evaluation uses all 8,000 NUTS states. These diagnostics come from finitely many NUTS states. They do not prove that the chains explored the whole posterior, and they do not give a certified error floor.

D.3 Metrics and broad endpoint comparisons

SW_2 uses prior-whitened coefficients and the scalar block-trace normalization of (C.3). Here the full-space implementation uses 256 fixed directions, sample sets capped at 1,000 evenly indexed states, and 1,000 midpoint quantiles per projection. The midpoint quantiles approximate the quantile integral. Each event block uses 64 directions in its four east/north/depth/time coordinates. The primary earthquake error is the mean of the normalized location–origin-time block distances for events E2–E5. Projection seed 42 is fixed for evaluation. Full-space SW_2 covers all 23 coordinates. Relative covariance error uses all retained metric states. Error magnitudes are not directly comparable between applications because the block dimensions and implementations differ. Table 13 retains full-space and covariance results alongside the primary block error. Annealing lowers the mean block SW_2 of GN-GAD by 50.6%, to 42.5% below direct GN-smMALA, the MCMC method with the lowest block error. Annealed GN-GAD has the lowest mean full-space SW_2 among the tested methods. It also has lower mean event and depth-interval errors than direct GN-GAD. Annealed GN-GAD has the lowest mean maximum event-probability error and

Table 13: **Complete selected earthquake geometric comparison.** Five-seed means ± sample SD, with actual spent E+G+H requests under allowance 27,000. Lower errors indicate closer agreement. Block and full SW_2 have the scalar NUTS-based normalization in (C.3). GAD continuations include inherited transport work. Settings are frozen on seed 42.

Method	Spent W	Block SW_2	Full SW_2	Covariance error
Annealed GN-GAD	27,000	0.154 ± 0.019	0.130 ± 0.019	0.150 ± 0.008
Direct GN-GAD	27,000	0.311 ± 0.104	0.273 ± 0.101	0.274 ± 0.037
MAP-pMALA	26,999	0.491 ± 0.025	0.424 ± 0.011	0.508 ± 0.072
Annealed MALA	26,880	0.653 ± 0.020	0.574 ± 0.016	0.778 ± 0.027
Annealed MAP-pMALA	26,863	1.581 ± 0.065	1.583 ± 0.088	3.521 ± 0.354
GN-smMALA	26,880	0.267 ± 0.028	0.247 ± 0.016	0.423 ± 0.052
Annealed GN-smMALA	26,880	0.374 ± 0.021	0.322 ± 0.016	0.462 ± 0.032
Annealed HMC	26,752	0.359 ± 0.028	0.310 ± 0.022	0.454 ± 0.051
Direct MALA	26,880	0.583 ± 0.024	0.540 ± 0.031	0.681 ± 0.046
Direct HMC	24,832	0.687 ± 0.024	0.850 ± 0.035	1.076 ± 0.086
GAD then MAP-pMALA	26,943	0.463 ± 0.090	0.439 ± 0.061	0.443 ± 0.049
GAD then GN-smMALA	26,832	0.455 ± 0.032	0.410 ± 0.025	0.466 ± 0.034
GAD then HMC	24,272	0.287 ± 0.035	0.268 ± 0.024	0.417 ± 0.008
MAP-start HMC	26,623	0.539 ± 0.182	0.475 ± 0.152	1.050 ± 0.609
Replica-exchange HMC	25,903	0.775 ± 0.216	1.002 ± 0.356	3.784 ± 1.473
Replica-exchange GN-smMALA	26,880	0.618 ± 0.117	0.594 ± 0.065	1.151 ± 0.233
SMC–MALA	26,912	1.369 ± 0.237	1.360 ± 0.171	0.952 ± 0.010
SMC–GN-smMALA	26,880	1.051 ± 0.170	0.992 ± 0.191	0.887 ± 0.274

depth-interval error among the tested methods (Table 14).

For each of the five events, we measure the probabilities of $z_k < 5$ km and $z_k > 10$ km. Maximum event error is the largest of the ten absolute probability errors, in percentage points. Depth-interval error is the root mean squared (RMS) discrepancy of the 5% and 95% depth quantiles for E2–E5, in km. The data-fit statistic uses at most 512 metric states and is the median of $\|r(\alpha)\|^2/24$, divided by the NUTS benchmark median. Table 14 reports these physical checks for the same fixed comparator settings.

The broad comparison includes direct fixed-metric methods, annealed fixed-metric methods, transport-to-MCMC continuations, replica exchange, and SMC. All reported seeds use the settings selected on seed 42. MAP-pMALA has higher block error with the likelihood schedule of annealed GN-GAD than it does at full power throughout (Tables 13 and 17). The prior-started GN-smMALA kernel has lower block error than MAP-pMALA. Its own block error increases with annealing at the tested allowance. The selected SMC–MALA and SMC–GN-smMALA runs reach $\beta = 1$. None of the 24 tested SMC–HMC settings reaches $\beta = 1$ within 27,000 requests on any of seeds 42–47. Table 13 reports endpoint errors for runs that reach full likelihood power.

D.4 Physical displays and seed robustness

Figure 8 uses experiment seed 43 throughout. For each event, the horizontal axis projects east/north displacement onto the leading eigenvector of its NUTS benchmark covariance, after subtracting the NUTS benchmark mean. Depth remains its physical coordinate. The two GAD panels transport 10,000 fresh Gaussian draws with the score field from their own final 1,500 reference samples. The NUTS benchmark has 8,000 states, and GN-smMALA uses its 4,480 saved metric states. Each event row uses common 66-by-66 bins and a shared logarithmic probability-per-bin scale. Bin counts are divided by the full sample count without smoothing.

Table 14: **Earthquake physical checks for the complete selected set.** Same five-seed endpoints as Table 13. Event and depth-interval errors are smaller when agreement is closer. The depth-interval error is a root mean squared error (RMSE) in km. The data-fit ratio targets one. Between-seed SD excludes NUTS benchmark uncertainty.

Method	Event error (pp)	Depth interval RMSE (km)	Fit ratio $\to 1$
Annealed GN-GAD	7.7 ± 1.2	0.39 ± 0.10	1.45 ± 0.05
Direct GN-GAD	8.9 ± 3.3	0.49 ± 0.12	1.36 ± 0.04
MAP-pMALA	15.3 ± 4.3	1.08 ± 0.16	1.00 ± 0.02
Annealed MALA	41.7 ± 4.5	1.68 ± 0.04	2.73 ± 0.07
Annealed MAP-pMALA	49.6 ± 5.5	3.21 ± 0.18	341.69 ± 14.91
GN-smMALA	16.5 ± 2.0	0.69 ± 0.11	3.76 ± 0.12
Annealed GN-smMALA	29.0 ± 2.7	0.96 ± 0.12	4.92 ± 0.25
Annealed HMC	28.6 ± 5.3	0.96 ± 0.08	1.28 ± 0.06
Direct MALA	37.8 ± 3.7	1.62 ± 0.15	2.26 ± 0.04
Direct HMC	43.8 ± 3.0	1.82 ± 0.10	36.97 ± 1.80
GAD then MAP-pMALA	35.1 ± 7.3	1.30 ± 0.41	2.16 ± 0.09
GAD then GN-smMALA	33.2 ± 5.8	1.25 ± 0.17	2.60 ± 0.16
GAD then HMC	17.3 ± 2.6	0.77 ± 0.15	1.71 ± 0.14
MAP-start HMC	16.7 ± 7.1	1.59 ± 0.96	4.78 ± 8.16
Replica-exchange HMC	31.6 ± 19.5	2.58 ± 1.49	1.61 ± 0.40
Replica-exchange GN-smMALA	23.1 ± 3.5	1.54 ± 0.36	1.60 ± 0.09
SMC–MALA	68.9 ± 21.7	3.77 ± 0.50	2.26 ± 0.65
SMC–GN-smMALA	60.4 ± 18.3	2.51 ± 0.71	1.70 ± 0.40

The additional GAD draws are used only in the display.

Figure 11 complements the seed-43 earthquake histograms in Figure 8 with the saved all-seed comparison. The density panels retain the original E4 histograms of 1,000 states per seed at their original physical coordinates, without rebinning or pooling. The paired block distances come from the saved metric rows.

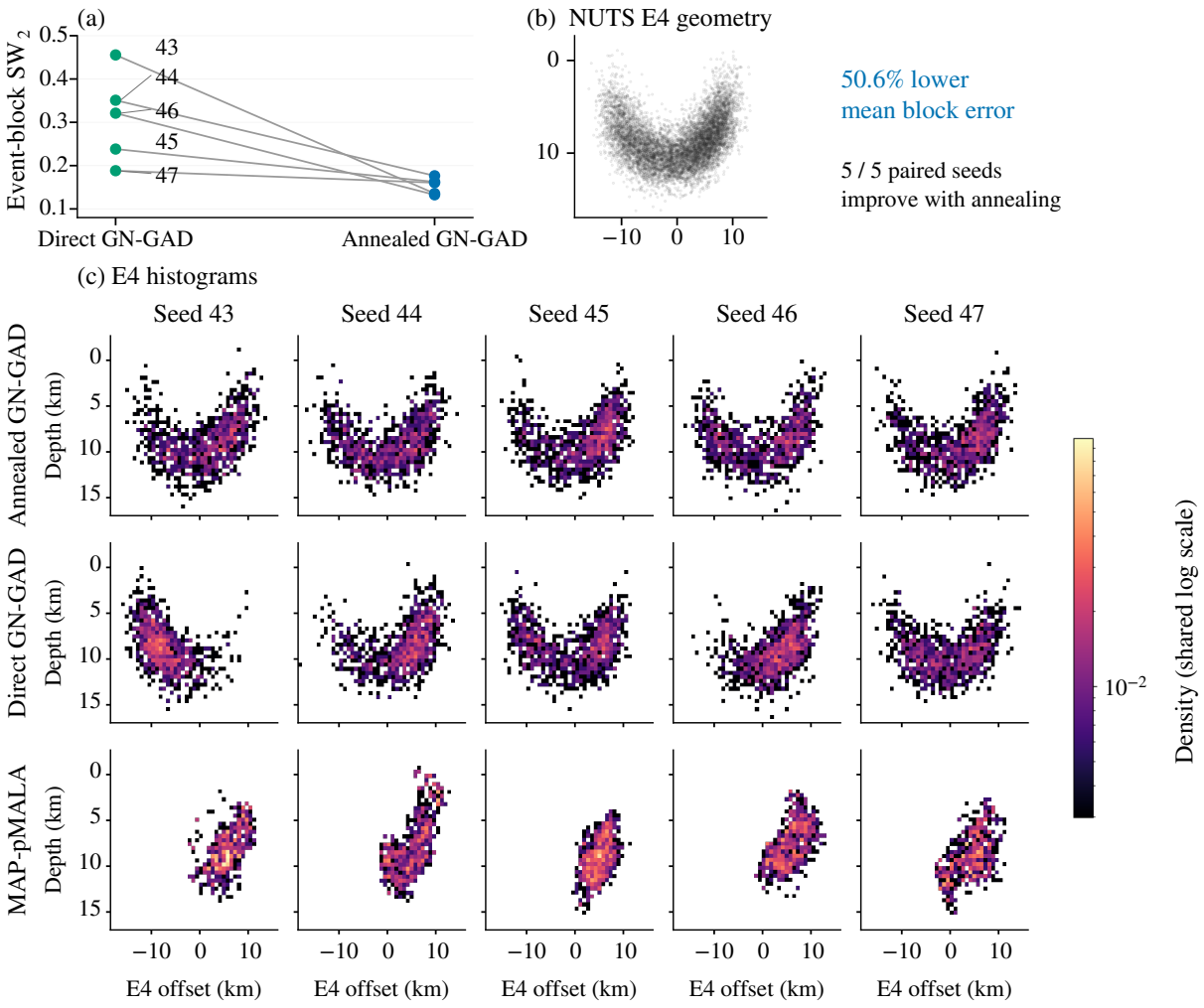


Figure 11: **Earthquake annealing comparison across the five seeds.** (a) Final E2–E5 block SW_2 for direct and annealed GN-GAD in each paired seed. Lower values are better. (b) NUTS benchmark for E4. (c) Saved E4 densities for annealed GN-GAD, direct GN-GAD, and MAP-pMALA, using 1,000 states per seed. Panels (b,c) show horizontal offset and depth in km, with depth positive downward. The histograms in (c) share a logarithmic density scale. Appendix D.4 describes how the saved panels and the probability-per-bin panels in Figure 8 were made.

E Gaussian, mixture-weight, and gate experiments

In the Gaussian and gate studies, we estimate conditional means from finitely many reference samples and summarize variation across seeds. The mixture-weight study instead evolves densities with exact conditional expectations and no mixture fit. Its numerical checks are deterministic.

E.1 Sliced Wasserstein distance for the synthetic tests

For two indexed samples $X = (x_i)_{i=1}^m$ and $Y = (y_i)_{i=1}^m$ in $\mathbb{R}^d$, let $v_1, \dots, v_L$ be fixed unit directions and use parenthesized indices for the sorted projections. The empirical distance and its scalar

normalization are

$$\widehat{\mathrm{SW}}_2(X,Y)=\left\{\frac{1}{Lm}\sum_{\ell=1}^{L}\sum_{i=1}^{m}\Big(\langle v_\ell,X\rangle_{(i)}-\langle v_\ell,Y\rangle_{(i)}\Big)^2\right\}^{1/2},\qquad \sigma_{\mathrm{tar}}=\sqrt{\frac{\mathrm{tr}(\mathrm{Cov}_{p_0}(X_0))}{d}}. \tag{E.1}$$

Here $X_0\sim p_0$ is a single target draw. Tables report $100\widehat{\mathrm{SW}}_2/\sigma_{\mathrm{tar}}$ using (E.1).

E.2 Gaussian score recovery

The anisotropic target in Section 11.1 is $\mathcal{N}((0.6,-0.4),\left(\begin{smallmatrix}1.6 & 0.5\\ 0.5 & 0.7\end{smallmatrix}\right))$. The first reference law is $\mathcal{N}((0.6,-0.4),4I_2)$. The second is an equal mixture with means $(-2,1)$ and $(2,-1)$, both with covariance $\mathrm{diag}(0.9^2,0.6^2)$. Each law supplies 4,096 reference samples for seeds 42–44. Each run produces 4,096 reverse endpoints and is compared with 4,096 independent target samples, shared by the two reference laws within the seed. The two runs in a seed also share terminal samples and Brownian increments.

Both runs use the two-dimensional field implementation in Appendix F.3: $T=5$, $t_{\min}=10^{-4}$, 240 Heun steps, 65 field times, and 129×129 reference and score-evaluation grids, with spatial half-width at least 9. Pre-interpolation residuals compare every tabulated score value with the analytic noisy Gaussian score. This analytic score is used only for evaluation and is not an input to the estimated field. The largest absolute residual over nodes, times, and seeds is below 1.3×10^{-10} (Table 2). The maximum relative RMS residuals over seeds are 4.70×10^{-14} and 1.07×10^{-13} for the Gaussian and bimodal reference laws. The largest difference between paired endpoints of the two runs is 7.95×10^{-14}. All endpoints are finite, and no gate safeguard activates. Sample discrepancies use the 64 fixed directions and scalar normalization in (E.1). The nonzero endpoint-to-target SW_2 values include numerical integration and finite-output sampling error, although the cancellation of Section 6 holds to numerical precision.

E.3 Exact-expectation calculation for the mixture-weight study

The mixture family in Section 11.2 is

$$\begin{aligned} p_{0,\Delta}&=\tfrac12\mathcal{N}(-\Delta\sigma/2,\sigma^2)+\tfrac12\mathcal{N}(\Delta\sigma/2,\sigma^2),\\ q_{0,\Delta}&=0.7\mathcal{N}(-\Delta\sigma/2,\sigma^2)+0.3\mathcal{N}(\Delta\sigma/2,\sigma^2),\qquad \sigma=0.7. \end{aligned}$$

The initial law q_0 and target p_0 share the Gaussian components and differ only in their component weights. Component overlap is $2\Pr\{Z<-\Delta/2\}$, $Z\sim\mathcal{N}(0,1)$. As separation grows, the initial KL tends to $-\frac12\log(4\cdot0.7\cdot0.3)$, while the clean-score Fisher discrepancy $F_{\mathrm{mix}}(q_0)$ becomes small. The target-weighted Fisher discrepancy F_{mix} in (11.1) has no factor of $1/2$. Its initial value comes from independent quadrature of the nonnegative integrand formed from the analytic score difference. KL is integrated separately under each target component. At separation 5, the first-round gain is 0.174 and the sensitivity is 0.141. The corresponding values are 0.00991 and 0.00981 at separation 7, then 0.000199 and 0.000233 at separation 9. The first-round gains and sensitivities differ by less than 25% relative to the sensitivities at these separations. The curves in Figure 2(a) show first-round fractional KL reduction and initial clean-score sensitivity without fitting or rescaling either quantity to match the other. After 30 rounds, forward KL is 6.78×10^{-6}, 0.0399, and 0.0852, respectively.

The density solver uses $[-10,10)$, horizon $T=8$, terminal density $\mathcal{N}(0,1)$, and endpoint time zero. A zoomed fast Fourier transform (ZoomFFT) with Gaussian Fourier multipliers computes the OU moments under the current law. For each round's fixed reverse SDE, we evolve the

density using a conservative Chang–Cooper finite-volume generator, zero boundary flux, and two-stage L-stable singly diagonally implicit Runge–Kutta integration (SDIRK2). The ascending time grid starts at zero, takes increments $\min\{\delta t, 0.065(t+0.003)\}$, shortens the last step at T, and is traversed in reverse. The primary trajectories use 3,072 cells and maximum step $\delta t = 0.01$, retaining all 30 rounds for $\Delta = 5, 7, 9$. The additional first-round calculations use the same 3,072-cell grid and maximum time step. No mixture projection or endpoint sampling is performed.

The raw scalar gate is $\alpha_t^2/(\alpha_t^2 + \gamma_t H_t^q)$. A denominator with magnitude below 10^{-8} retains its sign and is set to that magnitude. The gate is capped at 50 and floored at $10^{-3}\alpha_t^2/(\alpha_t^2 + \gamma_t/\sigma^2)$. Below marginal density 2×10^{-13}, the solver uses its inward Gaussian tail continuation. No safeguard activates on retained probability-weighted score-evaluation points, and no negative mass is repaired in the selected trajectories or numerical controls. Reconstructed KL values from saved densities agree with the recorded trajectories to 7.94×10^{-15} absolute error. The maximum mass drift in normalized saved trajectory densities is 9.14×10^{-14}.

Table 15 records the resolution check for the separation-9 first-round fractional KL reduction g_0. Medium-to-fine change is 0.3804% of the gain, within our 5% tolerance. For $\Delta = 9$, panel (a) of Figure 2 uses the finest-grid value. Panel (b) uses the base grid for the entire trajectory. Expanding the domain to $[-12.5, 12.5)$ with 3,840 cells at the same spacing and maximum step changes the base-grid gain by 1.04×10^{-11} relative. A Gaussian density-solver control gives one-round KL 3.06×10^{-11}. Separate controls driven by exact analytic noisy target scores give KL 2.05×10^{-10} at separation 5 and 1.08×10^{-9} at separation 9. Only the control runs use analytic noisy target scores. The grid and domain checks measure deterministic numerical error.

Table 15: **Refinement of the separation-9 first-step gain.** The initial component weights are 0.7/0.3, the domain is $[-10, 10)$, and g_0 is the fractional forward-KL reduction in (11.2). Successive rows refine the deterministic solver, and the final refinement changes the gain by 0.38%.

Cells	Maximum time step	g_0
3,072	0.0100	0.00020303335
6,144	0.0050	0.00019989021
12,288	0.0025	0.00019913279

E.4 Exact target for the gate-ablation experiment

The target in Section 11.3 and Figure 3 is the following fixed density on $\mathbb{R}^8$:

$$p_0(x) = \frac{|\det A|}{Z_\epsilon^8} \exp\left[-\sum_{j=1}^{8} v_\epsilon((Ax)_j)\right],$$
$$v_\epsilon(u) = \tfrac{1}{2}u^2 + 2\epsilon \log\cosh(u/\epsilon), \qquad \epsilon = 0.04, \qquad Z_\epsilon = \int_{\mathbb{R}} e^{-v_\epsilon(u)}\,\mathrm{d}u.$$

The transformed coordinates $z = Ax$ are independent under this density. The factor $|\det A|$ accounts for the change from z to the original x coordinates. All eight coordinates contribute non-Gaussian curvature. The same target is used for every gate and validation seed.

With $z = Ax$, the ridge target's exact derivatives are

$$s_0(x) = -A^\top\{z + 2\tanh(z/\epsilon)\}, \qquad H_0(x) = A^\top \operatorname{diag}\{1 + 50\operatorname{sech}^2(z_j/\epsilon)\}A.$$

The hyperbolic functions act componentwise. The fixed matrix A is constructed using one NumPy `default_rng(20261106)` generator. Draw two successive standard-normal 8×8 arrays in row

order, and take the orthogonal Q factors from NumPy's orthogonal–triangular factorizations, first U, then V, without changing the signs of their columns. Set

$$A = U\,\mathrm{diag}(1, 1, 1.5, 1.5, 2, 2, 3, 3)V^\top.$$

The target observed-information matrix satisfies $I_8 \preceq H_0(x) \preceq 459 I_8$. All eight ridge coordinates are non-Gaussian. To draw independent exact target samples, we sample each z_j by rejection from a Laplace law of rate 2, accepting with probability

$$\exp\{-z_j^2/2 - 2\epsilon \log(1 + e^{-2|z_j|/\epsilon})\},$$

and then apply $x = A^{-1}z$. One-dimensional quadrature gives $\mathrm{Var}(z_j) = 0.2543156962$ and target scale $\sigma_{\mathrm{tar}} = 0.3388145209$. Exact target samples, this transform, and normalization information are used for evaluation only.

E.5 Gate definitions and reconstruction settings

For the current reference samples $X_1, \ldots, X_N$, let $w_i(y,t)$ be the normalized OU weights in (I.1) and set

$$b_i = \frac{\alpha_t X_i - y}{\gamma_t}, \qquad c_i = \frac{s_0(X_i)}{\alpha_t}, \qquad \delta_i = c_i - b_i, \qquad \widehat{M}_q = \frac{1}{N}\sum_i s_0(X_i)s_0(X_i)^\top.$$

The matrix $\widehat{M}_q$ averages outer products of the target score over the current reference samples. Write $\bar{u} = \sum_i w_i u_i$ for a weighted mean. In particular, $\bar{b} = \widehat{b}^q(y,t)$ and $\bar{\delta} = \widehat{m}_t^q(y)$. With ridge scale η_{ridge}, the weighted covariance, empirical observed information, and ridge are

$$\widehat{C}_{uv} = \sum_i w_i(u_i - \bar{u})(v_i - \bar{v})^\top, \qquad \widehat{H}^q = \sum_i w_i H_0(X_i), \qquad \lambda_{\mathrm{ridge}} = 10^{-8} + \eta_{\mathrm{ridge}}\ \mathrm{tr}(\widehat{C}_{\delta\delta})/8.$$

Each gate G gives the score $\bar{b} + G\bar{\delta}$, computed from the same reference samples as the gate. The gates are

$$G_{\mathrm{Tweedie}} = 0, \qquad G_{\mathrm{TSI}} = I_8,$$

$$G_{\text{uniform scalar}} = \frac{\alpha_t^2}{\alpha_t^2 + \gamma_t\,\mathrm{tr}(\widehat{M}_q)/8} I_8, \qquad G_{\text{uniform matrix}} = \alpha_t^2(\alpha_t^2 I_8 + \gamma_t \widehat{M}_q)^{-1},$$

$$G_{\text{local scalar}} = \mathrm{clip}_{[0,1]}\left(\frac{-\mathrm{tr}(\widehat{C}_{b\delta})}{\max\{\mathrm{tr}(\widehat{C}_{\delta\delta}), 10^{-20}\}}\right) I_8, \qquad G_{\text{local matrix}} = -\widehat{C}_{b\delta}(\widehat{C}_{\delta\delta} + \lambda_{\mathrm{ridge}} I_8)^{-1},$$

$$\widehat{G}^q_{\mathrm{LFGI}} = \alpha_t^2(\alpha_t^2 I_8 + \gamma_t \widehat{H}^q)^{-1}.$$

Using moments under q in the spatially uniform control-variate schedules gives the rows labeled q-CV and q-DPSMC [23, 40]. Conditioning the scalar coefficient on the noisy state gives the local analogue q-CVSI [23]. For a local matrix coefficient, the comparison uses the centered matrix blend regression [14]. In the opposite interpolation convention used by DPSMC, its coefficient is $I - G$. Matrix regression and LFGI give the same gate with exact expectations under the target law and the regularity conditions in [14]. Substituting moments under $q \neq p_0$ can give different gates. The comparison tests how these gates affect the GAD update.

The matrix ridge uses $\eta_{\mathrm{ridge}} = 0.001$, selected on pilot seed 42 from $\{0.001, 0.01, 0.1\}$ before validation. Covariances are computed by stable two-pass centering with every OU weight retained. The local regression matrix is not symmetrized or spectrally clipped. The regularized covariance matrix inverted in the local matrix blend is positive definite. The shifted matrix inverted in the

uniform matrix gate q-DPSMC is positive definite because $\widehat{M}_q \succeq 0$. On this target, $\widehat{H}^q \succ 0$, so q-LFGI uses its raw Hessian resolvent without clipping or an inverse safeguard.

The reported runs start from $\mathcal{N}(0, I_8)$ and use $N = 1{,}024$, four rounds, $T = 5$, $t_{\min} = 10^{-4}$, and 300 stochastic Heun steps uniform in $\log(\mathrm{expm1}(2t))$. The predictor and corrector share the Brownian increment, and no final denoising or annealing is applied. Each round replaces the reference samples by all 1,024 endpoints. Within each seed from 43 to 47, all gates share the initial samples, terminal samples, and Brownian increments. Different seeds are independent. TSI stops on nonfinite states or a coordinate magnitude exceeding 10^6. All five TSI runs fail by round three. Table 3 records the failures in place of a round-three mean.

Metrics use 1,024 GAD endpoints, 1,024 independent target samples per seed, and 256 fixed unit directions obtained by normalizing Gaussian vectors from `default_rng(85411)`. The target samples are shared across gates and rounds within a run. Tables 3 and 16 report normalized SW_2 multiplied by 100. A second independent target sample gives the target/target comparison. Display draws are not used in the metrics or as reference samples. The displays use the score fields from seed 43 and the first two coordinates of $z = Ax$, with 10,000 fresh paths per panel. The target display sample is independent of the metric samples and is repeated in the first column of Figure 3. Histograms share axes and logarithmic probability-per-bin colors without smoothing. Bin counts are divided by the full sample count. Panels annotate off-panel percentages of at least 0.05%. Exact-target contours enclose 50% and 90% projected probability.

The round-three reporting cutoff was chosen after inspecting the original four-round results. Table 16 records all finite fourth-round results. q-LFGI improves in all five seeds, with paired mean reduction 0.391 ± 0.432 percentage points of target scale. The uniform scalar, uniform matrix, and local matrix blend gates reduce SW_2 more than q-LFGI between rounds three and four. The local scalar gate also improves less between these rounds, so q-LFGI is not the only gate whose progress has slowed. Sensitivity checks on the pilot seed for the number of reference samples, the step count, the matrix ridge, and a 2,048-direction metric are kept separate from the validation runs.

Table 16: **Fourth-round gate cutoff check.** Normalized full-dimensional SW_2 in percent, mean $\pm$ sample SD over the same five validation seeds and 1,024-point metric samples. Lower is better. TSI fails in all five runs by round three and has no fourth-round result. The target/target comparison scale is 7.87 ± 0.29.

Gate	Round 3	Round 4
Tweedie	199.41 ± 2.18	200.78 ± 3.07
Uniform scalar (q-CV)	71.06 ± 2.13	36.62 ± 0.83
Uniform matrix (q-DPSMC)	63.46 ± 2.53	36.47 ± 2.81
Local scalar (q-CVSI)	12.12 ± 0.75	11.18 ± 0.29
Local matrix blend	30.05 ± 3.47	23.48 ± 1.17
q-LFGI	8.56 ± 0.40	8.17 ± 0.46

F Low-dimensional sampler experiment details

We compare GAD, MALA, and HMC on four connected two-dimensional targets. For this comparison, we specify the targets, sampling settings, evaluation settings, configuration selection, and displays. The one-dimensional illustration is described separately in Appendix G.1. Inverse model and metric details remain in Appendices C and D.

F.1 Exact targets for the four sampler benchmarks

The four rows of Figure 4 use the following normalized densities on $\mathbb{R}^2$, in the same top-to-bottom order. Each has the Gaussian-mixture form $p_0(x) = \sum_j w_j \mathcal{N}(x; \mu_j, \Sigma_j)$, with means and covariances in the displayed physical coordinates.

- The Gaussian has $w_1 = 1$, $\mu_1 = (0.6, -0.4)$, and $\Sigma_1 = \left(\begin{smallmatrix} 1.6 & 0.5 \\ 0.5 & 0.7 \end{smallmatrix}\right)$.
- The unequal-lobes target has $w = (0.32, 0.68)$, means $(-1.25, 0)$ and $(1.25, 0.3)$, and both covariances $\mathrm{diag}(0.85^2, 0.70^2)$.
- The ring has 12 equal weights, means $2.1(\cos\phi_j, \sin\phi_j)$, $\phi_j = 2\pi j/12$, $j = 0, \ldots, 11$, and covariances $0.65^2 I_2$.
- The S-shaped chain has 13 equal weights, $u_j = -2.1 + 0.35j$, means $(u_j, 1.35\sin(1.3u_j))$, $j = 0, \ldots, 12$, and covariances $0.47^2 I_2$.

The neighboring components overlap. The three non-Gaussian targets have negative observed-information eigenvalues at some locations, so they are outside Assumption 8.1. For responsibilities $r_j(x) = w_j \mathcal{N}(x; \mu_j, \Sigma_j)/p_0(x)$ and component scores $a_j = \Sigma_j^{-1}(\mu_j - x)$, exact derivatives are

$$s_0(x) = \sum_j r_j a_j, \qquad H_0(x) = \sum_j r_j \Sigma_j^{-1} - \sum_j r_j (a_j - s_0)(a_j - s_0)^\top.$$

These derivatives are evaluated at the current reference samples. No exact noisy mixture scores or target samples enter the score field of Algorithm 1. Exact target sampling is used for evaluation and display only.

Sample errors use (E.1) with $v_\ell = (\cos(\pi\ell/64), \sin(\pi\ell/64))$, $\ell = 0, \ldots, 63$. The scalar σ_{tar} is computed from the analytic mixture covariance. No covariance whitening is applied. Final metrics select 1,536 samples from each method and compare them with the same 1,536 independent target samples within a seed. The GAD and target sample indices are spread evenly over their saved 4,096-point sample sets. MCMC selection retains 384 evenly spaced production states per chain, including rejection repeats. It creates no duplicate observations by resampling, although actual chain rejections can repeat coordinates. The target/target comparison uses another independent target sample. Seeds 42–44 define three complete runs, and the reported SD uses divisor $3 - 1$, rather than being a standard error or a confidence interval.

F.2 Display controls

The displays in this subsection use the preselected seed 42 and the window $[-5, 5]^2$. The main histograms in Figure 4 and equal-base histograms in Figure 12 use 128×128 bins. For the main display, the target panels start with 60,000 independent target draws. The GAD panels use 60,000 independent reverse-SDE draws transported by the third-round score field, which is computed from the saved round-two reference samples. No further round is run. All columns apply the same display procedure. For each display, we resample the method's base sample set uniformly and add independent Gaussian jitter of standard deviation 0.10 in physical coordinates, producing 60,000 displayed points. MALA and HMC use their original production states with empirical weights, including rejection repeats. The MCMC panels show resampled chain states. The colors give log probability per bin with the full sample count in the denominator, including points outside the window.

Figure 12 restricts every method to the 1,536 samples used in Table 4 and then applies the display procedure of Figure 4. The target and GAD histograms then also become granular, which shows how much of the smoothness in Figure 4 comes from the number of base draws. Figure 13 shows the metric samples in 48×48 bins. Figure 14 shows independent target and GAD draws

in 128×128 bins. Figures 13 and 14 display their original sample sets without resampling or jitter. A fixed GAD field allows us to draw many additional samples cheaply, which smooths the displays. The two-dimensional SW_2 comparison uses equal-size sample sets, as reported in Table 4.

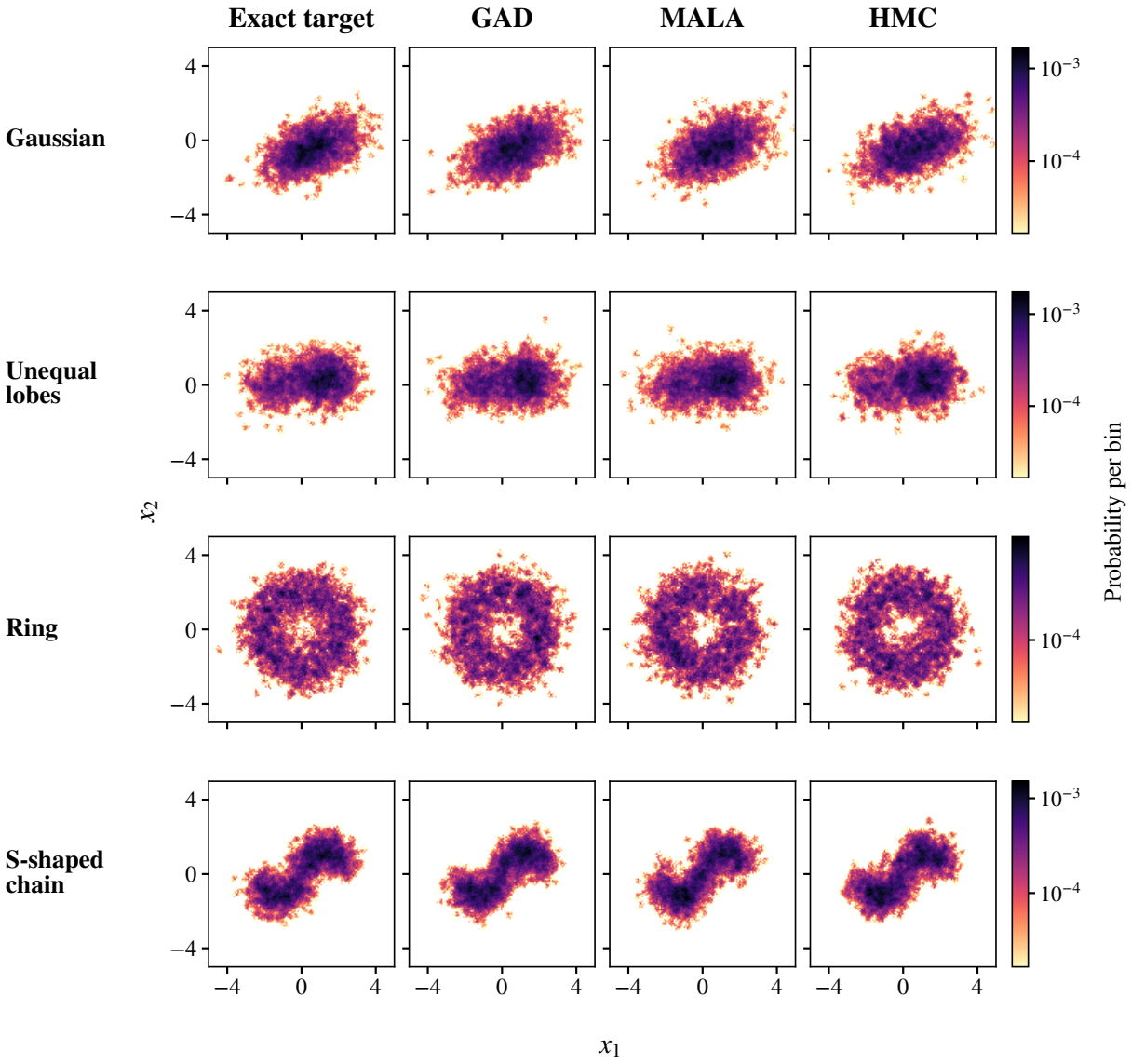


Figure 12: **High-resolution display with equal base samples.** Every panel starts with its 1,536 metric samples from seed 42. Resampling, jitter, and the color scale match Figure 4. With equal base counts, the target and GAD histograms also become granular. Appendix F.2 gives the display settings.

F.3 GAD field, transport, and request counts

The common initial law is $\mathcal{N}((0.6, -0.4), 4I_2)$. GAD uses 4,096 reference samples per round for three rounds. Each round starts with fresh terminal samples from $\mathcal{N}(0, I_2)$. Stochastic Heun integration uses 240 steps from $T = 5$ to $t_{\min} = 10^{-4}$, uniform in $u = \log(\mathrm{expm1}(2t) + 0.2)$, with

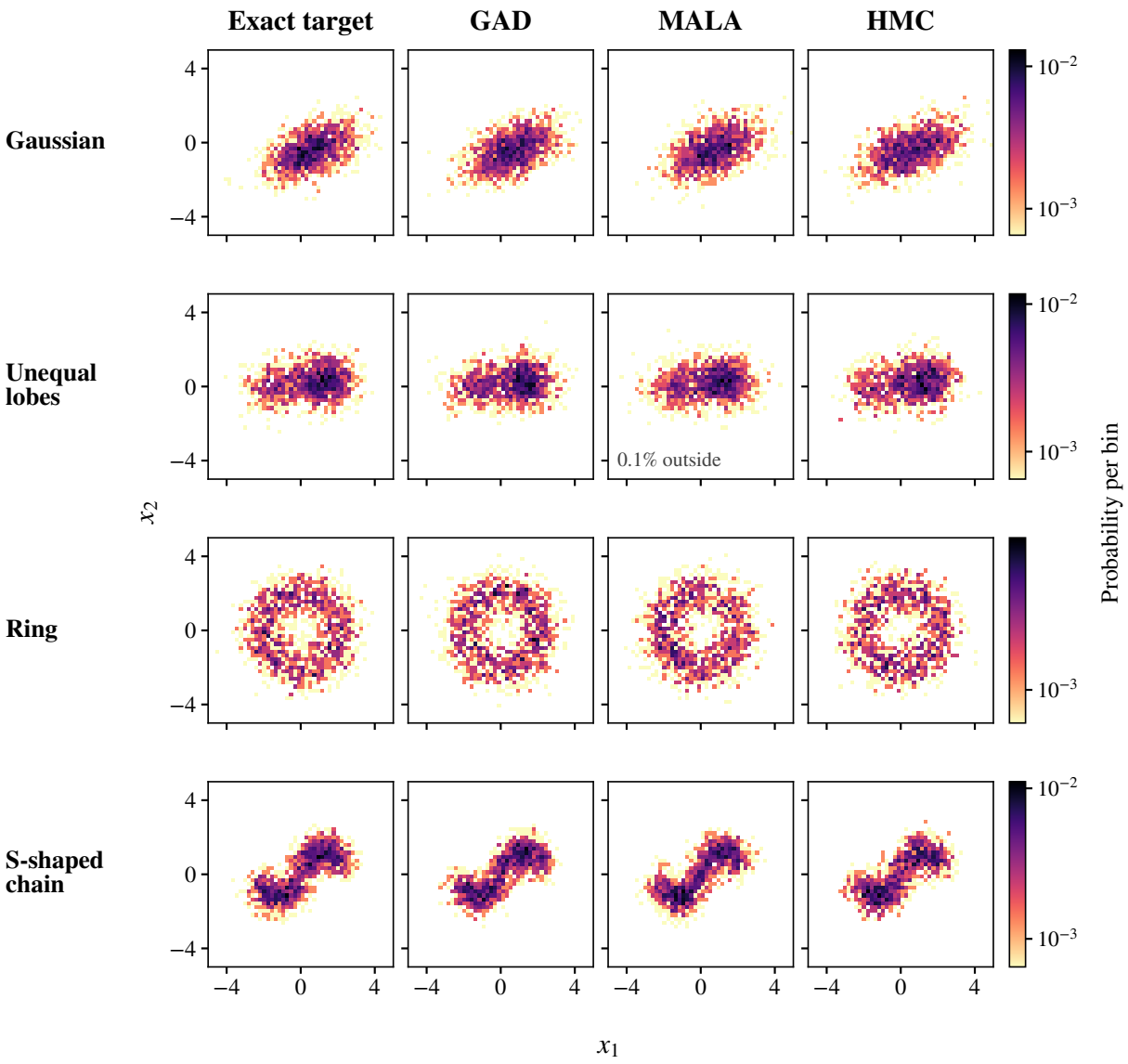


Figure 13: **Samples used by the two-dimensional metrics.** Rows show the four targets and columns show target, GAD, MALA, and HMC samples in the original coordinates. Each seed-42 panel uses the 1,536 metric samples directly. Axes and log probability-per-bin colors are shared within each row. Table 4 uses all points, including those outside the displayed window. Appendix F.2 gives the binning and normalization.

the same Brownian increment in predictor and corrector. There is no final denoising or density fit. The frozen score is tabulated at 65 times uniform in u, interpolated linearly in time and bilinearly in space. Beyond the spatial table the nearest boundary score is used. Transported samples can leave the grid.

For moments under the current law, we use a Cartesian reference grid with 129 nodes per coordinate. Its half-width is $B = \max\{9, \max_i \|X_i\|_\infty + 0.1\}$. Each occupied cell retains empirical mass, mean clean position, mean target gradient, and mean target observed-information matrix. The OU kernel is evaluated at the cell node. At time t, the 129×129 score-evaluation grid has half-width $\alpha_t B + 6\sqrt{\gamma_t}$. Separable Gaussian-kernel products are rescaled before normalization.

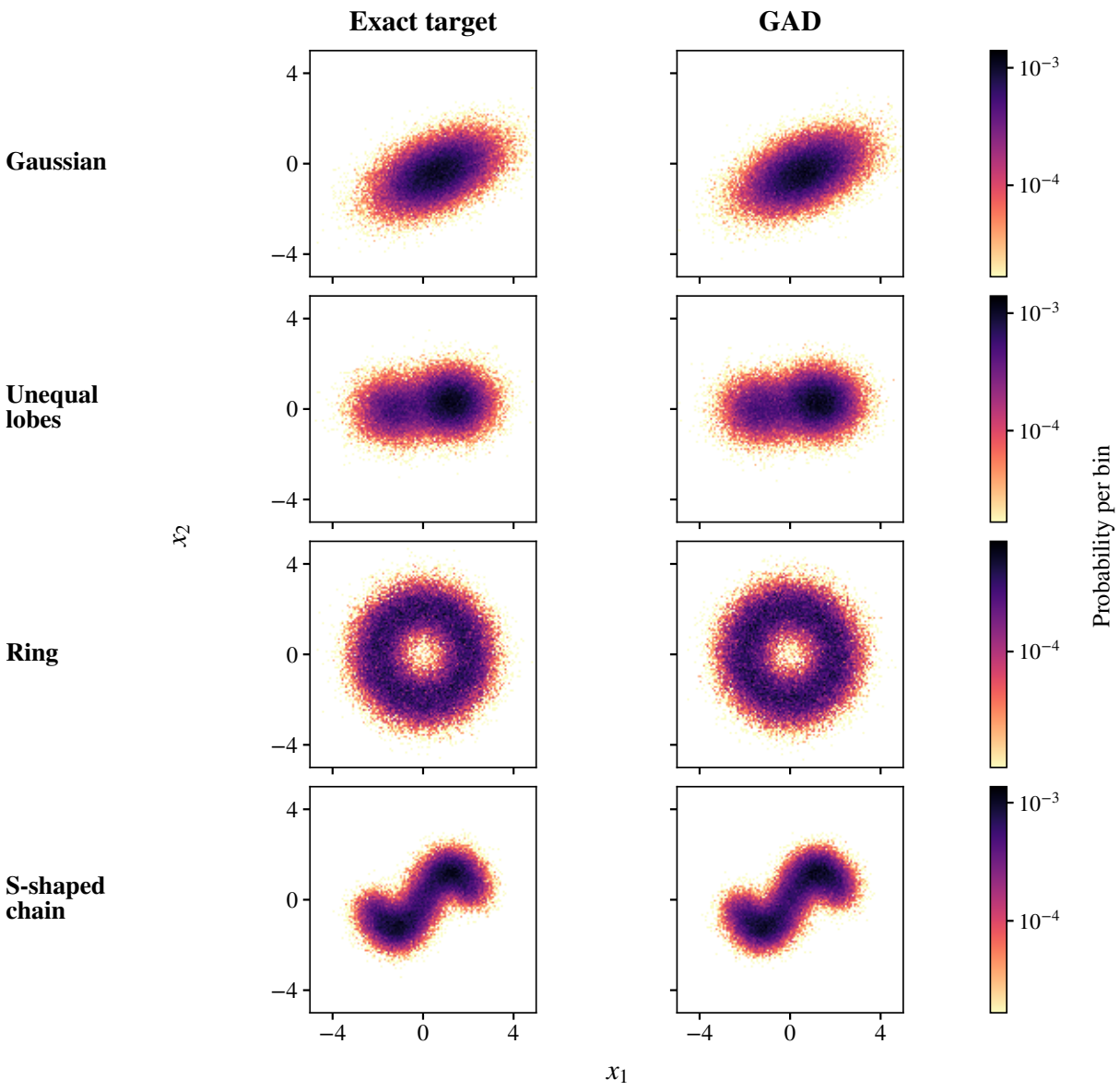


Figure 14: **Independent target and GAD draws.** Each target is paired with output from the third-round GAD field for seed 42. Each panel uses 60,000 independent draws in the original coordinates, without resampling or jitter. Log probability-per-bin colors are shared within each row. The additional GAD draws require no new target derivatives. Appendix F.2 gives the binning and normalization.

Near-underflow rows are recomputed by log-sum-exp over occupied cells. The log-sum-exp recomputation retains the occupied-cell masses and cached position, gradient, and Hessian means.

For an eigenvalue λ of the conditional mean observed information, the raw gate eigenvalue is $g_{\text{raw}} = \alpha_t^2/(\alpha_t^2 + \gamma_t\lambda)$. Shifted denominators with magnitude below 10^{-8} are replaced by that signed magnitude, with positive sign at zero. Nonpositive raw gates are replaced by 10^{-3}. Positive values are capped at 50, while positive values below 10^{-3} remain unchanged. These safeguards, cell quadrature, interpolation, finite horizon, and time discretization are approximations not covered by Theorem 8.4.

The request allowance is $W = n_E + n_G + n_H = 24{,}576$, where n_E, n_G, n_H count requested energy, gradient, and Hessian evaluations by derivative order. GAD requests a gradient and Hessian for each reference sample in each round, giving $(n_E, n_G, n_H) = (0, 12{,}288, 12{,}288)$. The count W counts each requested quantity once. Field tabulation, transport, and display generation reuse the derivatives already evaluated at the reference samples. The quantitative comparison uses the samples drawn for the metric.

F.4 MALA and HMC at the fixed allowance

MALA proposes $x' = x + \varepsilon^2 s_0(x)/2 + \varepsilon\xi$, $\xi \sim \mathcal{N}(0, I_2)$, with the exact Hastings correction. HMC refreshes momentum from $\mathcal{N}(0, I_2)$, takes leapfrog steps with unit mass, and applies the Hamiltonian acceptance test [29]. Both use the same target and start four chains from the saved initial GAD reference samples, giving them the same broad initial law. Neither uses covariance, Hessian, or MAP preconditioning. Both retain rejection repeats.

Warmup uses 20% of the total request allowance, including initial energy and gradient evaluations. During warmup, we adapt a common step across the four chains by dual averaging, starting from $\varepsilon = 0.2$. The target mean acceptance is 0.574 for MALA and 0.8 for HMC. The constants are $\gamma_{\mathrm{DA}} = 0.05$, $t_{\mathrm{DA}} = 10$, $\kappa_{\mathrm{DA}} = 0.75$, and $\mu_{\mathrm{DA}} = \log(10\varepsilon_{\mathrm{initial}})$, where DA denotes dual averaging. Adapted steps are constrained to $[10^{-4}, 5]$. The averaged log step is frozen for production. HMC trajectory lengths are drawn uniformly from $\{5, \ldots, 15\}$, with a final state-independent shortening if needed to fit the remaining allowance. No tuning uses independent target samples.

Each MALA proposal requests one new energy and gradient per chain, reusing the current values. The final MALA request count is $(12{,}288, 12{,}288, 0)$. HMC requests one gradient per leapfrog step and only the final energy, reusing the current values. The gradient and energy counts depend on the trajectory lengths. The runs spend 24,572–24,576 requests under the common allowance, with at most four requests left unspent and no curvature-matrix requests. MALA produces 9,832 production states per run. In seed 42, HMC produces 1,764–1,808 production states, depending on the target. The 1,536 states used in the metric fit within each run's production states, without extending a chain or resampling.

F.5 Configuration selection

We chose 4,096 reference samples and three rounds after an earlier study that varied the reference sample count and number of GAD rounds. In that study, with 4,096 samples in the metric, the mean normalized SW_2 of this setting across the four targets is 3.488, compared with 3.693 at round four and 3.604 at round eight. Three rounds gave the smallest average error in that study. Nearby settings gave similar values. All targets and seeds use this round count. Table 4 reports the separate evaluation with 1,536 samples.

G One-dimensional illustration details

The illustration in Section 7.1 fits a mixture to each current law. It differs from the exact-expectation calculation in Appendix E.3 and from the sample-based estimator in Algorithm 1. The following settings describe only that illustration.

G.1 Mixture fit, transport, and display

- **Target.** The target density is $p_0 = \sum_{j=1}^{3} w_j \mathcal{N}(\mu_j, \sigma_j^2)$ with $w = (0.40, 0.37, 0.23)$, $\mu = (-3.2, 0, 1.7)$, and $\sigma = (0.72, 0.85, 0.58)$. The score $s_0 = (\log p_0)'$ and the observed information $H_0 = -(\log p_0)''$ are evaluated in closed form from the mixture responsibilities.
- **Initialization.** The initial law is the analytic mixture $q_0 = 0.34\,\mathcal{N}(-0.25, 1.45^2) + 0.33\,\mathcal{N}(0, 1.45^2) + 0.33\,\mathcal{N}(0.25, 1.45^2)$. It has mean -0.0025 and variance 2.144. It is near-Gaussian, not exactly Gaussian, and is not represented by samples.
- **Conditional expectations and projection.** Each q_k is a three-component Gaussian mixture. The OU-posterior moments entering m_t^q and H^q are computed by 16-point Gauss–Hermite quadrature under the current mixture q_k conditioned on the noisy state, not by the sample-based weights (I.1). For this illustration, we tabulate the frozen score field at 1,201 equally spaced points on $y \in [-7, 5.5]$ and 91 times. Interpolation in y is linear during transport. The 5,000 endpoints of each round are fitted by unweighted expectation–maximization with three components (at most 100 iterations, tolerance 10^{-7}, variance regularization 10^{-4}). The fitted mixture is q_{k+1}.
- **Gate and indefinite observed information.** The exact H_0 is used, with no positive surrogate. The raw gate is $G_{\mathrm{raw}} = \alpha_t^2 / A^q$ with $A^q = \alpha_t^2 + \gamma_t H^q$. Values with $|A^q| < 10^{-8}$ are set to magnitude 10^{-8} with their sign kept, G_{raw} is clipped to $[-50, 50]$, and the transport uses $G = \max(G_{\mathrm{raw}}, 10^{-3})$. In round one, $A^q \le 0$ at 8.42% of the $91 \times 1{,}201$ tabulated score-evaluation points and at least one denominator or gate safeguard activates at 8.52%. In later rounds, A^q is positive at every tabulated score-evaluation point and none of the denominator or gate safeguards activates.
- **Reverse diffusion.** The stochastic reverse-OU Heun scheme starts from $\mathcal{N}(0, 1)$ at $T = 2.5$ and takes 90 steps on a uniform time grid to $t_{\min} = 5 \times 10^{-3}$. No final denoising step is applied. The terminal mismatch is $\mathrm{KL}(p_T \| \mathcal{N}(0, 1)) = 0.0028$.
- **Rounds and output.** The illustration uses five rounds, each drawing fresh terminal samples and Brownian increments to produce 5,000 endpoints. Rounds 0, 1, 2, 3, and 5 are displayed.
- **Displayed densities.** The histograms use 319 equal bins on $[-6.8, 5.0]$ (width 0.0370). A Gaussian filter with standard deviation 0.9 bins smooths the histograms, which are then renormalized by trapezoidal integration. The target curve uses 400,000 draws from p_0, and each q_k curve uses 300,000 draws from the fitted mixture q_k. These draws affect only the display and the KL estimate, not the iteration.
- **Forward-KL estimator.** With x_i the bin centres and $\widehat{q}_k$ the smoothed q_k histogram, we estimate forward KL by

$$\widehat{\mathrm{KL}}(p_0 \| q_k) = \sum_i{}^{\mathrm{trap}}\, p_0(x_i)\{\log[p_0(x_i) + 10^{-12}] - \log[\widehat{q}_k(x_i) + 10^{-12}]\},$$

  where $\sum^{\mathrm{trap}}$ is trapezoidal quadrature and p_0 is evaluated exactly. Rounds 0 to 5 give 0.5724, 0.0761, 0.0254, 0.0091, 0.0042, and 0.0035. The KL of the fitted mixtures by direct quadrature is 0.5691, 0.0756, 0.0255, 0.0086, 0.0041, and 0.0035.
- **Estimator floor.** With a 300,000-draw target sample in place of q_k, the plotted estimator gives $(2.06 \pm 0.38) \times 10^{-4}$ over 30 seeds. A nonparametric estimate from a single 5,000-sample output, which this experiment does not use, would have a floor of 0.0113 ± 0.0044 over 100 seed pairs (5–95% range 0.0054–0.0192). The iteration itself carries the sampling noise of fitting a three-component mixture to 5,000 endpoints.
- **Seeds.** The transport producing q_{k+1}, $k = 0, \ldots, 4$, uses seed $41 + k$ for its terminal sample

and Brownian path. The target display sample uses seed 999, and the q_k display samples use seed $2000 + k$.

H Allen–Cahn and Navier–Stokes inverse problems

The Allen–Cahn and Navier–Stokes initial-condition inversions qualify the acoustic and earthquake application results in Section 13. Both forward models evolve a nonlinear partial differential equation (PDE). Here we ask whether the resulting posterior differs enough from MAP–Laplace for further GAD rounds to improve on the MAP–Laplace approximation. The comparison examines coefficient distributions and spatial field moments against NUTS, then measures the improvements under a common request allowance.

H.1 Forward models and observations

Both problems use the posterior in (13.1) with $\alpha \sim \mathcal{N}(0, I_d)$, a periodic unit square, a 32×32 grid, and one synthetic dataset generated with seed 42. In the Allen–Cahn problem the order parameter satisfies

$$\partial_t u = 0.045^2 \Delta u + u - u^3, \qquad u(0) = B\alpha.$$

The columns of B are the leading 32 eigenvectors of the grid covariance $\exp(-\|x - x'\|/0.1)$, scaled by the square roots of their eigenvalues. The Fourier solver uses implicit diffusion and explicit reaction with step 0.0005. The observations consist of u at 40 fixed grid sites at time 0.33, with independent Gaussian noise of standard deviation 0.01.

In the Navier–Stokes problem the vorticity satisfies

$$\partial_t \omega + v \cdot \nabla \omega = 7.5 \times 10^{-4} \Delta \omega + f, \qquad v = (\partial_y \psi, -\partial_x \psi), \qquad \Delta \psi = \omega.$$

In this vorticity equation, $\omega = \partial_y v_x - \partial_x v_y$. The initial field is a mean-zero Fourier expansion with 36 coefficients. The basis uses a cosine and a sine for each of the first 18 nonzero integer wave vectors k on the half-plane $k_x > 0$ or $k_x = 0,\ k_y > 0$, ordered by $(\|k\|^2, \|k\|_1, k_x, k_y)$. Each mode has unit grid Euclidean norm before weighting by $1.2 \exp[-((2\pi\, 0.075 k_x)^2 + (2\pi\, 0.13 k_y)^2)/2]$. The forcing on the grid is

$$\begin{aligned} f(x, y) = {} & 0.14 \sin(2\pi(1.5x + 0.35y)) + 0.08 \cos(2\pi(0.25x + 2.5y)) \\ & + 0.04 \sin(2\pi(3x - 0.75y)). \end{aligned}$$

The solver uses Crank–Nicolson diffusion with explicit advection and forcing. The time step is 0.05. It keeps Fourier modes with $|k_x|, |k_y| \le 32/3$ for dealiasing and zeros the vorticity mean after each step. At time 10.5, we apply the Fourier multiplier $\exp[-((k_x/8)^2 + (k_y/4)^2)/2]$ and observe the filtered vorticity at 80 fixed grid sites with independent Gaussian noise of standard deviation 0.005. Appendix I.6 identifies the files containing the frozen sensor indices, observations, and basis arrays for both problems.

H.2 Sampling and evaluation

Both GN-GAD and MAP-pMALA start from MAP–Laplace. MAP-pMALA runs eight chains with 64 warmup steps per chain and initial step size $h = 0.25$ in the proposal of Section 10. GN-GAD uses 1,000 reference samples per round, Gauss–Newton curvature, and 300 flow steps on a geometric grid from 10 to 10^{-7}. For both PDE inversions, we report the first three rounds of the saved five-round Allen–Cahn and four-round Navier–Stokes runs. The five paired sampler seeds

are 43–47. At each round, the MAP-pMALA checkpoint is the last complete chain batch within the cumulative $E + G + H$ allowance of (13.2), including common MAP setup and MAP-pMALA warmup. MAP–Laplace is evaluated once per target at setup cost.

Each target has a fixed NUTS benchmark with four chains, each with 3,000 warmup and 5,000 retained draws. Target acceptance is 0.95 and maximum tree depth in sampling is 12. Neither run records a divergence. Maximum rank-normalized $\widehat{R}$ is 1.00099 for Allen–Cahn and 1.00100 for Navier–Stokes. For evaluation, we whiten coefficients using the mean and covariance of the 20,000 NUTS benchmark draws. In those coordinates, SW_2 is the root mean square of the one-dimensional Wasserstein-2 distances over 128 fixed unit directions. Each of MAP–Laplace, GN-GAD, and MAP-pMALA contributes at most 1,000 states to the NUTS-whitened SW_2 metric and the NUTS benchmark contributes all 20,000. The deterministic stratified MAP-pMALA subsample changes with the cumulative checkpoint. Covariance error is the relative Frobenius discrepancy in the whitened coordinates. Initial-field mean and pointwise standard deviation errors are relative Euclidean discrepancies from the corresponding NUTS fields. Moment estimates use all saved states, including MAP-pMALA rejection repeats. At the third checkpoint the moment estimates use 10,000 MAP–Laplace Gaussian draws, 1,000 GN-GAD samples, and 3,976 cumulative MAP-pMALA states.

H.3 Posterior shape and field uncertainty

The NUTS projections in Figure 15 show one roughly elliptical region in each view. MAP–Laplace and GN-GAD recover the broad shapes, including the strongly tilted Allen–Cahn projection in the prior-whitened coefficients.

MAP–Laplace, GN-GAD, and MAP-pMALA reproduce the main features of the NUTS mean in Figure 16. They also recover the regions of larger pointwise uncertainty. The projections and field maps use one sampler seed and probe selected features of the posterior.

H.4 Accuracy at matched request allowances

At round three, GN-GAD has lower SW_2 than MAP-pMALA in all five paired seeds on each problem (Figure 17). Mean SW_2 for Allen–Cahn is 0.0651, 0.0624, 0.1014 for MAP–Laplace, GN-GAD, and MAP-pMALA, respectively. The corresponding Navier–Stokes values are 0.0895, 0.0834, 0.1052. The gain over MAP–Laplace is about 0.003 and 0.006, respectively. In Allen–Cahn the MAP–Laplace and GN-GAD errors are close to the NUTS split discrepancy (0.0595), a descriptive finite-sample scale for SW_2. MAP–Laplace gives the smallest observed covariance and initial-field moment errors on both targets.

The MAP–Laplace relative error in pointwise field standard deviation is about 1.2% for each problem. GN-GAD gives 2.2% for Allen–Cahn and 2.4% for Navier–Stokes. Through the third checkpoint, mean recorded times on the NVIDIA T4 graphics processing unit (GPU) are 12.9 minutes for GN-GAD versus 5.9 minutes for MAP-pMALA on Allen–Cahn. On Navier–Stokes, GN-GAD takes 52.9 minutes versus 7.7 minutes for MAP-pMALA. These times include common MAP setup and MAP-pMALA warmup.

The 32-draw coupled check between 300 and 600 flow steps gave a MAP-whitened RMS discrepancy of 0.0518 against a tolerance of 0.05 for Navier–Stokes seed 47 at round two. For the reported results, we retain the 300-step samples, which initialized round three. All round-three coupled integration checks passed. GN-GAD also has lower round-three SW_2 than MAP-pMALA in each of the four Navier–Stokes seeds other than seed 47.

The Allen–Cahn and Navier–Stokes inversions condition low-dimensional initial fields on spatial

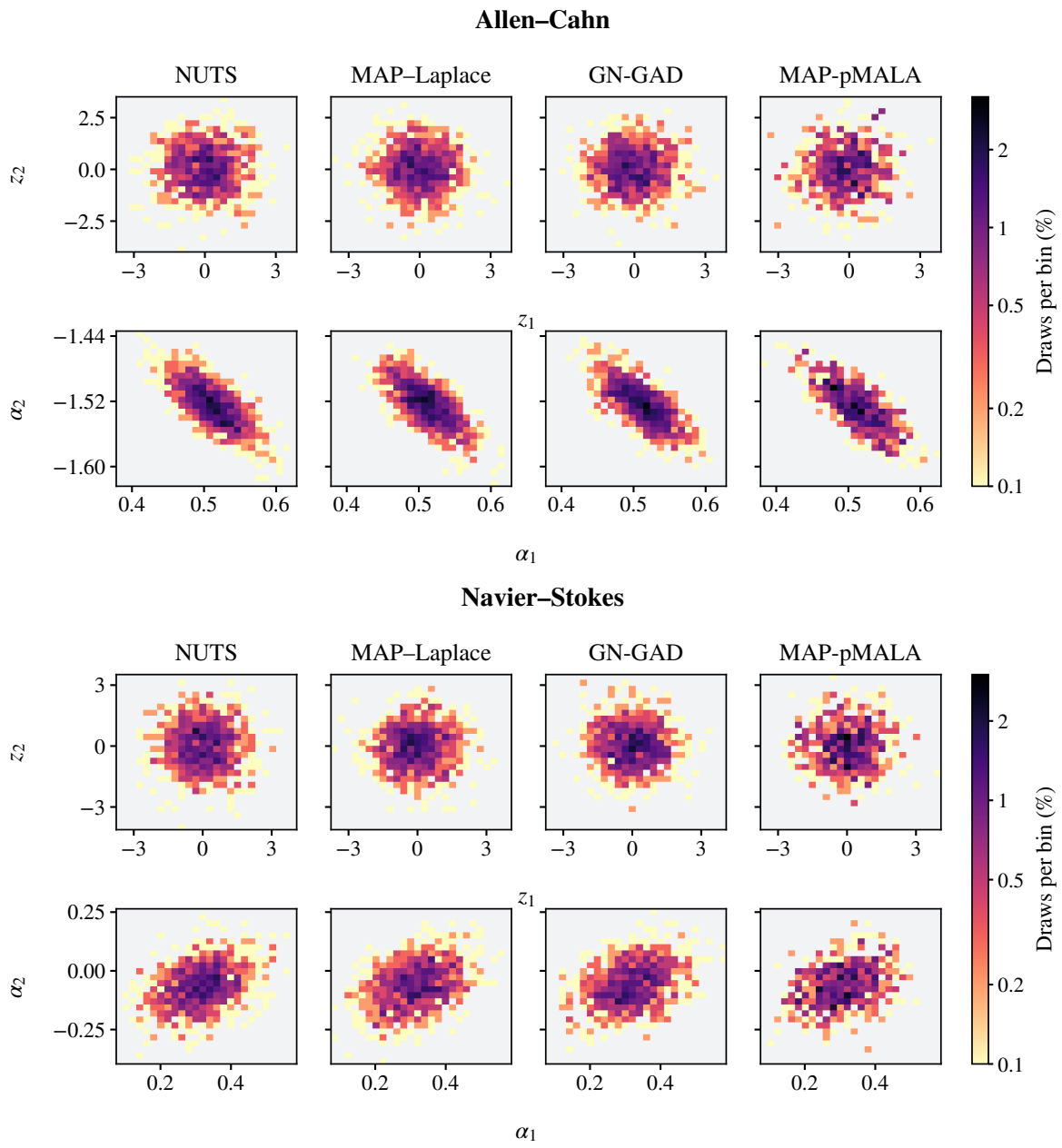


Figure 15: **Coefficient distributions for the two initial-condition inversions.** Allen–Cahn is above Navier–Stokes. Within each seed-43 block, the upper row gives coordinates (z_1, z_2) in the two lowest-curvature MAP Hessian directions. The projected coordinates are centered at the MAP and divided by the corresponding Laplace standard deviations. The lower row gives the first two prior-whitened coordinates (α_1, α_2). GN-GAD and MAP-pMALA use the round-three request checkpoint. Each panel has 1,000 states, with deterministic subsampling for NUTS, MAP–Laplace, and MAP-pMALA. Each row shares limits and 26×26 bins. Color shows the percentage of draws per bin on a common log scale within each problem. Gray bins are empty.

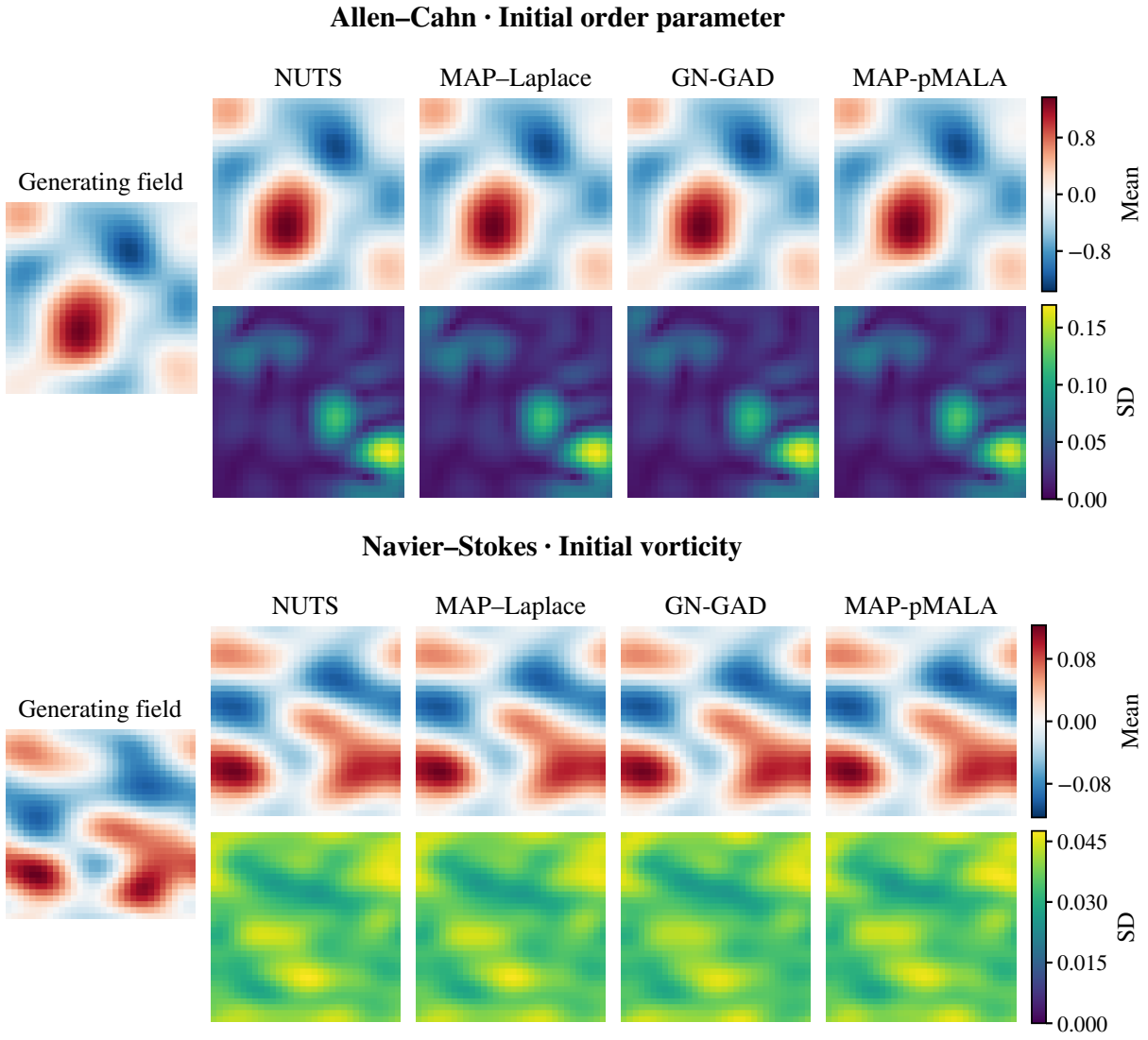


Figure 16: **Reconstructed initial fields and pointwise uncertainty.** The initial order parameter (Allen–Cahn) and vorticity (Navier–Stokes) are shown on the 32×32 spatial grid defined in Appendix H.1. Within each problem, synthetic truth and posterior means share a color scale. Posterior standard deviations share a separate scale. GN-GAD and MAP-pMALA are shown at the round-three request checkpoint for seed 43. Appendix H.2 gives the samples used for the maps.

measurements of the evolved order parameter or filtered vorticity. The resulting uncertainty in the displayed coefficients is compatible with fluctuations around a single reconstructed field. In acoustic inversion, each range difference confines the source near a curved locus at one pulse time, and the temporal prior couples those loci into a family of plausible routes. In earthquake location, the sparse picks allow lateral position and depth to trade off as origin times and shared velocities vary. The NUTS views in Figures 6 and 8 illustrate this curved uncertainty. The ellipsoidal contours of the MAP Gaussian keep the same principal directions throughout, so they capture only the local orientation of such a tradeoff. The annealed GN-GAD runs have the lowest mean primary sliced Wasserstein error in the acoustic and earthquake comparisons (Tables 5 and 6).

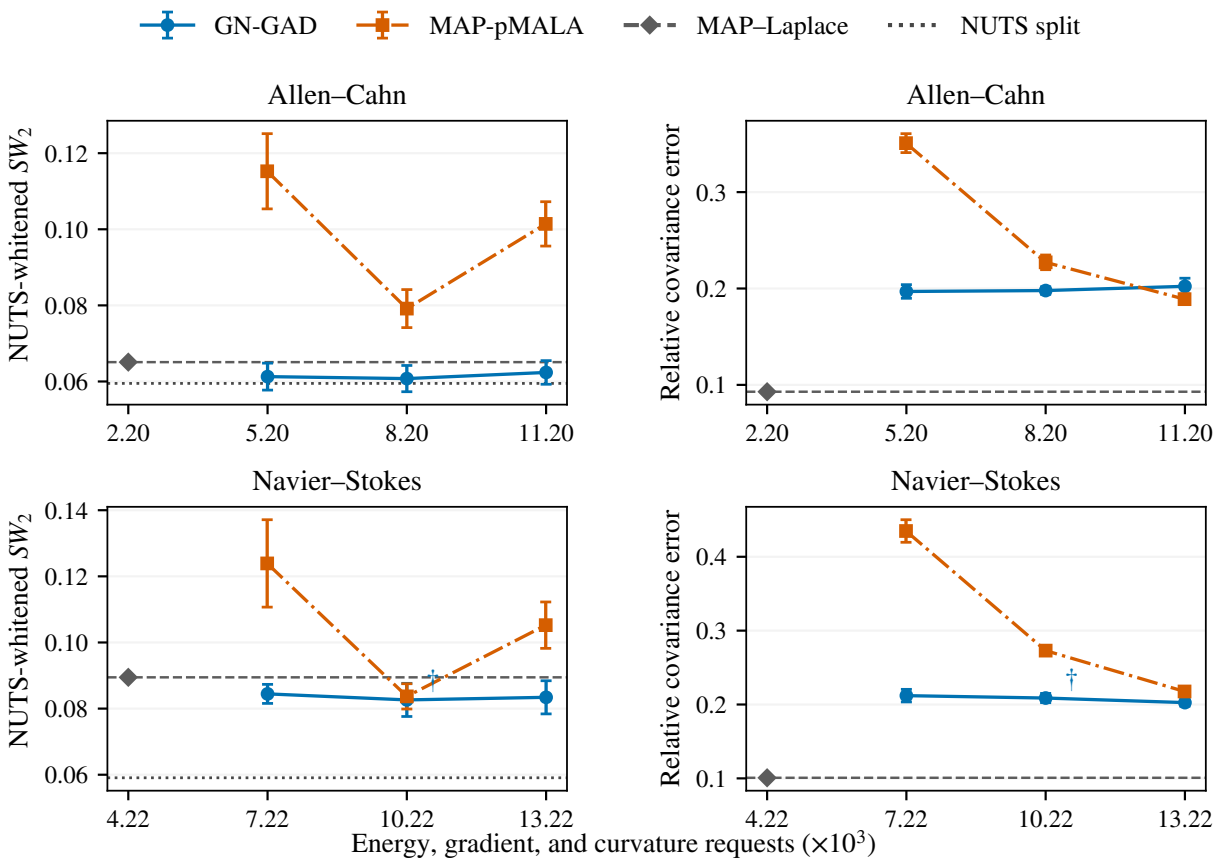


Figure 17: **Request-matched initial-condition inversions.** NUTS-whitened SW_2 (left) and relative Frobenius covariance error in NUTS-whitened coordinates (right) for Allen–Cahn and Navier–Stokes. Lower values indicate closer agreement with NUTS. The horizontal axis counts pointwise energy, gradient, and curvature-matrix requests, including common MAP setup and MAP-pMALA warmup. The three sampler checkpoints correspond to GN-GAD rounds one through three. Markers show means over five paired seeds, with bars indicating one sample SD. The MAP–Laplace diamond is at its setup cost, and the dashed line shows its error. The dotted line gives the mean SW_2 between a 1,000-draw NUTS subsample and its disjoint remainder over 100 splits. Appendix H.2 defines the metrics and sample sets. The dagger marks Navier–Stokes GN-GAD round two, where seed 47 exceeded the coupled integration tolerance. The mean includes all five seeds.

In the two field inversions, the gain in sliced Wasserstein distance over MAP–Laplace is small. MAP–Laplace already agrees closely with NUTS on the reported field summaries.

I Implementation and comparison details

I.1 The sample-based estimator

At GAD round k, let

$$X_{1:N}^{(k)} = (X_1^{(k)}, \dots, X_N^{(k)})$$

be the current unweighted reference samples. For a score-evaluation point (y,t), define the normalized OU weights

$$w_i(y,t) := \frac{K_t^{\mathrm{OU}}(y \mid X_i^{(k)})}{\sum_{j=1}^N K_t^{\mathrm{OU}}(y \mid X_j^{(k)})}. \tag{I.1}$$

The implementation evaluates these weights through equivalent Gaussian-kernel logits in log space. Averages using (I.1) are self-normalized importance-sampling (SNIS) averages when the reference samples are unweighted. The weight function is the OU kernel. The effective sample size (ESS) of these weights at (y,t) is

$$N_{\mathrm{eff}}(y,t) := \left(\sum_i w_i(y,t)^2\right)^{-1}.$$

For each reference sample, the inverse-problem implementations cache the exact clean score at the current likelihood power β and the Gauss–Newton approximation

$$H_{i,\beta}^{\mathrm{GN}} := H_{0,\beta}^{\mathrm{GN}}(X_i^{(k)}) = I_d + \beta DF(X_i^{(k)})^\top \Gamma^{-1} DF(X_i^{(k)}),$$

where DF is the Jacobian of the unwhitened forward map. Substituting $J = \Gamma^{-1/2}DF$ in (9.4) gives the cached matrix. The flow reuses the cached score and curvature at each integration stage. Study-specific safeguards are described in Appendices C and D.

Using the weights in (I.1), we average the Gauss–Newton matrices and form the shifted matrix as follows:

$$\widehat{H}_{\mathrm{GN}}^q(y,t) := \sum_i w_i(y,t) H_{i,\beta}^{\mathrm{GN}}, \qquad \widehat{A}_{\mathrm{GN}}^q(y,t) := \alpha_t^2 I_d + \gamma_t \widehat{H}_{\mathrm{GN}}^q(y,t).$$

The empirical Gauss–Newton gate is

$$\boxed{\widehat{G}_{\mathrm{GN}}^q(y,t) = \Psi_t(\widehat{H}_{\mathrm{GN}}^q(y,t)) = \alpha_t^2 (\widehat{A}_{\mathrm{GN}}^q(y,t))^{-1}.}$$

The symmetric resolvent uses linear solves or an equivalent eigendecomposition, with the numerical guards given in Appendices C and D. The sample-based score field is

$$\widehat{s}_t^{a,q}(y) = \widehat{b}^q(y,t) + \widehat{G}_{\mathrm{GN}}^q(y,t)\widehat{m}_t^q(y), \qquad \widehat{m}_t^q(y) := \widehat{c}^q(y,t) - \widehat{b}^q(y,t), \tag{I.2}$$

where $\widehat{b}^q$, $\widehat{c}^q$, and $\widehat{H}_{\mathrm{GN}}^q$ use the same reference samples and the same pointwise OU weights in the reported GAD implementation.

The acoustic and earthquake experiments integrate the flow (10.1) with the score estimate (I.2). Integration uses the classical fourth-order Runge–Kutta method on logarithmic OU-time grids. The synthetic studies use the reverse-SDE integrators stated in their appendices. Finite terminal time, nonzero endpoint time, conditional estimation from finitely many samples, and time discretization are numerical approximations. Theorem 8.4, which concerns the exact-expectation reverse SDE, does not cover them.

I.2 GN consistency and likelihood changes

The GN gate uses the following averages under the current OU posterior:

$$H_{\mathrm{GN}}^q(y,t) = \mathbb{E}_{\rho_{y,t}^q}[H_0^{\mathrm{GN}}(X_0)],$$
$$A_{\mathrm{GN}}^q(y,t) = \alpha_t^2 I_d + \gamma_t H_{\mathrm{GN}}^q(y,t), \qquad G_{\mathrm{GN}}^q(y,t) = \alpha_t^2 (A_{\mathrm{GN}}^q(y,t))^{-1}.$$

Table 17: **Inverse-problem GN-GAD settings.** Direct and annealed GN-GAD use the same number of samples per round, number of rounds, initial law, and allowance. The acoustic allowance includes 2,941 MAP setup requests. The earthquake runs start from the prior and require no MAP setup.

Problem	Dimension	Samples / rounds	Final allowance	Initial law
Acoustic	16	2,048 / 4	27,517	Equal mixture of prior and MAP–Laplace
Earthquakes	23	1,500 / 6	27,000	Prior $\mathcal{N}(0, I_{23})$

Here $A^q_{\mathrm{GN}} \succ 0$ by (9.1). The GN score error is $e^q_{t,\mathrm{GN}} = d_t - G^q_{\mathrm{GN}} m^q_t$. Where the Poisson moments are defined, (5.2) gives

$$\boxed{e^q_{t,\mathrm{GN}} = \left(I_d - (A^q_{\mathrm{GN}})^{-1} A^q\right) d_t - \alpha_t (A^q_{\mathrm{GN}})^{-1} r_H.} \tag{I.3}$$

The term multiplying d_t in (I.3) measures the effect of replacing the exact-curvature matrix A^q by the Gauss–Newton matrix A^q_{GN}. The second term is the flux covariance r_H from (5.1) multiplied by $-\alpha_t (A^q_{\mathrm{GN}})^{-1}$.

At a fixed likelihood power β, A^q_β denotes the shifted conditional mean of the exact observed information, and $A^q_{\mathrm{GN},\beta}$ denotes its Gauss–Newton counterpart. The exact observed information is $H_{0,\beta} = H^{\mathrm{GN}}_{0,\beta} + \beta R_{\mathrm{nl}}$, with R_{nl} as in (9.2). Wherever the conditional moments and Poisson representation are defined, the matrix factor multiplying d_t in (I.3) satisfies

$$I_d - (A^q_{\mathrm{GN},\beta})^{-1} A^q_\beta = -\gamma_t (A^q_{\mathrm{GN},\beta})^{-1} \mathbb{E}_{\rho^q_{y,t}}[\beta R_{\mathrm{nl}}(X_0)].$$

The matrix $A^q_{\mathrm{GN},\beta}$ is positive definite. Bounding the curvature term in (I.3) also requires a bound on the conditional mean of βR_{nl}. The experiments evaluate the combined sampling error of the Gauss–Newton implementation.

When the likelihood power changes, the previous endpoints supply reference samples for a new target. The score and Gauss–Newton matrix must both use the new power.

I.3 Inverse-problem comparison settings

Each problem uses one synthetic dataset chosen in an exploratory study. The MCMC settings are selected by their final primary sliced Wasserstein error on seed 42 and are fixed for seeds 43–47. The evaluation seeds share the dataset and NUTS samples. The physical histograms in Figures 6 and 8 use seed 43.

Direct and annealed GN-GAD share their initial samples and terminal Gaussian draws within each seed. Direct transport uses full likelihood power at every round. The annealed powers are $(0.07, 0.45, 1, 1)$ for acoustic inversion and $(0.02, 0.07, 0.2, 0.45, 1, 1)$ for earthquake location. The settings in Table 17 are shared by the two GAD variants within each problem. MAP-pMALA instead starts from MAP–Laplace. Other MCMC methods use the corresponding GAD initial law, except where explicitly identified as MAP starts or continuations.

The count W includes setup, initialization, warmup or pilot runs, and rejected proposals. It excludes the NUTS runs, metric evaluation, and display draws. It also excludes the work of evaluating the GAD score field from cached derivatives and integrating the reverse dynamics. The first acoustic allowance in Figure 9 is $9{,}085 = 2{,}941 + 6{,}144$ requests, including MAP setup and three requests per initial reference sample. The first earthquake allowance is 4,500 requests. A snapshot can use fewer requests than its allowance when the next complete step would exceed it. The per-problem records in Appendices C and D give the actual expenditures.

I.4 MCMC kernels, calibration, and retained laws

For $U_\beta(\alpha) = \|\alpha\|^2/2 + \beta U_{\text{like}}(\alpha)$, fixed-metric MALA, run as a population of parallel chains, proposes

$$\alpha' = \alpha - \frac{\epsilon^2}{2} C \nabla U_\beta(\alpha) + \epsilon C^{1/2} \xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

with the full Hastings correction. Here C is identity, the inverse exact full-posterior MAP Hessian, or a separately learned pilot covariance. GN-smMALA instead uses $C(\alpha) = (I_d + \beta J(\alpha)^\top J(\alpha))^{-1}$. Its MH correction includes the forward and reverse state-dependent Gaussian densities with their log determinants [17]. It does not add a metric-derivative drift. For HMC, C is the fixed inverse mass. The momentum law $p \sim \mathcal{N}(0, C^{-1})$ gives kinetic energy $p^\top C p/2$ and position update $\alpha \leftarrow \alpha + \epsilon C p$. After L leapfrog steps, a Hamiltonian MH correction completes the proposal. For MAP-pMALA, the step h in Section 10 satisfies $h = \epsilon^2/2$.

Each population method has a fixed production step and metric. Rejections produce repeated states. Invalid proposals are rejected. Annealed families change likelihood power at the GAD checkpoints specified in Appendix I.3. At each checkpoint, the reported empirical posterior uses the latter half of the $\beta = 1$ production states. Direct families retain the latter half of their production states. MAP-pMALA retains all post-warmup production states. The acoustic sensitivity check is in Table 10. Intermediate-power snapshots in Figure 9 are progress diagnostics, not posterior-production endpoints.

For the acoustic problem we evaluate 48 settings on seed 42, using the same dataset as the GAD runs. The population size is $N_{\text{pop}} \in \{64, 128\}$. For annealed identity MALA, we test $\epsilon \in \{0.003, 0.01, 0.03\}$, compared with $\{0.03, 0.1, 0.3\}$ for annealed MAP-pMALA. Both direct and annealed GN-smMALA use $\{0.1, 0.3, 0.8\}$. The identity-MALA and MAP-pMALA step ranges are also tested in annealed HMC with the matching identity or MAP inverse mass and $L \in \{8, 24\}$. For each method we select the setting with the lowest final joint SW_2 among those that produce full-power samples. The acoustic configuration search uses 1,251,696 requests in total, reported separately from each selected run. The acoustic powers $(0.07, 0.45, 1, 1)$ are obtained from the earthquake schedule by removing 0.02 and 0.2.

For the earthquake problem we evaluate 240 configurations per seed in addition to direct GN-GAD, annealed GN-GAD, and MAP-pMALA. Direct, annealed, and GAD-continuation MCMC families use the same ranges of population size and step, with an additional independently learned pilot mass for MALA. MAP-start HMC uses $N_{\text{pop}} \in \{4, 8\}$ and $L \in \{8, 24, 64\}$. Replica exchange uses four independent ladders with 8 or 16 geometrically spaced powers from 0.01 to one, and adjacent swaps. Only the cold chain contributes posterior states. SMC uses conditional ESS fractions 0.5 or 0.8, adaptive powers, systematic resampling at each stage, and three MALA/GN-smMALA moves or two HMC moves per stage. The pilot covariance is estimated from a separate set of 32 states, each moved ten times by GN-smMALA at step 0.3. The pilot covariance has 5% diagonal shrinkage and a positive trace-scaled ridge. The request count includes the pilot. Evaluation uses only the production states. GAD-continuation runs start from the annealed GN-GAD samples after round four in the same seed, count the 18,000 requests spent on the four preceding GAD rounds, and then run at $\beta = 1$. SMC retains all of its full-target production states. The other population methods retain the latter half of their full-target production states.

For each earthquake family and kernel, we select one configuration by its final E2–E5 block SW_2 on seed 42, considering all available mass choices. An annealed MAP-pMALA configuration is also selected from the settings that use the MAP preconditioner. The isotropic configuration has the lowest calibration error among the annealed fixed-metric MALA settings for both problems. Table 18 lists the selected settings. The NUTS samples used for selection never enter the transition

Table 18: **Selected MCMC settings.** Chosen by final primary SW_2 on calibration seed 42, then reused at every checkpoint for seeds 43–47. N_{pop} is population size, ϵ is the stated proposal/leapfrog step, and L is HMC trajectory length in leapfrog steps. MAP-pMALA adapts only during its warmup, which counts toward the allowance. A dash means not applicable. SMC additionally lists conditional ESS fraction c. Replica exchange lists the number of temperatures n_{temp}.

Method	Metric / inverse mass	N_{pop}	ϵ	L or extra
Acoustic				
Annealed HMC	Identity	128	0.01	8
Annealed MALA	Identity	128	0.03	–
Annealed MAP-pMALA	MAP	128	0.03	–
Annealed GN-smMALA	Local GN	64	0.8	–
GN-smMALA	Local GN	64	0.8	–
Earthquakes				
Direct MALA	Identity	128	0.03	–
Direct HMC	Identity	128	0.003	24
Annealed MALA	Identity	128	0.03	–
Annealed MAP-pMALA	MAP	128	0.03	–
GN-smMALA	Local GN	128	0.8	–
Annealed GN-smMALA	Local GN	128	0.8	–
Annealed HMC	Identity	64	0.03	8
GAD then MAP-pMALA	MAP	64	0.1	–
GAD then GN-smMALA	Local GN	64	0.3	–
GAD then HMC	Identity	64	0.03	24
MAP-start HMC	MAP	8	0.1	24
Replica-exchange HMC	MAP	32	0.1	24, $n_{temp} = 8$
Replica-exchange GN-smMALA	Local GN	64	0.8	–, $n_{temp} = 16$
SMC–MALA	Pilot	64	0.1	–, $c = 0.8$
SMC–GN-smMALA	Local GN	64	0.3	–, $c = 0.8$

kernels, initial states, or within-run adaptation. If the remaining allowance cannot cover one more complete step of all chains, or one more GAD round, a run stops and we record the number of requests actually used. The allowance applies to each selected run. We report the cost of the configuration search separately.

I.5 Numerical error and work between target requests

Storing a dense observed-information matrix requires $O(d^2)$ entries per reference sample, and factoring the shifted dense matrix requires $O(d^3)$ work per score-field evaluation point. Evaluating the field from cached derivatives and integrating the dynamics add work beyond the target requests in (13.2).

The iteration in Theorem 8.4 uses exact conditional expectations and observed information in the reverse SDE (3.1). In the score estimate (I.2), weighted averages using (I.1) introduce sampling error. The ESS measures weight concentration at a score-evaluation point. The first term of (I.3) gives the error from substituting Gauss–Newton curvature in the exact-expectation gate. A numerical implementation also incurs errors from shifted linear solves and transport discretization. It stops at positive t_{min} and computes each new score field from finitely many endpoints. The inverse problems use (10.1). In annealed runs, the target also changes with the likelihood power. The nonlinear-target sampling results measure the combined effect of

finite-reference score estimation, Gauss–Newton curvature, transport discretization, terminal and endpoint times, and any likelihood annealing.

The Gaussian checks for annealed acoustic transport and both earthquake variants evolve the Gaussian mean and variance through the same RK4 recursion as the samples. The resulting moments are compared with both the exact flow from the finite terminal start and the desired OU marginal at the endpoint. Each comparison must satisfy a 1% tolerance for the maximum relative variance error and the root mean square of the mean errors standardized by the marginal standard deviations. These checks concern the Gaussians specified in Appendices C.2 and D.2. The comparisons with NUTS assess the combined sampling error for the nonlinear targets.

I.6 Reproducibility records

The acoustic record combines the original `arrival120` case of the four-round, five-seed study with `acoustic120_extended_run_RESULTS.zip`. It preserves the exact observations, covariance basis, shared MAP estimate, and NUTS benchmark samples. It also contains the original GAD histories, original MAP-pMALA histories, annealed GAD reference samples, MCMC trajectories, frozen calibration choices, metrics at each request allowance, output-window checks, request counts, and separate display records. The NUTS benchmark samples, direct GAD runs, and MAP-pMALA runs are shared with that four-round study.

The earthquake record is `Quake_Integration_Handoff.zip`. It contains the 23-dimensional model, fixed data, source, settings for six rounds, NUTS diagnostics, all 7,596 metric rows, and configurations selected on seed 42. The record includes selected physical projections for one seed, all-seed figures, and data for the figures regenerated from the record. Reanalysis of the full trajectories requires the original campaign outputs.

Both studies use float64 sampling and differentiation. Their generating environments record JAX/jaxlib 0.6.2, NumPyro 0.19.0, and NumPy 2.2.6, together with hardware and run metadata. The acoustic NUTS benchmark and selected-run timing records originate from different stages.

The Allen–Cahn and Navier–Stokes records are `GAD_ALLEN_CAHN_FIVE_SEED_RESULTS.zip` and `GAD_NAVIER_STOKES_R4_S300_FIVE_SEED_RESULTS.zip`. They preserve the fixed data, forward solvers, NUTS draws, and sampler checkpoints. The fixed data include training and holdout sensor indices. For each target, the `target_arrays.npz` and `target_constants.npz` files in its directory under `code/frozen/` contain the arrays used by the solver. The three-round report uses the saved checkpoints.

The generating parameters are used to simulate the data. The starting distributions and preconditioners are obtained from the prior, MAP fit, or separately generated pilot samples as specified for each method in Section 10 and Appendix I.4. NUTS benchmark draws are used for evaluation and the configuration selection criterion. The request totals follow the saved run records. Configuration search and extra display draws are recorded separately.

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